2.0: Prelude to Limits

Science fiction writers often imagine spaceships that can travel to far-off planets in distant galaxies. However, back in 1905, Albert Einstein showed that a limit exists to how fast any object can travel. The problem is that the faster an object moves, the more mass it attains (in the form of energy), according to the equation

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

where m_0 is the object's mass at rest, v is its speed, and c is the speed of light. What is this speed limit? (We explore this problem further in the chapter)

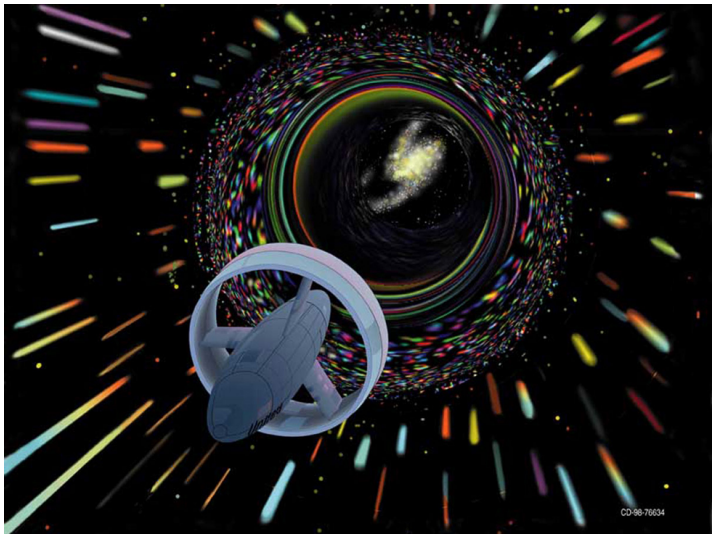


Figure 2.0.1:The vision of human exploration by the National Aeronautics and Space Administration (NASA) to distant parts of the universe illustrates the idea of space travel at high speeds. But, is there a limit to how fast a spacecraft can go? (credit: NASA)

The idea of a limit is central to all of calculus. We begin this chapter by examining why limits are so important. Then, we go on to describe how to find the limit of a function at a given point. Not all functions have limits at all points, and we discuss what this means and how we can tell if a function does or does not have a limit at a particular value. This chapter has been created in an informal, intuitive fashion, but this is not always enough if we need to prove a mathematical statement involving limits. The last section of this chapter presents the more precise definition of a limit and shows how to prove whether a function has a limit.

This page titled 2.0: Prelude to Limits is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

2.1: A Preview of Calculus

Learning Objectives

- Describe the tangent problem and how it led to the idea of a derivative.
- Explain how the idea of a limit is involved in solving the tangent problem.
- Recognize a tangent to a curve at a point as the limit of secant lines.
- Identify instantaneous velocity as the limit of average velocity over a small time interval.
- Describe the area problem and how it was solved by the integral.
- Explain how the idea of a limit is involved in solving the area problem.
- Recognize how the ideas of limit, derivative, and integral led to the studies of infinite series and multivariable calculus.

As we embark on our study of calculus, we shall see how its development arose from common solutions to practical problems in areas such as engineering physics—like the space travel problem posed in the chapter opener. Two key problems led to the initial formulation of calculus: (1) the tangent problem, or how to determine the slope of a line tangent to a curve at a point; and (2) the area problem, or how to determine the area under a curve.

The Tangent Problem and Differential Calculus

Rate of change is one of the most critical concepts in calculus. We begin our investigation of rates of change by looking at the graphs of the three lines $f(x) = -2x - 3,\ g(x) = \frac{1}{2}x + 1$, and $h(x) = 2$, shown in Figure 2.1.1.

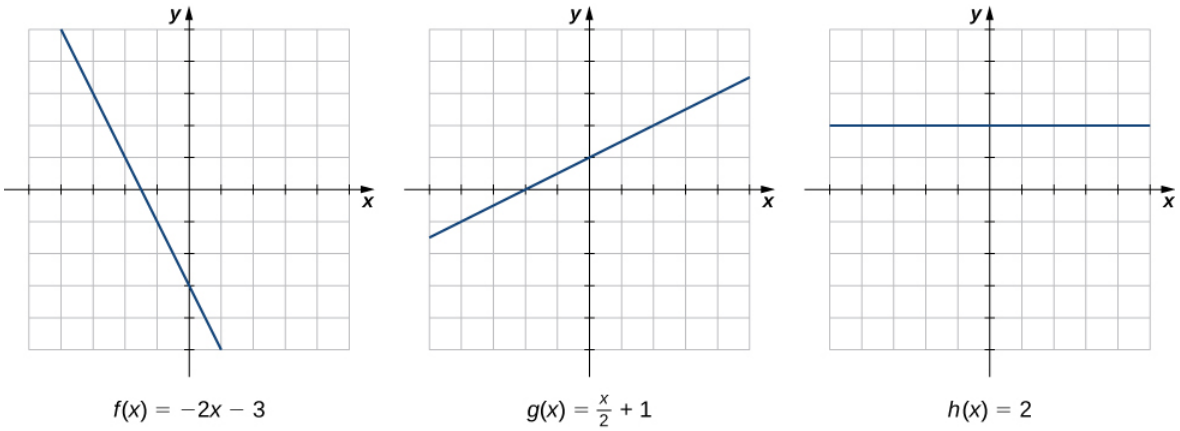


Figure 2.1.1: The rate of change of a linear function is constant in each of these three graphs, with the constant determined by the slope.

As we move from left to right along the graph of $f(x) = -2x - 3$, we see that the graph decreases at a constant rate. For every 1 unit we move to the right along the x-axis, the y-coordinate decreases by 2 units. This **rate of change** is determined by the slope (-2) of the line. Similarly, the slope of $1/2$ in the function $g(x)$ tells us that for every change in x of 1 unit there is a corresponding change in y of $1/2$ unit. The function $h(x) = 2$ has a slope of zero, indicating that the values of the function remain constant. We see that the slope of each linear function indicates the rate of change of the function.

Compare the graphs of these three functions with the graph of $k(x) = x^2$ (Figure 2.1.2). The graph of $k(x) = x^2$ starts from the left by decreasing rapidly, then begins to decrease more slowly and level off, and then finally begins to increase—slowly at first, followed by an increasing rate of increase as it moves toward the right. Unlike a linear function, no single number represents the rate of change for this function. We quite naturally ask: How do we measure the rate of change of a nonlinear function?

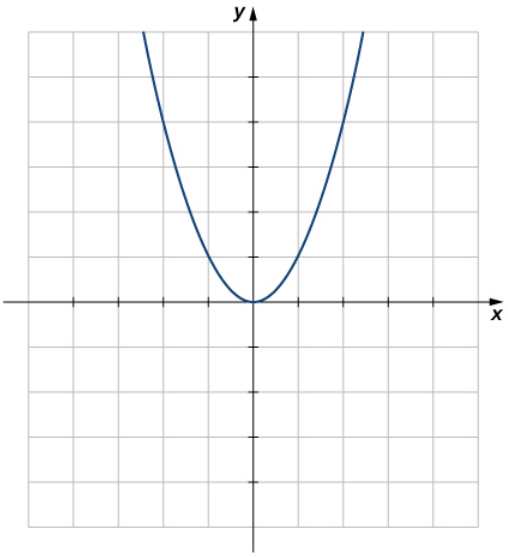


Figure 2.1.2: The function $k(x) = x^2$ does not have a constant rate of change.

We can approximate the rate of change of a function $f(x)$ at a point $(a, f(a))$ on its graph by taking another point $(x, f(x))$ on the graph of $f(x)$, drawing a line through the two points, and calculating the slope of the resulting line. Such a line is called a secant line. Figure 2.1.3 shows a *secant line* to a function $f(x)$ at a point $(a, f(a))$.

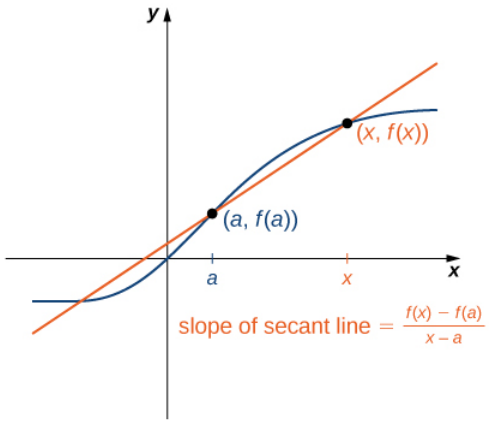


Figure 2.1.3: The slope of a secant line through a point $(a, f(a))$ estimates the rate of change of the function at the point $(a, f(a))$.

We formally define a secant line as follows:

Definition: Secant Line

The *secant* to the function $f(x)$ through the points $(a, f(a))$ and $(x, f(x))$ is the line passing through these points. Its slope is given by

$$m_{sec} = \frac{f(x) - f(a)}{x - a}. \tag{2.1.1}$$

The accuracy of approximating the rate of change of the function with a secant line depends on how close x is to a. As we see in Figure 2.1.4, if x is closer to a, the slope of the secant line is a better measure of the rate of change of $f(x)$ at a.

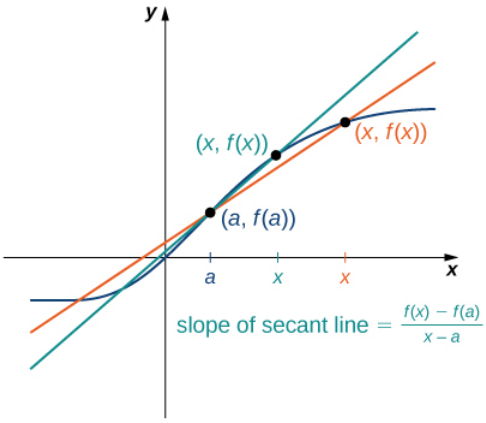


Figure 2.1.4: As x gets closer to a, the slope of the secant line becomes a better approximation to the rate of change of the function $f(x)$ at a.

The secant lines themselves approach a line that is called the **tangent** to the function $f(x)$ at a (Figure 2.1.5). The slope of the tangent line to the graph at a measures the rate of change of the function at a. This value also represents the derivative of the function $f(x)$ at a, or the rate of change of the function at a. This derivative is denoted by $f'(a)$. **Differential calculus** is the field of calculus concerned with the study of derivatives and their applications.

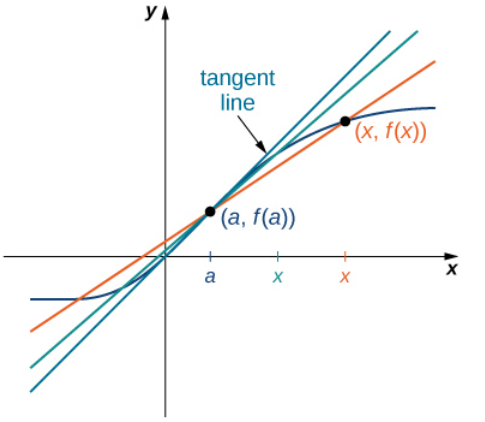


Figure 2.1.5: Solving the Tangent Problem: As x approaches a, the secant lines approach the tangent line.

Example 2.1.1 illustrates how to find slopes of secant lines. These slopes estimate the slope of the tangent line or, equivalently, the rate of change of the function at the point at which the slopes are calculated.

Example 2.1.1: Finding Slopes of Secant Lines

Estimate the slope of the tangent line (rate of change) to $f(x) = x^2$ at $x = 1$ by finding slopes of secant lines through $(1, 1)$ and each of the following points on the graph of $f(x) = x^2$.

a. $(2, 4)$
b. $\left(\frac{3}{2}, \frac{9}{4}\right)$

Solution:

Use the formula for the slope of a secant line (Equation 2.1.1).

a. $m_{sec} = \frac{4-1}{2-1} = 3$

b. $m_{sec} = \frac{\frac{9}{4}-1}{\frac{3}{2}-1} = \frac{5}{2} = 2.5$

The point in part b. is closer to the point $(1, 1)$, so the slope of 2.5 is closer to the slope of the tangent line. A good estimate for the slope of the tangent would be in the range of 2 to 2.5 (Figure 2.1.6).

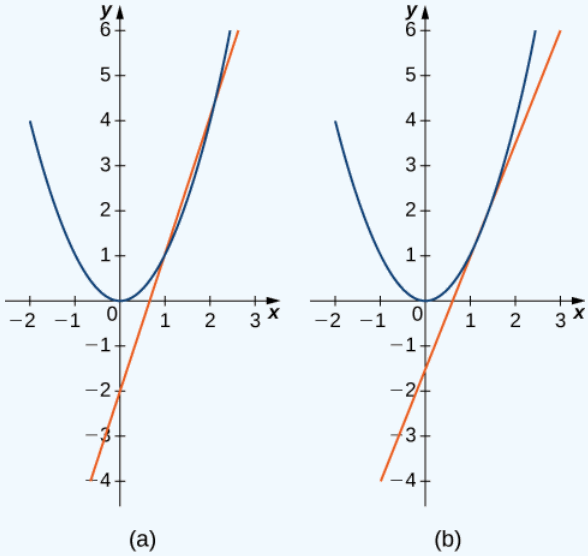


Figure 2.1.6: The secant lines to $f(x) = x^2$ at $(1, 1)$ through (a) $(2, 4)$ and (b) $(\frac{3}{2}, \frac{9}{4})$ provide successively closer approximations to the tangent line to $f(x) = x^2$ at $(1, 1)$.

? Exercise 2.1.1

Estimate the slope of the tangent line (rate of change) to $f(x) = x^2$ at $x = 1$ by finding slopes of secant lines through $(1, 1)$ and the point $(\frac{5}{4}, \frac{25}{16})$ on the graph of $f(x) = x^2$.

Answer

2.25

We continue our investigation by exploring a related question. Keeping in mind that velocity may be thought of as the rate of change of position, suppose that we have a function, $s(t)$, that gives the position of an object along a coordinate axis at any given time t. Can we use these same ideas to create a reasonable definition of the instantaneous velocity at a given time $t = a$? We start by approximating the instantaneous velocity with an average velocity. First, recall that the speed of an object traveling at a constant rate is the ratio of the distance traveled to the length of time it has traveled. We define the *average velocity* of an object over a time period to be the change in its position divided by the length of the time period.

Definition: Average Velocity

Let $s(t)$ be the position of an object moving along a coordinate axis at time t. The *average velocity* of the object over a time interval $[a, t]$ where $a < t$ (or $[t, a]$ if $t < a$) is

$$v_{ave} = \frac{s(t) - s(a)}{t - a}. \tag{2.1.2}$$

As t is chosen closer to a, the average velocity becomes closer to the instantaneous velocity. Note that finding the average velocity of a position function over a time interval is essentially the same as finding the slope of a secant line to a function. Furthermore, to find the slope of a tangent line at a point a, we let the x-values approach a in the slope of the secant line. Similarly, to find the instantaneous velocity at time a, we let the t-values approach a in the average velocity. This process of letting x or t approach a in an expression is called taking a **limit**. Thus, we may define the *instantaneous velocity* as follows.

Definition: Instantaneous Velocity

For a position function $s(t)$, the *instantaneous velocity* at a time $t = a$ is the value that the average velocities approach on intervals of the form $[a, t]$ and $[t, a]$ as the values of t become closer to a, provided such a value exists.

Example 2.1.2 illustrates this concept of limits and average velocity.

Example 2.1.2: Finding Average Velocity

A rock is dropped from a height of 64 ft. It is determined that its height (in feet) above ground t seconds later (for $0 \le t \le 2$) is given by $s(t) = -16t^2 + 64$. Find the average velocity of the rock over each of the given time intervals. Use this information to guess the instantaneous velocity of the rock at time $t = 0.5$.

a. $[0.49, 0.5]$
b. $[0.5, 0.51]$

Solution

Substitute the data into Equation 2.1.2 for the definition of average velocity.

a. $$v_{ave} = \frac{s(0.49) - s(0.5)}{0.49 - 0.5} = -15.84$$

b. $$v_{ave} = \frac{s(0.51) - s(0.5)}{0.51 - 0.5} = -16.016$$

The instantaneous velocity is somewhere between −15.84 and −16.16 ft/sec. A good guess might be −16 ft/sec.

Exercise 2.1.2

An object moves along a coordinate axis so that its position at time t is given by $s(t) = t^3$. Estimate its instantaneous velocity at time $t = 2$ by computing its average velocity over the time interval $[2, 2.001]$.

Hint

Use Equation 2.1.2 with $v_{ave} = \frac{s(2.001) - s(2)}{2.001 - 2}$.

Answer

12.006001

The Area Problem and Integral Calculus

We now turn our attention to a classic question from calculus. Many quantities in physics—for example, quantities of work—may be interpreted as the area under a curve. This leads us to ask the question: How can we find the area between the graph of a function and the x-axis over an interval (Figure 2.1.7)?

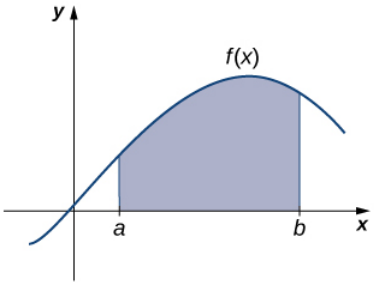


Figure 2.1.7: The Area Problem: How do we find the area of the shaded region?

As in the answer to our previous questions on velocity, we first try to approximate the solution. We approximate the area by dividing up the interval $[a, b]$ into smaller intervals in the shape of rectangles. The approximation of the area comes from adding up

the areas of these rectangles (Figure 2.1.8).

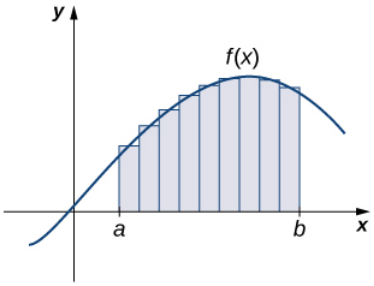


Figure 2.1.8: The area of the region under the curve is approximated by summing the areas of thin rectangles.

As the widths of the rectangles become smaller (approach zero), the sums of the areas of the rectangles approach the area between the graph of $f(x)$ and the x-axis over the interval $[a, b]$. Once again, we find ourselves taking a limit. Limits of this type serve as a basis for the definition of the definite integral. **Integral calculus** is the study of integrals and their applications.

Example 2.1.3: Estimation Using Rectangles

Estimate the area between the x-axis and the graph of $f(x) = x^2 + 1$ over the interval $[0, 3]$ by using the three rectangles shown in Figure 2.1.9.

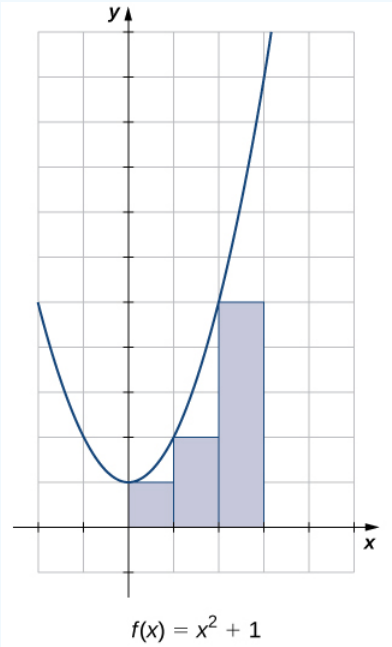


Figure 2.1.9: The area of the region under the curve of $f(x) = x^2 + 1$ can be estimated using rectangles.

Solution

The areas of the three rectangles are 1 unit^2, 2 unit^2, and 5 unit^2. Using these rectangles, our area estimate is 8 unit^2.

Exercise 2.1.3

Estimate the area between the x-axis and the graph of $f(x) = x^2 + 1$ over the interval $[0, 3]$ by using the three rectangles shown in Figure 2.1.10.

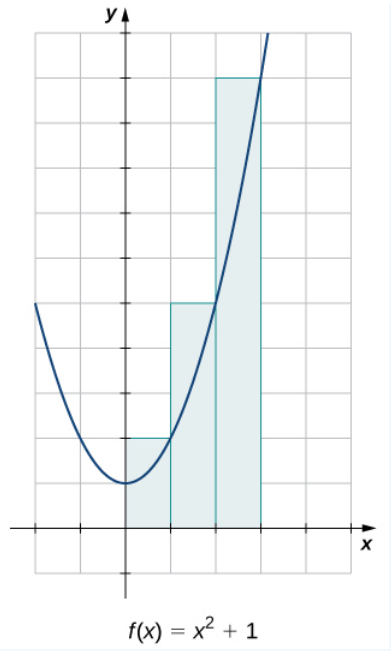


Figure 2.1.10: The area of the region under the curve of $f(x) = x^2 + 1$ can be estimated using rectangles.

Hint

Use Example 2.1.3 as a guide

Answer

17 unit^2

Other Aspects of Calculus

So far, we have studied functions of one variable only. Such functions can be represented visually using graphs in two dimensions; however, there is no good reason to restrict our investigation to two dimensions. Suppose, for example, that instead of determining the velocity of an object moving along a coordinate axis, we want to determine the velocity of a rock fired from a catapult at a given time, or of an airplane moving in three dimensions. We might want to graph real-value functions of two variables or determine volumes of solids of the type shown in Figure 2.1.11. These are only a few of the types of questions that can be asked and answered using *multivariable calculus*. Informally, multivariable calculus can be characterized as the study of the calculus of functions of two or more variables. However, before exploring these and other ideas, we must first lay a foundation for the study of calculus in one variable by exploring the concept of a limit.

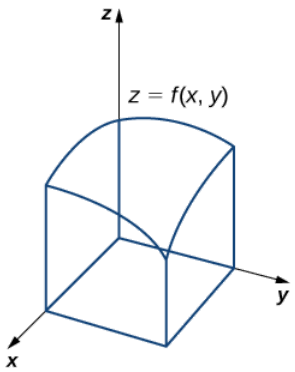


Figure 2.1.11: We can use multivariable calculus to find the volume between a surface defined by a function of two variables and a plane.

Key Concepts

- Differential calculus arose from trying to solve the problem of determining the slope of a line tangent to a curve at a point. The slope of the tangent line indicates the rate of change of the function, also called the *derivative*. Calculating a derivative requires finding a limit.
- Integral calculus arose from trying to solve the problem of finding the area of a region between the graph of a function and the x -axis. We can approximate the area by dividing it into thin rectangles and summing the areas of these rectangles. This summation leads to the value of a function called the *integral*. The integral is also calculated by finding a limit and, in fact, is related to the derivative of a function.
- Multivariable calculus enables us to solve problems in three-dimensional space, including determining motion in space and finding volumes of solids.

Key Equations

- **Slope of a Secant Line**

$$m_{sec} = \frac{f(x) - f(a)}{x - a}$$

- **Average Velocity over Interval [a,t]**

$$v_{ave} = \frac{s(t) - s(a)}{t - a}$$

Glossary

average velocity

the change in an object's position divided by the length of a time period; the average velocity of an object over a time interval $[t, a]$ (if $t < a$ or $[a, t]$ if $t > a$), with a position given by $s(t)$, that is $v_{ave} = \frac{s(t) - s(a)}{t - a}$

differential calculus

the field of calculus concerned with the study of derivatives and their applications

instantaneous velocity

The instantaneous velocity of an object with a position function that is given by $s(t)$ is the value that the average velocities on intervals of the form $[t, a]$ and $[a, t]$ approach as the values of t move closer to a, provided such a value exists

integral calculus

the study of integrals and their applications

limit

the process of letting x or t approach a in an expression; the limit of a function $f(x)$ as x approaches a is the value that $f(x)$ approaches as x approaches a

multivariable calculus

the study of the calculus of functions of two or more variables

secant

A secant line to a function $f(x)$ at a is a line through the point $(a, f(a))$ and another point on the function; the slope of the secant line is given by $m_{sec} = \frac{f(x) - f(a)}{x - a}$

tangent

A tangent line to the graph of a function at a point $(a, f(a))$ is the line that secant lines through $(a, f(a))$ approach as they are taken through points on the function with x-values that approach a; the slope of the tangent line to a graph at a measures the rate of change of the function at a

This page titled 2.1: A Preview of Calculus is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

2.1E: Exercises for Section 2.1

For exercises 1 - 3 , points $P(1,2)$ and $Q(x,y)$ are on the graph of the function $f(x)=x^2+1$.

1) [T] Complete the following table with the appropriate values: y-coordinate of Q, the point $Q(x,y)$, and the slope of the secant line passing through points P and Q. Round your answer to eight significant digits.

x	y	$Q(x,y)$	m_{sec}
1.1	a.	e.	i.
1.01	b.	f.	j.
1.001	c.	g.	k.
1.0001	d.	h.	l.

Answer

a. 2.2100000
b. 2.0201000
c. 2.0020010
d. 2.0002000
e. (1.1000000, 2.2100000)
f. (1.0100000, 2.0201000)
g. (1.0010000, 2.0020010)
h. (1.0001000, 2.0002000)
i. 2.1000000
j. 2.0100000
k. 2.0010000
l. 2.0001000

2) Use the values in the right column of the table in the preceding exercise to guess the value of the slope of the line tangent to f at $x=1$.

3) Use the value in the preceding exercise to find the equation of the tangent line at point P. Graph $f(x)$ and the tangent line.

Answer

$y=2x$

For the exercises 4-6, points $P(1,1)$ and $Q(x,y)$ are on the graph of the function $f(x)=x^3$.

4) [T] Complete the following table with the appropriate values: y-coordinate of Q, the point $Q(x,y)$, and the slope of the secant line passing through points P and Q. Round your answer to eight significant digits.

x	y	$Q(x,y)$	m_{sec}
1.1	a.	e.	i.
1.01	b.	f.	j.
1.001	c.	g.	k.
1.0001	d.	h.	l.2

5) Use the values in the right column of the table in the preceding exercise to guess the value of the slope of the tangent line to f at $x=1$.

Answer

3

6) Use the value in the preceding exercise to find the equation of the tangent line at point P. Graph $f(x)$ and the tangent line.

For the exercises 7 - 9, points $P(4,2)$ and $Q(x,y)$ are on the graph of the function $f(x)=\sqrt{x}$.

7) [T] Complete the following table with the appropriate values: y-coordinate of Q, the point $Q(x,y)$, and the slope of the secant line passing through points P and Q. Round your answer to eight significant digits.

x	y	$Q(x,y)$	m_{sec}
4.1	a.	e.	i.
4.01	b.	f.	j.
4.001	c.	g.	k.
4.0001	d.	h.	l.

Answer

a. 2.0248457
b. 2.0024984
c. 2.0002500
d. 2.0000250
e. (4.1000000,2.0248457)
f. (4.0100000,2.0024984)
g. (4.0010000,2.0002500)
h. (4.00010000,2.0000250)
i. 0.24845673
j. 0.24984395
k. 0.24998438
l. 0.24999844

8) Use the values in the right column of the table in the preceding exercise to guess the value of the slope of the tangent line to f at $x=4$.

9) Use the value in the preceding exercise to find the equation of the tangent line at point P.

Answer

$y=\frac{x}{4}+1$

For exercises 10 - 12, points $P(1.5,0)$ and $Q(\phi,y)$ are on the graph of the function $f(\phi)=\cos(\pi\phi)$.

10) [T] Complete the following table with the appropriate values: y-coordinate of Q, the point $Q(\phi,y)$, and the slope of the secant line passing through points P and Q. Round your answer to eight significant digits.

x	y	$Q(\phi,y)$	m_{sec}
1.4	a.	e.	i.
1.49	b.	f.	j.
1.499	c.	g.	k.
1.4999	d.	h.	l.

11) Use the values in the right column of the table in the preceding exercise to guess the value of the slope of the tangent line to f at $\phi=1.5$.

Answer

π

12) Use the value in the preceding exercise to find the equation of the tangent line at point P.

For exercises 13 - 15, points $P(-1,-1)$ and $Q(x,y)$ are on the graph of the function $f(x)=\frac{1}{x}$.

13) [T] Complete the following table with the appropriate values: y-coordinate of Q, the point $Q(x,y)$, and the slope of the secant line passing through points P and Q. Round your answer to eight significant digits.

x	y	$Q(x,y)$	m_{sec}
-1.05	a.	e.	i.
-1.01	b.	f.	j.
-1.005	c.	g.	k.
-1.001	d.	h.	l.

Answer

a. −0.95238095
b. −0.99009901
c. −0.99502488
d. −0.99900100
e. (−1;.0500000,−0;.95238095)
f. (−1;.0100000,−0;.9909901)
g. (−1;.0050000,−0;.99502488)
h. (1.0010000,−0;.99900100)
i. −0.95238095
j. −0.99009901
k. −0.99502488
l. −0.99900100

14) Use the values in the right column of the table in the preceding exercise to guess the value of the slope of the line tangent to f at $x=-1$.

15) Use the value in the preceding exercise to find the equation of the tangent line at point P.

Answer

$y=-x-2$

For exercises 16 - 17, the position function of a ball dropped from the top of a 200-meter tall building is given by $s(t)=200-4.9t^2$, where position s is measured in meters and time t is measured in seconds. Round your answer to eight significant digits.

16) [T] Compute the average velocity of the ball over the given time intervals.

a. [4.99,5]

b. [5,5.01]

c. [4.999,5]

d. [5,5.001]

17) Use the preceding exercise to guess the instantaneous velocity of the ball at $t=5$ sec.

Answer

-49 m/sec (velocity of the ball is 49 m/sec downward)

For exercises 18 - 19, consider a stone tossed into the air from ground level with an initial velocity of 15 m/sec. Its height in meters at time t seconds is $h(t) = 15t - 4.9t^2$.

18) [T] Compute the average velocity of the stone over the given time intervals.

a. [1,1.05]

b. [1,1.01]

c. [1,1.005]

d. [1,1.001]

19) Use the preceding exercise to guess the instantaneous velocity of the stone at $t = 1$ sec.

Answer

5.2 m/sec

For exercises 20 - 21, consider a rocket shot into the air that then returns to Earth. The height of the rocket in meters is given by $h(t) = 600 + 78.4t - 4.9t^2$, where t is measured in seconds.

20) [T] Compute the average velocity of the rocket over the given time intervals.

a. [9,9.01]

b. [8.99,9]

c. [9,9.001]

d. [8.999,9]

21) Use the preceding exercise to guess the instantaneous velocity of the rocket at $t = 9$ sec.

Answer

-9.8 m/sec

For exercises, consider an athlete running a 40-m dash. The position of the athlete is given by $d(t) = \frac{t^3}{6} + 4t$, where d is the position in meters and t is the time elapsed, measured in seconds.

22) [T] Compute the average velocity of the runner over the given time intervals.

a. [1.95,2.05]

b. [1.995,2.005]

c. [1.9995,2.0005]

d. [2,2.00001]

23) Use the preceding exercise to guess the instantaneous velocity of the runner at $t = 2$ sec.

Answer

6 m/sec

For exercises 24 - 25, consider the function $f(x) = |x|$.

24) Sketch the graph of f over the interval $[-1, 2]$ and shade the region above the x-axis.

25) Use the preceding exercise to find the exact value of the area between the x-axis and the graph of f over the interval $[-1, 2]$ using rectangles. For the rectangles, use the square units, and approximate both above and below the lines. Use geometry to find the exact answer.

Answer

Under, 1 $unit^2$; over: 4 $unit^2$.
The exact area of the two triangles is $\frac{1}{2}(1)(1) + \frac{1}{2}(2)(2) = 2.5 units^2$.

For exercises 26 - 27, consider the function $f(x) = \sqrt{1 - x^2}$. (Hint: This is the upper half of a circle of radius 1 positioned at $(0, 0)$.)

26) Sketch the graph of f over the interval $[-1, 1]$.

27) Use the preceding exercise to find the exact area between the x-axis and the graph of f over the interval $[-1, 1]$ using rectangles. For the rectangles, use squares 0.4 by 0.4 units, and approximate both above and below the lines. Use geometry to find the exact answer.

Answer

Under, 0.96 units^2; over, 1.92 units^2).
The exact area of the semicircle with radius 1 is $\frac{\pi(1)^2}{2} = \frac{\pi}{2}$ units^2

For exercises 28 - 29, consider the function $f(x) = -x^2 + 1$.

28) Sketch the graph of f over the interval $[-1, 1]$.

29) Approximate the area of the region between the x-axis and the graph of f over the interval $[-1, 1]$.

Answer

Approximately 1.3333333 units^2

This page titled 2.1E: Exercises for Section 2.1 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

2.2: The Limit of a Function

Learning Objectives

- Using correct notation, describe the limit of a function.
- Use a table of values to estimate the limit of a function or to identify when the limit does not exist.
- Use a graph to estimate the limit of a function or to identify when the limit does not exist.
- Define one-sided limits and provide examples.
- Explain the relationship between one-sided and two-sided limits.
- Using correct notation, describe an infinite limit.
- Define a vertical asymptote.

The concept of a limit or limiting process, essential to the understanding of calculus, has been around for thousands of years. In fact, early mathematicians used a limiting process to obtain better and better approximations of areas of circles. Yet, the formal definition of a limit—as we know and understand it today—did not appear until the late 19th century. We therefore begin our quest to understand limits, as our mathematical ancestors did, by using an intuitive approach. At the end of this chapter, armed with a conceptual understanding of limits, we examine the formal definition of a limit.

We begin our exploration of limits by taking a look at the graphs of the functions

- $f(x) = \dfrac{x^2 - 4}{x - 2}$,
- $g(x) = \dfrac{|x - 2|}{x - 2}$, and
- $h(x) = \dfrac{1}{(x - 2)^2}$,

which are shown in Figure 2.2.1. In particular, let's focus our attention on the behavior of each graph at and around $x = 2$.

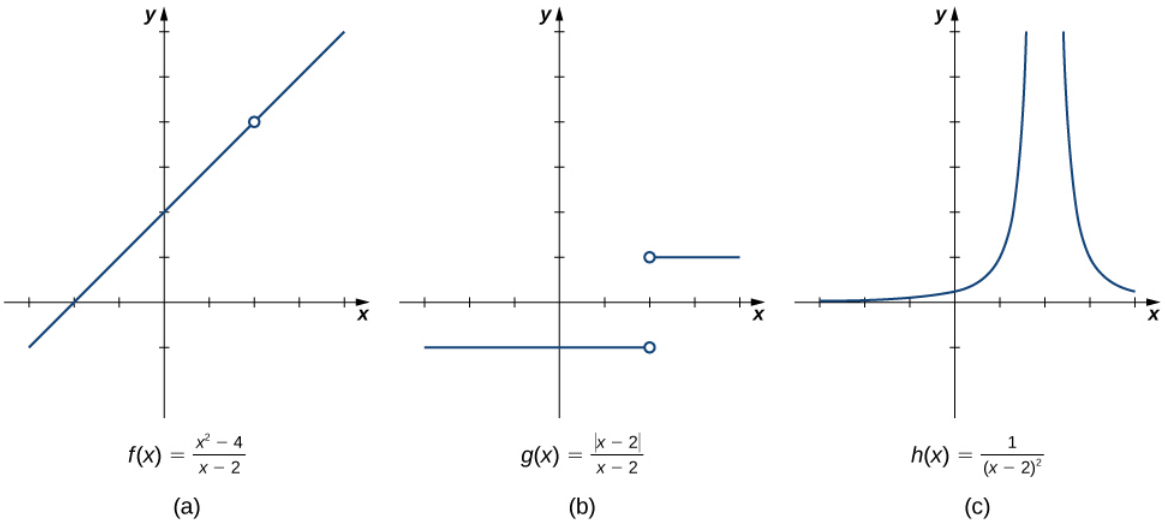


Figure 2.2.1: These graphs show the behavior of three different functions around $x = 2$.

Each of the three functions is undefined at $x = 2$, but if we make this statement and no other, we give a very incomplete picture of how each function behaves in the vicinity of $x = 2$. To express the behavior of each graph in the vicinity of 2 more completely, we need to introduce the concept of a limit.

Intuitive Definition of a Limit

Let's first take a closer look at how the function $f(x) = (x^2 - 4)/(x - 2)$ behaves around $x = 2$ in Figure 2.2.1. As the values of x approach 2 from either side of 2, the values of $y = f(x)$ approach 4. Mathematically, we say that the limit of $f(x)$ as x approaches 2 is 4. Symbolically, we express this limit as

$$\lim_{x \to 2} f(x) = 4$$

From this very brief informal look at one limit, let's start to develop an *intuitive definition of the limit*. We can think of the limit of a function at a number a as being the one real number L that the functional values approach as the x-values approach a, provided such a real number L exists. Stated more carefully, we have the following definition:

Definition (Intuitive): Limit

Let $f(x)$ be a function defined at all values in an open interval containing a, with the possible exception of a itself, and let L be a real number. If all values of the function $f(x)$ approach the real number L as the values of $x (\neq a)$ approach the number a, then we say that the limit of $f(x)$ as x approaches a is L. (More succinctly, as x gets closer to a, $f(x)$ gets closer and stays close to L.) Symbolically, we express this idea as

$$\lim_{x \to a} f(x) = L. \tag{2.2.1}$$

We can estimate limits by constructing tables of functional values and by looking at their graphs. This process is described in the following Problem-Solving Strategy.

Problem-Solving Strategy: Evaluating a Limit Using a Table of Functional Values

1. To evaluate $\lim_{x \to a} f(x)$, we begin by completing a table of functional values. We should choose two sets of x-values—one set of values approaching a and less than a, and another set of values approaching a and greater than a. Table 2.2.1 demonstrates what your tables might look like.

Table 2.2.1

x	$f(x)$	x	$f(x)$
$a - 0.1$	$f(a - 0.1)$	$a + 0.1$	$f(a + 0.1)$
$a - 0.01$	$f(a - 0.01)$	$a + 0.01$	$f(a + 0.01)$
$a - 0.001$	$f(a - 0.001)$	$a + 0.001$	$f(a + 0.001)$
$a - 0.0001$	$f(a - 0.0001)$	$a + 0.0001$	$f(a + 0.0001)$
Use additional values as necessary.		Use additional values as necessary.	

2. Next, let's look at the values in each of the $f(x)$ columns and determine whether the values seem to be approaching a single value as we move down each column. In our columns, we look at the sequence $f(a - 0.1)$, $f(a - 0.01)$, $f(a - 0.001)$, $f(a - 0.0001)$, and so on, and $f(a + 0.1)$, $f(a + 0.01)$, $f(a + 0.001)$, $f(a + 0.0001)$, and so on. (Note: Although we have chosen the x-values $a \pm 0.1$, $a \pm 0.01$, $a \pm 0.001$, $a \pm 0.0001$, and so forth, and these values will probably work nearly every time, on very rare occasions we may need to modify our choices.)

3. If both columns approach a common y-value L, we state $\lim_{x \to a} f(x) = L$. We can use the following strategy to confirm the result obtained from the table or as an alternative method for estimating a limit.

4. Using a graphing calculator or computer software that allows us graph functions, we can plot the function $f(x)$, making sure the functional values of $f(x)$ for x-values near a are in our window. We can use the trace feature to move along the graph of the function and watch the y-value readout as the x-values approach a. If the y-values approach L as our x-values approach a from both directions, then $\lim_{x \to a} f(x) = L$. We may need to zoom in on our graph and repeat this process several times.

We apply this Problem-Solving Strategy to compute a limit in Examples 2.2.1A and 2.2.1B.

Example 2.2.1A: Evaluating a Limit Using a Table of Functional Values

Evaluate $\lim_{x\to 0}\frac{\sin x}{x}$ using a table of functional values.

Solution

We have calculated the values of $f(x)=\frac{\sin x}{x}$ for the values of x listed in Table 2.2.2.

Table 2.2.2

x	$\frac{\sin x}{x}$	x	$\frac{\sin x}{x}$
-0.1	0.998334166468	0.1	0.998334166468
-0.01	0.999983333417	0.01	0.999983333417
-0.001	0.999999833333	0.001	0.999999833333
-0.0001	0.999999998333	0.0001	0.999999998333

Note: The values in this table were obtained using a calculator and using all the places given in the calculator output.

As we read down each $\frac{\sin x}{x}$ column, we see that the values in each column appear to be approaching one. Thus, it is fairly reasonable to conclude that $\lim_{x\to 0}\frac{\sin x}{x}=1$. A calculator-or computer-generated graph of $f(x)=\frac{\sin x}{x}$ would be similar to that shown in Figure 2.2.2, and it confirms our estimate.

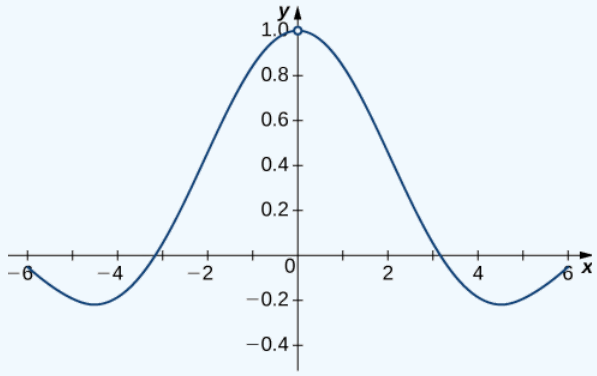


Figure 2.2.2: The graph of $f(x)=(\sin x)/x$ confirms the estimate from Table 2.2.2.

Example 2.2.1B: Evaluating a Limit Using a Table of Functional Values

Evaluate $\lim_{x\to 4}\frac{\sqrt{x}-2}{x-4}$ using a table of functional values.

Solution

As before, we use a table—in this case, Table 2.2.3—to list the values of the function for the given values of x.

Table 2.2.3

x	$\frac{\sqrt{x}-2}{x-4}$	x	$\frac{\sqrt{x}-2}{x-4}$
3.9	0.251582341869	4.1	0.248456731317
3.99	0.25015644562	4.01	0.24984394501
3.999	0.250015627	4.001	0.249984377
3.9999	0.250001563	4.0001	0.249998438
3.99999	0.25000016	4.00001	0.24999984

After inspecting this table, we see that the functional values less than 4 appear to be decreasing toward 0.25 whereas the functional values greater than 4 appear to be increasing toward 0.25. We conclude that $\lim\limits_{x\to 4}\dfrac{\sqrt{x}-2}{x-4}=0.25$. We confirm this estimate using the graph of $f(x)=\dfrac{\sqrt{x}-2}{x-4}$ shown in Figure 2.2.3.

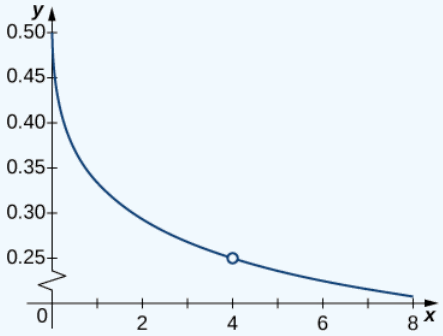


Figure 2.2.3: The graph of $\frac{\sqrt{x}-2}{x-4}$ confirms the estimate from Table 2.2.3.

Exercise 2.2.1

Estimate $\lim\limits_{x\to 1}\dfrac{\frac{1}{x}-1}{x-1}$ using a table of functional values. Use a graph to confirm your estimate.

Hint

Use 0.9, 0.99, 0.999, 0.9999, 0.99999 and 1.1, 1.01, 1.001, 1.0001, 1.00001 as your table values.

Answer

$$\lim_{x\to 1}\frac{\frac{1}{x}-1}{x-1}=-1$$

At this point, we see from Examples 2.2.1A and 2.2.1B that it may be just as easy, if not easier, to estimate a limit of a function by inspecting its graph as it is to estimate the limit by using a table of functional values. In Example 2.2.2, we evaluate a limit exclusively by looking at a graph rather than by using a table of functional values.

Example 2.2.2: Evaluating a Limit Using a Graph

For $g(x)$ shown in Figure 2.2.4, evaluate $\lim\limits_{x\to -1} g(x)$.

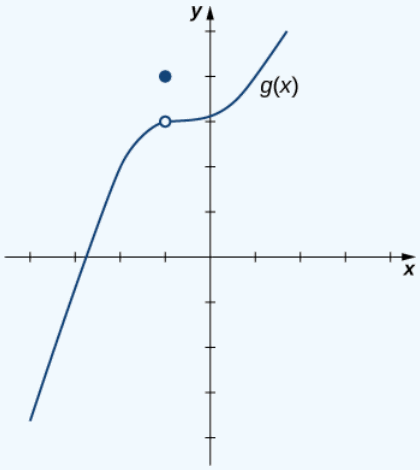


Figure 2.2.4: The graph of $g(x)$ includes one value not on a smooth curve.

Solution:

Despite the fact that $g(-1) = 4$, as the x-values approach -1 from either side, the $g(x)$ values approach 3. Therefore, $\lim_{x \to -1} g(x) = 3$. Note that we can determine this limit without even knowing the algebraic expression of the function.

Based on Example 2.2.2, we make the following observation: It is possible for the limit of a function to exist at a point, and for the function to be defined at this point, but the limit of the function and the value of the function at the point may be different.

Exercise 2.2.2

Use the graph of $h(x)$ in Figure 2.2.5 to evaluate $\lim_{x \to 2} h(x)$, if possible.

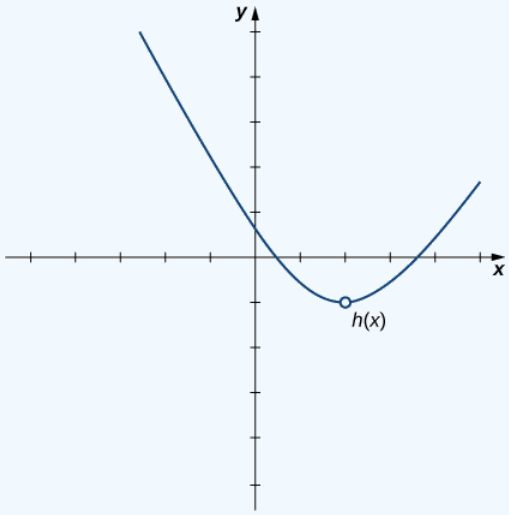


Figure 2.2.5:

Hint

What y-value does the function approach as the x-values approach 2?

Solution

$\lim_{x \to 2} h(x) = -1$.

Looking at a table of functional values or looking at the graph of a function provides us with useful insight into the value of the limit of a function at a given point. However, these techniques rely too much on guesswork. We eventually need to develop alternative methods of evaluating limits. These new methods are more algebraic in nature and we explore them in the next section; however, at this point we introduce two special limits that are foundational to the techniques to come.

Two Important Limits

Let a be a real number and c be a constant.

i. $\lim_{x \to a} x = a$
ii. $\lim_{x \to a} c = c$

We can make the following observations about these two limits.

i. For the first limit, observe that as x approaches a, so does $f(x)$, because $f(x) = x$. Consequently, $\lim_{x \to a} x = a$.
ii. For the second limit, consider Table 2.2.4.

Table 2.2.4

x	$f(x)=c$	x	$f(x)=c$
$a-0.1$	c	$a+0.1$	c
$a-0.01$	c	$a+0.01$	c
$a-0.001$	c	$a+0.001$	c
$a-0.0001$	c	$a+0.0001$	c

Observe that for all values of x (regardless of whether they are approaching a), the values $f(x)$ remain constant at c. We have no choice but to conclude $\lim\limits_{x\to a} c = c$.

The Existence of a Limit

As we consider the limit in the next example, keep in mind that for the limit of a function to exist at a point, the functional values must approach a single real-number value at that point. If the functional values do not approach a single value, then the limit does not exist.

Example 2.2.3: Evaluating a Limit That Fails to Exist

Evaluate $\lim\limits_{x\to 0} \sin(1/x)$ using a table of values.

Solution

Table 2.2.5 lists values for the function $\sin(1/x)$ for the given values of x.

Table 2.2.5

x	$\sin(1/x)$	x	$\sin(1/x)$
-0.1	0.544021110889	0.1	−0.544021110889
-0.01	0.50636564111	0.01	−0.50636564111
-0.001	−0.8268795405312	0.001	0.8268795405312
-0.0001	0.305614388888	0.0001	−0.305614388888
-0.00001	−0.035748797987	0.00001	0.035748797987
-0.000001	0.349993504187	0.000001	−0.349993504187

After examining the table of functional values, we can see that the y-values do not seem to approach any one single value. It appears the limit does not exist. Before drawing this conclusion, let's take a more systematic approach. Take the following sequence of x-values approaching 0:

$$\frac{2}{\pi}, \frac{2}{3\pi}, \frac{2}{5\pi}, \frac{2}{7\pi}, \frac{2}{9\pi}, \frac{2}{11\pi}, \ldots.$$

The corresponding y-values are

$$1, -1, 1, -1, 1, -1, \ldots.$$

At this point we can indeed conclude that $\lim\limits_{x\to 0} \sin(1/x)$ does not exist. (Mathematicians frequently abbreviate "does not exist" as DNE. Thus, we would write $\lim\limits_{x\to 0} \sin(1/x)$ DNE.) The graph of $f(x) = \sin(1/x)$ is shown in Figure 2.2.6 and it gives a clearer picture of the behavior of $\sin(1/x)$ as x approaches 0. You can see that $\sin(1/x)$ oscillates ever more wildly between -1 and 1 as x approaches 0.

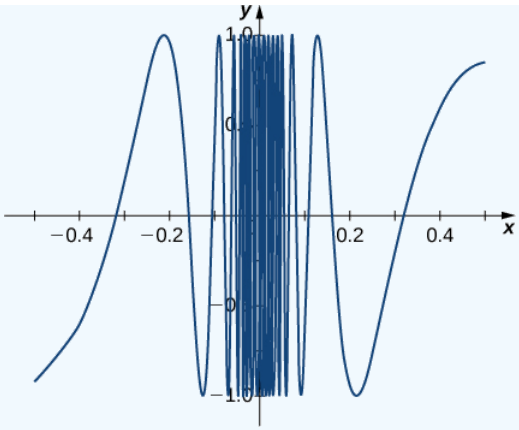


Figure 2.2.6: The graph of $f(x) = \sin(1/x)$ oscillates rapidly between -1 and 1 as x approaches 0.

? Exercise 2.2.3

Use a table of functional values to evaluate $\lim\limits_{x\to 2} \dfrac{|\, x^2 - 4\,|}{x-2}$, if possible.

Hint

Use x-values 1.9, 1.99, 1.999, 1.9999, 1.99999 and 2.1, 2.01, 2.001, 2.0001, 2.00001 in your table.

Answer

$\lim\limits_{x\to 2} \dfrac{|\, x^2 - 4\,|}{x-2}$ does not exist.

One-Sided Limits

Sometimes indicating that the limit of a function fails to exist at a point does not provide us with enough information about the behavior of the function at that particular point. To see this, we now revisit the function $g(x) = |x-2|/(x-2)$ introduced at the beginning of the section (see Figure 2.2.1(b)). As we pick values of x close to 2, $g(x)$ does not approach a single value, so the limit as x approaches 2 does not exist—that is, $\lim\limits_{x\to 2} g(x)$ DNE. However, this statement alone does not give us a complete picture of the behavior of the function around the x-value 2. To provide a more accurate description, we introduce the idea of a **one-sided limit**. For all values to the left of 2 (or the negative side of 2), $g(x) = -1$. Thus, as x approaches 2 from the left, $g(x)$ approaches -1. Mathematically, we say that the limit as x approaches 2 from the left is -1. Symbolically, we express this idea as

$$\lim_{x\to 2^-} g(x) = -1.$$

Similarly, as x approaches 2 from the right (or from the positive side), $g(x)$ approaches 1. Symbolically, we express this idea as

$$\lim_{x\to 2^+} g(x) = 1.$$

We can now present an informal definition of one-sided limits.

Definition: One-sided Limits

We define two types of one-sided limits.

Limit from the left:

Let $f(x)$ be a function defined at all values in an open interval of the form (z, a), and let L be a real number. If the values of the function $f(x)$ approach the real number L as the values of x (where $x < a$) approach the number a, then we say that L is the limit of $f(x)$ as x approaches a from the left. Symbolically, we express this idea as

$$\lim_{x\to a^-} f(x) = L.$$

Limit from the right:

Let $f(x)$ be a function defined at all values in an open interval of the form (a, c), and let L be a real number. If the values of the function $f(x)$ approach the real number L as the values of x (where $x > a$) approach the number a, then we say that L is the limit of $f(x)$ as x approaches a from the right. Symbolically, we express this idea as

$$\lim_{x \to a^+} f(x) = L.$$

Example 2.2.4: Evaluating One-Sided Limits

For the function $f(x) = \begin{cases} x+1, & \text{if } x < 2 \\ x^2 - 4, & \text{if } x \geq 2 \end{cases}$, evaluate each of the following limits.

a. $\lim_{x \to 2^-} f(x)$
b. $\lim_{x \to 2^+} f(x)$

Solution

We can use tables of functional values again. Observe in Table 2.2.6 that for values of x less than 2, we use $f(x) = x + 1$ and for values of x greater than 2, we use $f(x) = x^2 - 4$.

Table 2.2.6

x	$f(x) = x + 1$	x	$f(x) = x^2 - 4$
1.9	2.9	2.1	0.41
1.99	2.99	2.01	0.0401
1.999	2.999	2.001	0.004001
1.9999	2.9999	2.0001	0.00040001
1.99999	2.99999	2.00001	0.0000400001

Based on this table, we can conclude that a. $\lim_{x \to 2^-} f(x) = 3$ and b. $\lim_{x \to 2^+} f(x) = 0$. Therefore, the (two-sided) limit of $f(x)$ does not exist at $x = 2$. Figure 2.2.7 shows a graph of $f(x)$ and reinforces our conclusion about these limits.

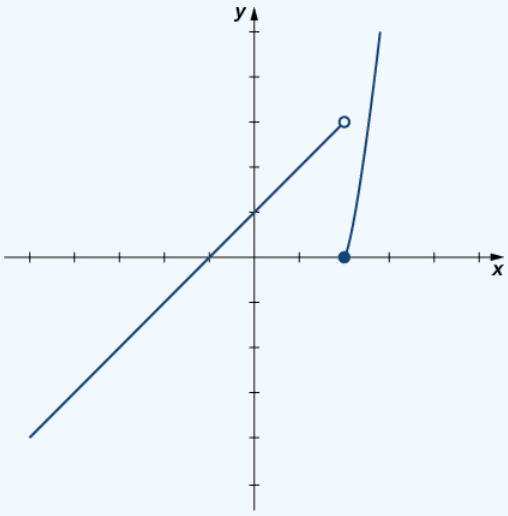


Figure 2.2.7: The graph of $f(x) = \begin{cases} x+1, & \text{if } x < 2 \\ x^2 - 4, & \text{if } x \geq 2 \end{cases}$ has a break at $x = 2$.

Exercise 2.2.4

Use a table of functional values to estimate the following limits, if possible.

a. $\lim_{x \to 2^-} \dfrac{|\, x^2 - 4 \,|}{x - 2}$

b. $\lim\limits_{x\to 2^+} \dfrac{|\,x^2-4\,|}{x-2}$

Hint

Use x-values 1.9, 1.99, 1.999, 1.9999, 1.99999 to estimate $\lim\limits_{x\to 2^-} \dfrac{|\,x^2-4\,|}{x-2}$.

Use x-values 2.1, 2.01, 2.001, 2.0001, 2.00001 to estimate $\lim\limits_{x\to 2^+} \dfrac{|\,x^2-4\,|}{x-2}$.

(These tables are available from a previous Checkpoint problem.)

Solution a

a. $\lim\limits_{x\to 2^-} \dfrac{|\,x^2-4\,|}{x-2} = -4$

Solution b

$\lim\limits_{x\to 2^+} \dfrac{|\,x^2-4\,|}{x-2} = 4$

Let us now consider the relationship between the limit of a function at a point and the limits from the right and left at that point. It seems clear that if the limit from the right and the limit from the left have a common value, then that common value is the limit of the function at that point. Similarly, if the limit from the left and the limit from the right take on different values, the limit of the function does not exist. These conclusions are summarized in Note.

Relating One-Sided and Two-Sided Limits

Let $f(x)$ be a function defined at all values in an open interval containing a, with the possible exception of a itself, and let L be a real number. Then,

$$\lim_{x\to a} f(x) = L$$

if and only if $\lim\limits_{x\to a^-} f(x) = L$ and $\lim\limits_{x\to a^+} f(x) = L$.

Infinite Limits

Evaluating the limit of a function at a point or evaluating the limit of a function from the right and left at a point helps us to characterize the behavior of a function around a given value. As we shall see, we can also describe the behavior of functions that do not have finite limits.

We now turn our attention to $h(x) = 1/(x-2)^2$, the third and final function introduced at the beginning of this section (see Figure 2.2.1(c)). From its graph we see that as the values of x approach 2, the values of $h(x) = 1/(x-2)^2$ become larger and larger and, in fact, become infinite. Mathematically, we say that the limit of $h(x)$ as x approaches 2 is positive infinity. Symbolically, we express this idea as

$$\lim_{x\to 2} h(x) = +\infty.$$

More generally, we define *infinite limits* as follows:

Definitions: Infinite Limits

We define three types of *infinite limits*.

Infinite limits from the left: Let $f(x)$ be a function defined at all values in an open interval of the form (b, a).

i. If the values of $f(x)$ increase without bound as the values of x (where $x < a$) approach the number a, then we say that the limit as x approaches a from the left is positive infinity and we write

$$\lim_{x\to a^-} f(x) = +\infty.$$

ii. If the values of $f(x)$ decrease without bound as the values of x (where $x < a$) approach the number a, then we say that the limit as x approaches a from the left is negative infinity and we write

$$\lim_{x \to a^-} f(x) = -\infty.$$

Infinite limits from the right: Let $f(x)$ be a function defined at all values in an open interval of the form (a, c).

i. If the values of $f(x)$ increase without bound as the values of x (where $x > a$) approach the number a, then we say that the limit as x approaches a from the right is positive infinity and we write

$$\lim_{x \to a^+} f(x) = +\infty.$$

ii. If the values of $f(x)$ decrease without bound as the values of x (where $x > a$) approach the number a, then we say that the limit as x approaches a from the right is negative infinity and we write

$$\lim_{x \to a^+} f(x) = -\infty.$$

Two-sided infinite limit: Let $f(x)$ be defined for all $x \neq a$ in an open interval containing a

i. If the values of $f(x)$ increase without bound as the values of x (where $x \neq a$) approach the number a, then we say that the limit as x approaches a is positive infinity and we write

$$\lim_{x \to a} f(x) = +\infty.$$

ii. If the values of $f(x)$ decrease without bound as the values of x (where $x \neq a$) approach the number a, then we say that the limit as x approaches a is negative infinity and we write

$$\lim_{x \to a} f(x) = -\infty.$$

It is important to understand that when we write statements such as $\lim_{x \to a} f(x) = +\infty$ or $\lim_{x \to a} f(x) = -\infty$ we are describing the behavior of the function, as we have just defined it. We are not asserting that a limit exists. For the limit of a function $f(x)$ to exist at a, it must approach a real number L as x approaches a. That said, if, for example, $\lim_{x \to a} f(x) = +\infty$, we always write $\lim_{x \to a} f(x) = +\infty$ rather than $\lim_{x \to a} f(x)$ DNE.

Example 2.2.5: Recognizing an Infinite Limit

Evaluate each of the following limits, if possible. Use a table of functional values and graph $f(x) = 1/x$ to confirm your conclusion.

a. $\lim_{x \to 0^-} \frac{1}{x}$
b. $\lim_{x \to 0^+} \frac{1}{x}$
c. $\lim_{x \to 0} \frac{1}{x}$

Solution

Begin by constructing a table of functional values.

Table 2.2.7

x	$\frac{1}{x}$	x	$\frac{1}{x}$
-0.1	-10	0.1	10
-0.01	-100	0.01	100
-0.001	-1000	0.001	1000
-0.0001	-10,000	0.0001	10,000
-0.00001	-100,000	0.00001	100,000
-0.000001	-1,000,000	0.000001	1,000,000

a. The values of $1/x$ decrease without bound as x approaches 0 from the left. We conclude that

$$\lim_{x\to 0^-} \frac{1}{x} = -\infty.$$

b. The values of $1/x$ increase without bound as x approaches 0 from the right. We conclude that

$$\lim_{x\to 0^+} \frac{1}{x} = +\infty.$$

c. Since $\lim_{x\to 0^-} \frac{1}{x} = -\infty$ and $\lim_{x\to 0^+} \frac{1}{x} = +\infty$ have different values, we conclude that

$$\lim_{x\to 0} \frac{1}{x} \quad \text{DNE.}$$

The graph of $f(x) = 1/x$ in Figure 2.2.8 confirms these conclusions.

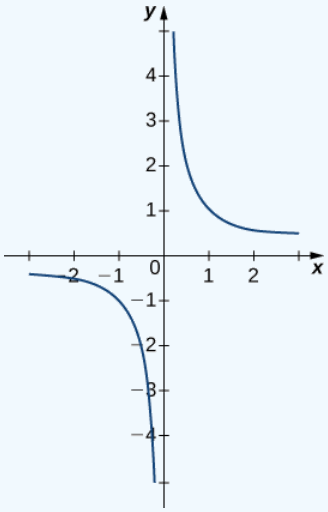


Figure 2.2.8: The graph of $f(x) = 1/x$ confirms that the limit as x approaches 0 does not exist.

Exercise 2.2.5

Evaluate each of the following limits, if possible. Use a table of functional values and graph $f(x) = 1/x^2$ to confirm your conclusion.

a. $\lim_{x\to 0^-} \frac{1}{x^2}$
b. $\lim_{x\to 0^+} \frac{1}{x^2}$
c. $\lim_{x\to 0} \frac{1}{x^2}$

Hint

Follow the procedures from Example 2.2.5.

Answer

a. $\lim\limits_{x\to 0^-} \dfrac{1}{x^2} = +\infty$;

b. $\lim\limits_{x\to 0^+} \dfrac{1}{x^2} = +\infty$;

c. $\lim\limits_{x\to 0} \dfrac{1}{x^2} = +\infty$

It is useful to point out that functions of the form $f(x) = 1/(x-a)^n$, where n is a positive integer, have infinite limits as x approaches a from either the left or right (Figure 2.2.9). These limits are summarized in the above definitions.

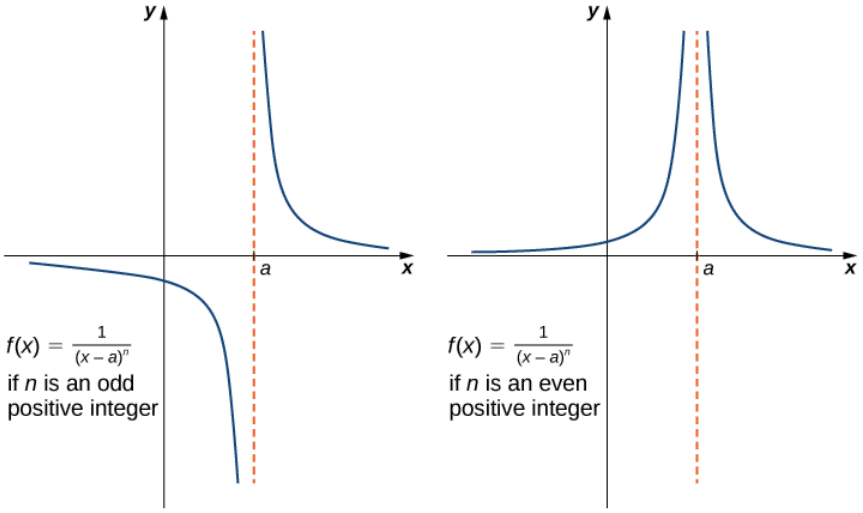


Figure 2.2.9: The function $f(x) = 1/(x-a)^n$ has infinite limits at a.

Infinite Limits from Positive Integers

If n is a positive even integer, then

$$\lim_{x\to a} \frac{1}{(x-a)^n} = +\infty. \tag{2.2.2}$$

If n is a positive odd integer, then

$$\lim_{x\to a^+} \frac{1}{(x-a)^n} = +\infty \tag{2.2.3}$$

and

$$\lim_{x\to a^-} \frac{1}{(x-a)^n} = -\infty. \tag{2.2.4}$$

We should also point out that in the graphs of $f(x) = 1/(x-a)^n$, points on the graph having x-coordinates very near to a are very close to the vertical line $x = a$. That is, as x approaches a, the points on the graph of $f(x)$ are closer to the line $x = a$. The line $x = a$ is called a vertical asymptote of the graph. We formally define a *vertical asymptote* as follows:

Definition: Vertical Asymptotes

Let $f(x)$ be a function. If any of the following conditions hold, then the line $x = a$ is a vertical asymptote of $f(x)$.

$$\lim_{x\to a^-} f(x) = +\infty$$

$$\lim_{x\to a^-} f(x) = -\infty$$

$$\lim_{x \to a^+} f(x) = +\infty$$

$$\lim_{x \to a^+} f(x) = -\infty$$

$$\lim_{x \to a} f(x) = +\infty$$

$$\lim_{x \to a} f(x) = -\infty$$

Example 2.2.6: Finding a Vertical Asymptote

Evaluate each of the following limits using Equations 2.2.2, 2.2.3, and 2.2.4 above. Identify any vertical asymptotes of the function $f(x) = 1/(x+3)^4$.

a. $\lim_{x \to -3^-} \frac{1}{(x+3)^4}$
b. $\lim_{x \to -3^+} \frac{1}{(x+3)^4}$
c. $\lim_{x \to -3} \frac{1}{(x+3)^4}$

Solution

We can use the above equations directly.

a. $\lim_{x \to -3^-} \frac{1}{(x+3)^4} = +\infty$
b. $\lim_{x \to -3^+} \frac{1}{(x+3)^4} = +\infty$
c. $\lim_{x \to -3} \frac{1}{(x+3)^4} = +\infty$

The function $f(x) = 1/(x+3)^4$ has a vertical asymptote of $x = -3$.

Exercise 2.2.6

Evaluate each of the following limits. Identify any vertical asymptotes of the function $f(x) = \frac{1}{(x-2)^3}$.

a. $\lim_{x \to 2^-} \frac{1}{(x-2)^3}$
b. $\lim_{x \to 2^+} \frac{1}{(x-2)^3}$
c. $\lim_{x \to 2} \frac{1}{(x-2)^3}$

Answer a

$$\lim_{x \to 2^-} \frac{1}{(x-2)^3} = -\infty$$

Answer b

$$\lim_{x \to 2^+} \frac{1}{(x-2)^3} = +\infty$$

Answer c

$\lim_{x \to 2} \frac{1}{(x-2)^3}$ DNE. The line $x = 2$ is the vertical asymptote of $f(x) = 1/(x-2)^3$.

In the next example we put our knowledge of various types of limits to use to analyze the behavior of a function at several different points.

Example 2.2.7: Behavior of a Function at Different Points

Use the graph of $f(x)$ in Figure 2.2.10 to determine each of the following values:

a. $\lim\limits_{x\to -4^-} f(x)$; $\lim\limits_{x\to -4^+} f(x)$; $\lim\limits_{x\to -4} f(x)$; $f(-4)$
b. $\lim\limits_{x\to -2^-} f(x)$; $\lim\limits_{x\to -2^+} f(x)$; $\lim\limits_{x\to -2} f(x)$; $f(-2)$
c. $\lim\limits_{x\to 1^-} f(x)$; $\lim\limits_{x\to 1^+} f(x)$; $\lim\limits_{x\to 1} f(x)$; $f(1)$
d. $\lim\limits_{x\to 3^-} f(x)$; $\lim\limits_{x\to 3^+} f(x)$; $\lim\limits_{x\to 3} f(x)$; $f(3)$

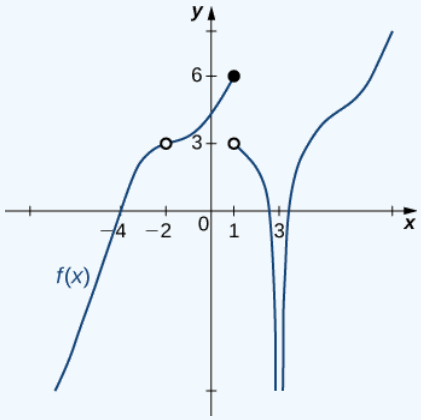


Figure 2.2.10: The graph shows $f(x)$.

Solution

Using the definitions above and the graph for reference, we arrive at the following values:

a. $\lim\limits_{x\to -4^-} f(x) = 0$; $\lim\limits_{x\to -4^+} f(x) = 0$; $\lim\limits_{x\to -4} f(x) = 0$; $f(-4) = 0$
b. $\lim\limits_{x\to -2^-} f(x) = 3$; $\lim\limits_{x\to -2^+} f(x) = 3$; $\lim\limits_{x\to -2} f(x) = 3$; $f(-2)$ is undefined
c. $\lim\limits_{x\to 1^-} f(x) = 6$; $\lim\limits_{x\to 1^+} f(x) = 3$; $\lim\limits_{x\to 1} f(x)$ DNE; $f(1) = 6$
d. $\lim\limits_{x\to 3^-} f(x) = -\infty$; $\lim\limits_{x\to 3^+} f(x) = -\infty$; $\lim\limits_{x\to 3} f(x) = -\infty$; $f(3)$ is undefined

Exercise 2.2.7

Evaluate $\lim\limits_{x\to 1} f(x)$ for $f(x)$ shown here:

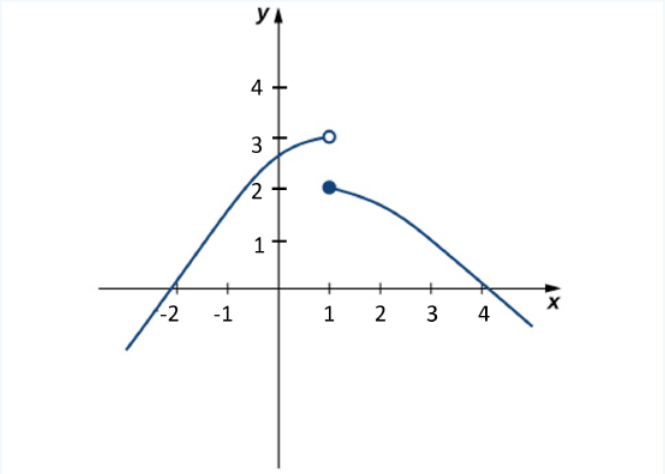


Figure 2.2.11. The graph of a piecewise function f.

Hint

Compare the limit from the right with the limit from the left.

Answer

$\lim_{x \to 1} f(x)$ does not exist

Example 2.2.8: Einstein's Equation

In the Chapter opener we mentioned briefly how Albert Einstein showed that a limit exists to how fast any object can travel. Given Einstein's equation for the mass of a moving object

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}},$$

what is the value of this bound?

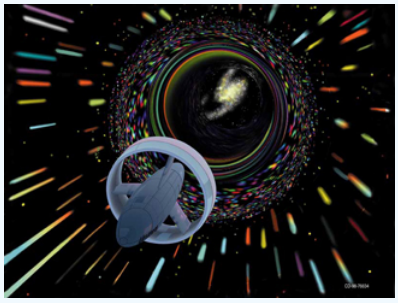


Figure 2.2.12. (Credit:NASA)

Solution

Our starting point is Einstein's equation for the mass of a moving object,

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}},$$

where m_0 is the object's mass at rest, v is its speed, and c is the speed of light. To see how the mass changes at high speeds, we can graph the ratio of masses m/m_0 as a function of the ratio of speeds, v/c (Figure 2.2.13).

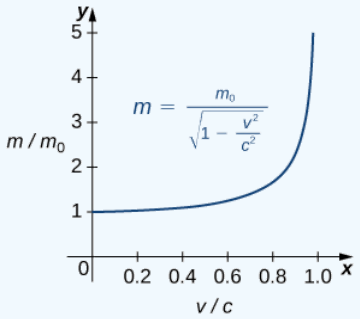


Figure 2.2.13: This graph shows the ratio of masses as a function of the ratio of speeds in Einstein's equation for the mass of a moving object.

We can see that as the ratio of speeds approaches 1—that is, as the speed of the object approaches the speed of light—the ratio of masses increases without bound. In other words, the function has a vertical asymptote at $v/c = 1$. We can try a few values of this ratio to test this idea.

Table 2.2.8

v/c	$\sqrt{1 - \frac{v^2}{c^2}}$	m/m_o
0.99	0.1411	7.089
0.999	0.0447	22.37
0.9999	0.0141	70.7

Thus, according to Table 2.2.8:, if an object with mass 100 kg is traveling at 0.9999c, its mass becomes 7071 kg. Since no object can have an infinite mass, we conclude that no object can travel at or more than the speed of light.

Key Concepts

- A table of values or graph may be used to estimate a limit.
- If the limit of a function at a point does not exist, it is still possible that the limits from the left and right at that point may exist.
- If the limits of a function from the left and right exist and are equal, then the limit of the function is that common value.
- We may use limits to describe infinite behavior of a function at a point.

Key Equations

- **Intuitive Definition of the Limit**

$$\lim_{x \to a} f(x) = L$$

- **Two Important Limits**

$$\lim_{x \to a} x = a \qquad \lim_{x \to a} c = c$$

- **One-Sided Limits**

$$\lim_{x \to a^-} f(x) = L \qquad \lim_{x \to a^+} f(x) = L$$

- **Infinite Limits from the Left**

$$\lim_{x \to a^-} f(x) = +\infty \qquad \lim_{x \to a^-} f(x) = -\infty$$

- **Infinite Limits from the Right**

$$\lim_{x \to a^+} f(x) = +\infty \qquad \lim_{x \to a^+} f(x) = -\infty$$

- **Two-Sided Infinite Limits**

$$\lim_{x \to a} f(x) = +\infty\text{: } \lim_{x \to a^-} f(x) = +\infty \text{ and } \lim_{x \to a^+} f(x) = +\infty$$

$$\lim_{x \to a} f(x) = -\infty\text{: } \lim_{x \to a^-} f(x) = -\infty \text{ and } \lim_{x \to a^+} f(x) = -\infty$$

Glossary

infinite limit

A function has an infinite limit at a point a if it either increases or decreases without bound as it approaches a

intuitive definition of the limit

If all values of the function $f(x)$ approach the real number L as the values of $x(\neq a)$ approach a, $f(x)$ approaches L

one-sided limit

A one-sided limit of a function is a limit taken from either the left or the right

vertical asymptote

A function has a vertical asymptote at $x = a$ if the limit as x approaches a from the right or left is infinite

This page titled 2.2: The Limit of a Function is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

2.2E: Exercises for Section 2.2

Intuitive Definition of Limits

For exercises 1 - 2, consider the function $f(x) = \dfrac{x^2 - 1}{|x - 1|}$.

1) [T] Complete the following table for the function. Round your solutions to four decimal places.

x	$f(x)$	x	$f(x)$
0.9	a.	1.1	e.
0.99	b.	1.01	f.
0.999	c.	1.001	g.
0.9999	d.	1.0001	h.

2) What do your results in the preceding exercise indicate about the two-sided limit $\lim\limits_{x \to 1} f(x)$? Explain your response.

Answer

$\lim\limits_{x \to 1} f(x)$ does not exist because $\lim\limits_{x \to 1^-} f(x) = -2 \neq \lim\limits_{x \to 1^+} f(x) = 2$.

For exercises 3 - 5, consider the function $f(x) = (1 + x)^{1/x}$.

3) [T] Make a table showing the values of f for $x = -0.01,\ -0.001,\ -0.0001,\ -0.00001$ and for $x = 0.01,\ 0.001,\ 0.0001,\ 0.00001$. Round your solutions to five decimal places.

x	$f(x)$	x	$f(x)$
-0.01	a.	0.01	e.
-0.001	b.	0.001	f.
-0.0001	c.	0.0001	g.
-0.00001	d.	0.00001	h.

4) What does the table of values in the preceding exercise indicate about the function $f(x) = (1 + x)^{1/x}$?

Answer

$\lim\limits_{x \to 0} (1 + x)^{1/x} \approx 2.7183$.

5) To which mathematical constant do the values in the preceding exercise appear to be approaching? This is the actual limit here.

In exercises 6 - 8, use the given values to set up a table to evaluate the limits. Round your solutions to eight decimal places.

6) [T] $\lim\limits_{x \to 0} \dfrac{\sin 2x}{x}$; $\pm 0.1,\ \pm 0.01,\ \pm 0.001,\ \pm .0001$

x	$\frac{\sin 2x}{x}$	x	$\frac{\sin 2x}{x}$
-0.1	a.	0.1	e.
-0.01	b.	0.01	f.
-0.001	c.	0.001	g.
-0.0001	d.	0.0001	h.

Answer

a. 1.98669331; b. 1.99986667; c. 1.99999867; d. 1.99999999; e. 1.98669331; f. 1.99986667; g. 1.99999867; h. 1.99999999;

$\lim_{x\to 0}\frac{\sin 2x}{x}=2$

7) [T] $\lim_{x\to 0}\frac{\sin 3x}{x}$ $\pm 0.1,\ \pm 0.01,\ \pm 0.001,\ \pm 0.0001$

x	$\frac{\sin 3x}{x}$	x	$\frac{\sin 3x}{x}$
-0.1	a.	0.1	e.
-0.01	b.	0.01	f.
-0.001	c.	0.001	g.
-0.0001	d.	0.0001	h.

8) Use the preceding two exercises to conjecture (guess) the value of the following limit: $\lim_{x\to 0}\frac{\sin ax}{x}$ for a, a positive real value.

Answer

$\lim_{x\to 0}\frac{\sin ax}{x}=a$

[T] In exercises 9 - 14, set up a table of values to find the indicated limit. Round to eight significant digits.

9) $\lim_{x\to 2}\frac{x^2-4}{x^2+x-6}$

x	$\frac{x^2-4}{x^2+x-6}$	x	$\frac{x^2-4}{x^2+x-6}$
1.9	a.	2.1	e.
1.99	b.	2.01	f.
1.999	c.	2.001	g.
1.9999	d.	2.0001	h.

10) $\lim_{x\to 1}(1-2x)$

x	$1-2x$	x	$1-2x$
0.9	a.	1.1	e.
0.99	b.	1.01	f.
0.999	c.	1.001	g.
0.9999	d.	1.0001	h.

Answer

a. −0.80000000; b. −0.98000000; c. −0.99800000; d. −0.99980000; e. −1.2000000; f. −1.0200000; g. −1.0020000; h. −1.0002000;

$\lim_{x\to 1}(1-2x)=-1$

11) $\lim_{x\to 0}\frac{5}{1-e^{1/x}}$

x	$\frac{5}{1-e^{1/x}}$	x	$\frac{5}{1-e^{1/x}}$
-0.1	a.	0.1	e.
-0.01	b.	0.01	f.
-0.001	c.	0.001	g.
-0.0001	d.	0.0001	h.

12) $\lim_{z\to 0} \dfrac{z-1}{z^2(z+3)}$

z	$\frac{z-1}{z^2(z+3)}$	z	$\frac{z-1}{z^2(z+3)}$
-0.1	a.	0.1	e.
-0.01	b.	0.01	f.
-0.001	c.	0.001	g.
-0.0001	d.	0.0001	h.

Answer

a. −37.931034; b. −3377.9264; c. −333,777.93; d. −33,337,778; e. −29.032258; f. −3289.0365; g. −332,889.04; h. −33,328,889

$\lim_{x\to 0} \dfrac{z-1}{z^2(z+3)} = -\infty$

13) $\lim_{t\to 0^+} \dfrac{\cos t}{t}$

t	$\frac{\cos t}{t}$
0.1	a.
0.01	b.
0.001	c.
0.0001	d.

14) $\lim_{x\to 2} \dfrac{1-\frac{2}{x}}{x^2-4}$

x	$\frac{1-\frac{2}{x}}{x^2-4}$	x	$\frac{1-\frac{2}{x}}{x^2-4}$
1.9	a.	2.1	e.
1.99	b.	2.01	f.
1.999	c.	2.001	g.
1.9999	d.	2.0001	h.

Answer

a. 0.13495277; b. 0.12594300; c. 0.12509381; d. 0.12500938; e. 0.11614402; f. 0.12406794; g. 0.12490631; h. 0.12499063;

$\therefore \lim_{x\to 2} \dfrac{1-\frac{2}{x}}{x^2-4} = 0.1250 = \dfrac{1}{8}$

[T] In exercises 15 - 16, set up a table of values and round to eight significant digits. Based on the table of values, make a guess about what the limit is. Then, use a calculator to graph the function and determine the limit. Was the conjecture correct? If not, why does the method of tables fail?

15) $\lim_{\theta \to 0} \sin\left(\frac{\pi}{\theta}\right)$

θ	$\sin\left(\frac{\pi}{\theta}\right)$	θ	$\sin\left(\frac{\pi}{\theta}\right)$
-0.1	a.	0.1	e.
-0.01	b.	0.01	f.
-0.001	c.	0.001	g.
-0.0001	d.	0.0001	h.

16) $\lim_{\alpha \to 0^+} \frac{1}{\alpha}\cos\left(\frac{\pi}{\alpha}\right)$

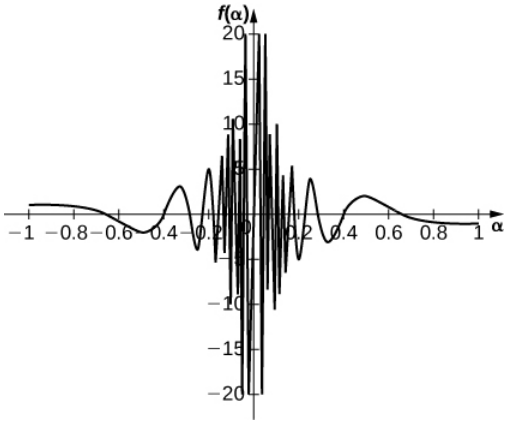
a	$\frac{1}{\alpha}\cos\left(\frac{\pi}{\alpha}\right)$
0.1	a.
0.01	b.
0.001	c.
0.0001	d.

Answer

a. 10.00000; b. 100.00000; c. 1000.0000; d. 10,000.000;

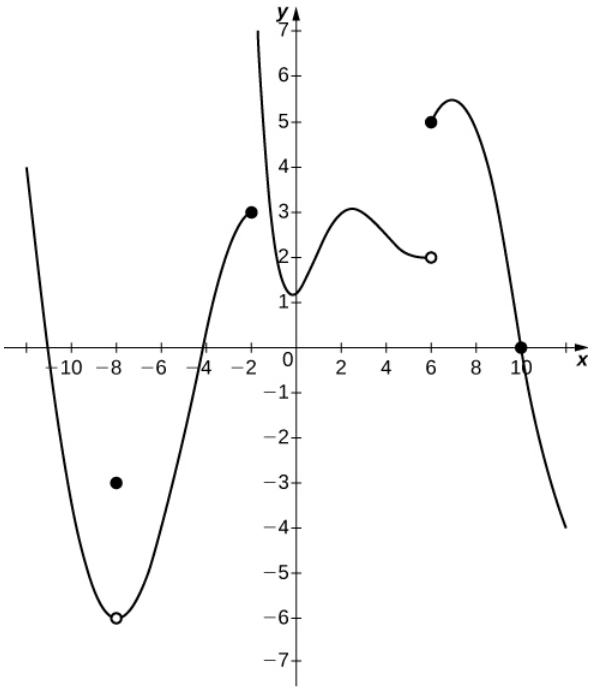
Guess: $\lim_{\alpha \to 0^+} \frac{1}{\alpha}\cos\left(\frac{\pi}{\alpha}\right) = \infty$;

Actual: DNE , since the graph shows the function oscillates wildly between values approaching positive infinity and values approaching negative infinity, as the value of α approaches 0 from the positive side.



In exercises 17 - 20, consider the graph of the function $y = f(x)$ shown here. Which of the statements about $y = f(x)$ are true and which are false? Explain why a statement is false.

https://math.libretexts.org/@go/page/49549



17) $\lim_{x\to 10} f(x) = 0$

18) $\lim_{x\to -2^+} f(x) = 3$

Answer

False; $\lim_{x\to -2^+} f(x) = +\infty$

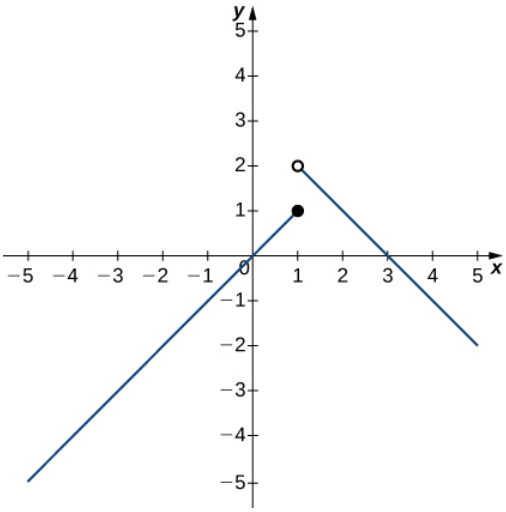
19) $\lim_{x\to -8} f(x) = f(-8)$

20) $\lim_{x\to 6} f(x) = 5$

Answer

False; $\lim_{x\to 6} f(x)$ DNE since $\lim_{x\to 6^-} f(x) = 2$ and $\lim_{x\to 6^+} f(x) = 5$.

In exercises 21 - 25, use the following graph of the function $y = f(x)$ to find the values, if possible. Estimate when necessary.



21) $\lim_{x\to 1^-} f(x)$

22) $\lim_{x \to 1^+} f(x)$

Answer

2

23) $\lim_{x \to 1} f(x)$

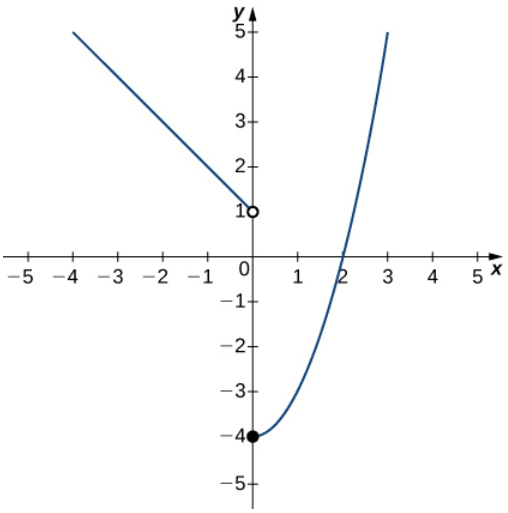
24) $\lim_{x \to 2} f(x)$

Answer

1

25) $f(1)$

In exercises 26 - 29, use the graph of the function $y = f(x)$ shown here to find the values, if possible. Estimate when necessary.



26) $\lim_{x \to 0^-} f(x)$

Answer

1

27) $\lim_{x \to 0^+} f(x)$

28) $\lim_{x \to 0} f(x)$

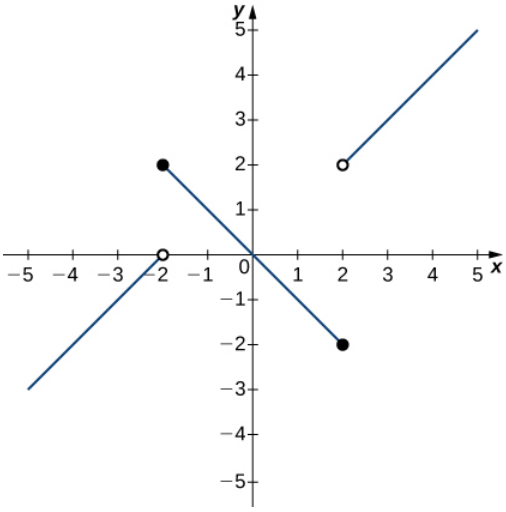
Answer

DNE

29) $\lim_{x \to 2} f(x)$

In exercises 30 - 35, use the graph of the function $y = f(x)$ shown here to find the values, if possible. Estimate when necessary.

https://math.libretexts.org/@go/page/49549



30) $\lim_{x \to -2^-} f(x)$

Answer

0

31) $\lim_{x \to -2^+} f(x)$

32) $\lim_{x \to -2} f(x)$

Answer

DNE

33) $\lim_{x \to 2^-} f(x)$

34) $\lim_{x \to 2^+} f(x)$

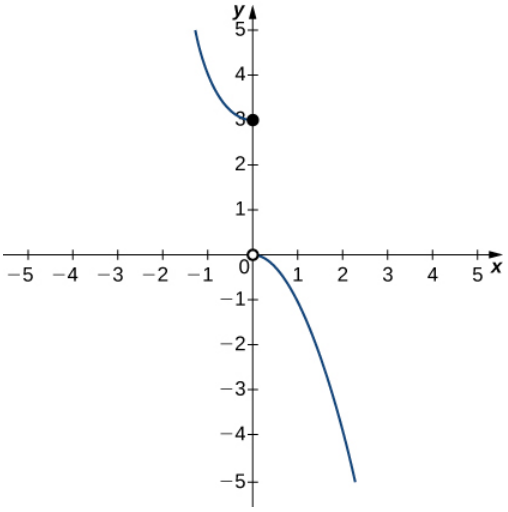
Answer

2

35) $\lim_{x \to 2} f(x)$

In exercises 36 - 38, use the graph of the function $y = g(x)$ shown here to find the values, if possible. Estimate when necessary.

https://math.libretexts.org/@go/page/49549



36) $\lim_{x \to 0^-} g(x)$

Answer

3

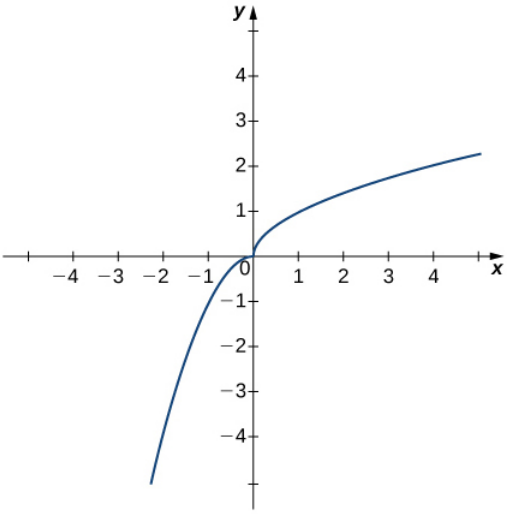
37) $\lim_{x \to 0^+} g(x)$

38) $\lim_{x \to 0} g(x)$

Answer

DNE

In exercises 39 - 41, use the graph of the function $y = h(x)$ shown here to find the values, if possible. Estimate when necessary.



39) $\lim_{x \to 0^-} h(x)$

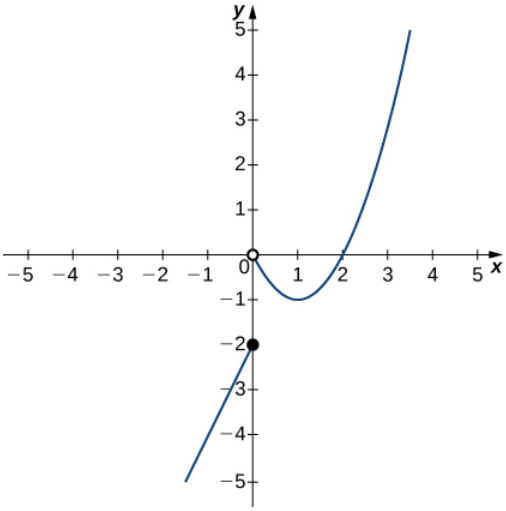
40) $\lim_{x \to 0^+} h(x)$

Answer

0

41) $\lim_{x\to 0} h(x)$

In exercises 42 - 46, use the graph of the function $y = f(x)$ shown here to find the values, if possible. Estimate when necessary.



42) $\lim_{x\to 0^-} f(x)$

Answer

-2

43) $\lim_{x\to 0^+} f(x)$

44) $\lim_{x\to 0} f(x)$

Answer

DNE

45) $\lim_{x\to 1} f(x)$

46) $\lim_{x\to 2} f(x)$

Answer

0

Infinite Limits

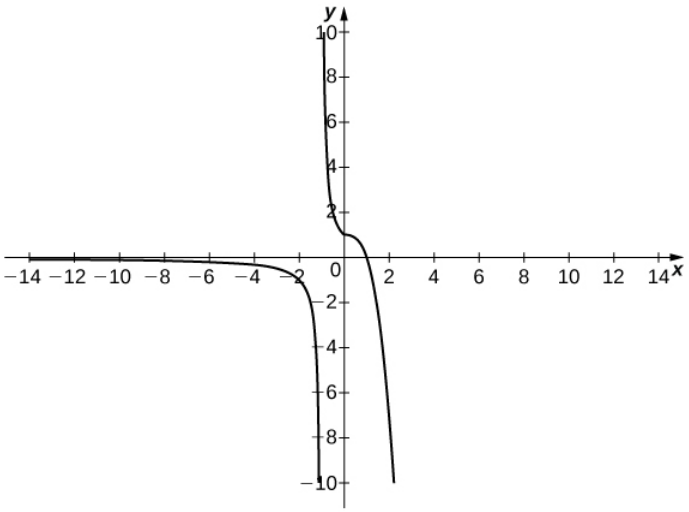
In exercises 47 - 51, sketch the graph of a function with the given properties.

47) $\lim_{x\to 2} f(x) = 1, \quad \lim_{x\to 4^-} f(x) = 3, \quad \lim_{x\to 4^+} f(x) = 6, \quad x = 4$ is not defined.

48) $\lim_{x\to -\infty} f(x) = 0, \quad \lim_{x\to -1^-} f(x) = -\infty, \quad \lim_{x\to -1^+} f(x) = \infty, \quad \lim_{x\to 0} f(x) = f(0), \quad f(0) = 1, \quad \lim_{x\to \infty} f(x) = -\infty$

Answer

Answers may vary

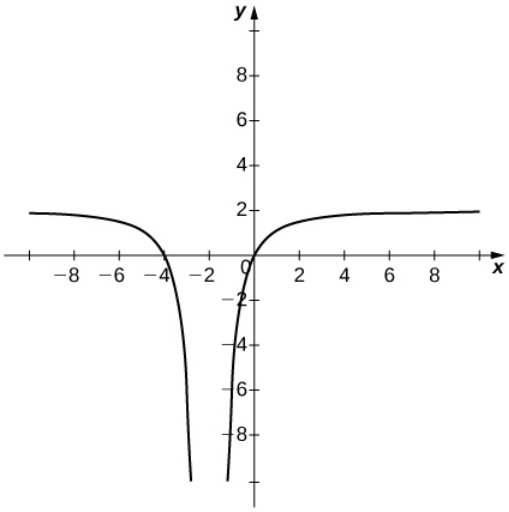


49) $\lim_{x\to-\infty} f(x) = 2, \quad \lim_{x\to3^-} f(x) = -\infty, \quad \lim_{x\to3^+} f(x) = \infty, \quad \lim_{x\to\infty} f(x) = 2, \quad f(0) = -\frac{1}{3}$

50) $\lim_{x\to-\infty} f(x) = 2, \quad \lim_{x\to-2} f(x) = -\infty, \quad \lim_{x\to\infty} f(x) = 2, \quad f(0) = 0$

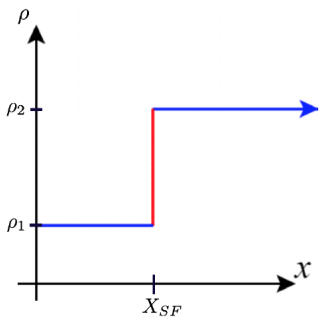
Answer

Answer may vary



51) $\lim_{x\to-\infty} f(x) = 0, \quad \lim_{x\to-1^-} f(x) = \infty, \quad \lim_{x\to-1^+} f(x) = -\infty, \quad f(0) = -1, \quad \lim_{x\to1^-} f(x) = -\infty, \quad \lim_{x\to1^+} f(x) = \infty, \quad \lim_{x\to\infty} f(x) = 0$

52) Shock waves arise in many physical applications, ranging from supernovas to detonation waves. A graph of the density of a shock wave with respect to distance, x, is shown here. We are mainly interested in the location of the front of the shock, labeled X_{SF} in the diagram.



a. Evaluate $\lim_{x \to X_{SF}^{+}} \rho(x)$.

b. Evaluate $\lim_{x \to X_{SF}^{-}} \rho(x)$.

c. Evaluate $\lim_{x \to X_{SF}} \rho(x)$. Explain the physical meanings behind your answers.

Answer

a. ρ_2 b. ρ_1 c. DNE unless $\rho_1 = \rho_2$. As you approach X_{SF} from the right, you are in the high-density area of the shock. When you approach from the left, you have not experienced the "shock" yet and are at a lower density.

53) A track coach uses a camera with a fast shutter to estimate the position of a runner with respect to time. A table of the values of position of the athlete versus time is given here, where x is the position in meters of the runner and t is time in seconds. What is $\lim_{t \to 2} x(t)$? What does it mean physically?

$t(sec)$	$x(m)$
1.75	4.5
1.95	6.1
1.99	6.42
2.01	6.58
2.05	6.9
2.25	8.5

This page titled 2.2E: Exercises for Section 2.2 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

2.3: The Limit Laws

Learning Objectives

- Recognize the basic limit laws.
- Use the limit laws to evaluate the limit of a function.
- Evaluate the limit of a function by factoring.
- Use the limit laws to evaluate the limit of a polynomial or rational function.
- Evaluate the limit of a function by factoring or by using conjugates.
- Evaluate the limit of a function by using the squeeze theorem.

In the previous section, we evaluated limits by looking at graphs or by constructing a table of values. In this section, we establish laws for calculating limits and learn how to apply these laws. In the Student Project at the end of this section, you have the opportunity to apply these limit laws to derive the formula for the area of a circle by adapting a method devised by the Greek mathematician Archimedes. We begin by restating two useful limit results from the previous section. These two results, together with the limit laws, serve as a foundation for calculating many limits.

Evaluating Limits with the Limit Laws

The first two limit laws were stated previously and we repeat them here. These basic results, together with the other limit laws, allow us to evaluate limits of many algebraic functions.

Theorem 2.3.1: Basic Limit Results

For any real number a and any constant c,

I. $\lim\limits_{x \to a} x = a$
II. $\lim\limits_{x \to a} c = c$

Example 2.3.1: Evaluating a Basic Limit

Evaluate each of the following limits using "Basic Limit Results."

a. $\lim\limits_{x \to 2} x$
b. $\lim\limits_{x \to 2} 5$

Solution

a. The limit of x as x approaches a is a: $\lim\limits_{x \to 2} x = 2$.
b. The limit of a constant is that constant: $\lim\limits_{x \to 2} 5 = 5$.

We now take a look at the **limit laws**, the individual properties of limits. The proofs that these laws hold are omitted here.

Theorem 2.3.2: Limit Laws

Let $f(x)$ and $g(x)$ be defined for all $x \neq a$ over some open interval containing a. Assume that L and M are real numbers such that $\lim\limits_{x \to a} f(x) = L$ and $\lim\limits_{x \to a} g(x) = M$. Let c be a constant. Then, each of the following statements holds:

- **Sum law for limits**:

$$\lim_{x \to a}(f(x) + g(x)) = \lim_{x \to a} f(x) + \lim_{x \to a} g(x) = L + M$$

- **Difference law for limits**:

$$\lim_{x \to a}(f(x) - g(x)) = \lim_{x \to a} f(x) - \lim_{x \to a} g(x) = L - M$$

- **Constant multiple law for limits**:

$$\lim_{x \to a} cf(x) = c \cdot \lim_{x \to a} f(x) = cL$$

- **Product law for limits**:

$$\lim_{x \to a} (f(x) \cdot g(x)) = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x) = L \cdot M$$

- **Quotient law for limits**:

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)} = \frac{L}{M}$$

for $M \neq 0$.

- **Power law for limits**:

$$\lim_{x \to a} \big(f(x)\big)^n = \big(\lim_{x \to a} f(x)\big)^n = L^n$$

for every positive integer n.

- **Root law for limits**:

$$\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)} = \sqrt[n]{L}$$

for all L if n is odd and for $L \geq 0$ if n is even.

We now practice applying these limit laws to evaluate a limit.

Example 2.3.2A: Evaluating a Limit Using Limit Laws

Use the limit laws to evaluate

$$\lim_{x \to -3} (4x + 2).$$

Solution

Let's apply the limit laws one step at a time to be sure we understand how they work. We need to keep in mind the requirement that, at each application of a limit law, the new limits must exist for the limit law to be applied.

$$\begin{aligned} \lim_{x \to -3} (4x + 2) &= \lim_{x \to -3} 4x + \lim_{x \to -3} 2 && \text{Apply the sum law.} \\ &= 4 \cdot \lim_{x \to -3} x + \lim_{x \to -3} 2 && \text{Apply the constant multiple law.} \\ &= 4 \cdot (-3) + 2 = -10. && \text{Apply the basic limit results and simplify.} \end{aligned}$$

Example 2.3.2B: Using Limit Laws Repeatedly

Use the limit laws to evaluate

$$\lim_{x \to 2} \frac{2x^2 - 3x + 1}{x^3 + 4}.$$

Solution

To find this limit, we need to apply the limit laws several times. Again, we need to keep in mind that as we rewrite the limit in terms of other limits, each new limit must exist for the limit law to be applied.

$$\lim_{x\to 2}\frac{2x^2-3x+1}{x^3+4}=\frac{\lim_{x\to 2}(2x^2-3x+1)}{\lim_{x\to 2}(x^3+4)}$$ Apply the quotient law, make sure that $(2)^3+4\neq 0$.

$$=\frac{2\cdot\lim_{x\to 2}x^2-3\cdot\lim_{x\to 2}x+\lim_{x\to 2}1}{\lim_{x\to 2}x^3+\lim_{x\to 2}4}$$ Apply the sum law and constant multiple law.

$$=\frac{2\cdot\left(\lim_{x\to 2}x\right)^2-3\cdot\lim_{x\to 2}x+\lim_{x\to 2}1}{\left(\lim_{x\to 2}x\right)^3+\lim_{x\to 2}4}$$ Apply the power law.

$$=\frac{2(4)-3(2)+1}{(2)^3+4}=\frac{1}{4}.$$ Apply the basic limit laws and simplify.

? Exercise 2.3.2

Use the limit laws to evaluate $\lim_{x\to 6}(2x-1)\sqrt{x+4}$. In each step, indicate the limit law applied.

Hint

Begin by applying the product law.

Answer

$11\sqrt{10}$

Limits of Polynomial and Rational Functions

By now you have probably noticed that, in each of the previous examples, it has been the case that $\lim_{x\to a}f(x)=f(a)$. This is not always true, but it does hold for all polynomials for any choice of a and for all rational functions at all values of a for which the rational function is defined.

Theorem 2.3.3: Limits of Polynomial and Rational Functions

Let $p(x)$ and $q(x)$ be polynomial functions. Let a be a real number. Then,

$$\lim_{x\to a}p(x)=p(a)$$

$$\lim_{x\to a}\frac{p(x)}{q(x)}=\frac{p(a)}{q(a)}$$

when $q(a)\neq 0$.

To see that this theorem holds, consider the polynomial

$$p(x)=c_nx^n+c_{n-1}x^{n-1}+\cdots+c_1x+c_0.$$

By applying the sum, constant multiple, and power laws, we end up with

$$\begin{aligned}\lim_{x\to a}p(x)&=\lim_{x\to a}(c_nx^n+c_{n-1}x^{n-1}+\cdots+c_1x+c_0)\\&=c_n\left(\lim_{x\to a}x\right)^n+c_{n-1}\left(\lim_{x\to a}x\right)^{n-1}+\cdots+c_1\left(\lim_{x\to a}x\right)+\lim_{x\to a}c_0\\&=c_na^n+c_{n-1}a^{n-1}+\cdots+c_1a+c_0\\&=p(a)\end{aligned}$$

It now follows from the quotient law that if $p(x)$ and $q(x)$ are polynomials for which $q(a)\neq 0$,

then

$$\lim_{x \to a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}.$$

Example 2.3.3: Evaluating a Limit of a Rational Function

Evaluate the $\lim_{x \to 3} \frac{2x^2 - 3x + 1}{5x + 4}$.

Solution

Since 3 is in the domain of the rational function $f(x) = \frac{2x^2 - 3x + 1}{5x + 4}$, we can calculate the limit by substituting 3 for x into the function. Thus,

$$\lim_{x \to 3} \frac{2x^2 - 3x + 1}{5x + 4} = \frac{10}{19}.$$

Exercise 2.3.3

Evaluate $\lim_{x \to -2} (3x^3 - 2x + 7)$.

Hint

Use LIMITS OF POLYNOMIAL AND RATIONAL FUNCTIONS as reference

Answer

-13

Additional Limit Evaluation Techniques

As we have seen, we may evaluate easily the limits of polynomials and limits of some (but not all) rational functions by direct substitution. However, as we saw in the introductory section on limits, it is certainly possible for $\lim_{x \to a} f(x)$ to exist when $f(a)$ is undefined. The following observation allows us to evaluate many limits of this type:

If for all $x \neq a,\ f(x) = g(x)$ over some open interval containing a, then

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x).$$

To understand this idea better, consider the limit $\lim_{x \to 1} \frac{x^2 - 1}{x - 1}$.

The function

$$f(x) = \frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1}$$

and the function $g(x) = x + 1$ are identical for all values of $x \neq 1$. The graphs of these two functions are shown in Figure 2.3.1.

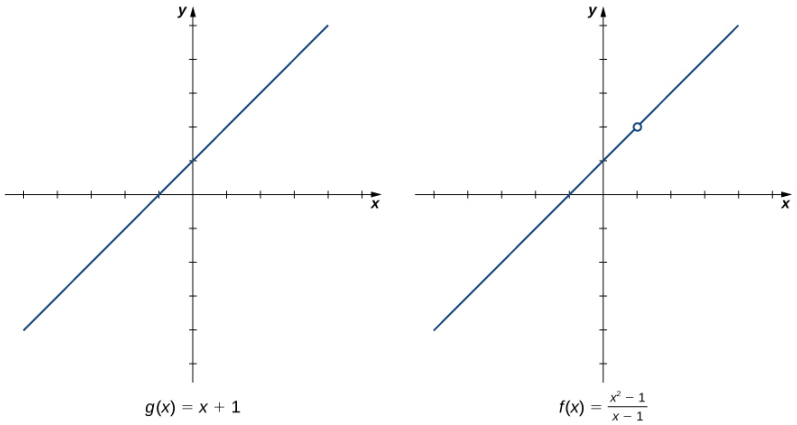


Figure 2.3.1: The graphs of $f(x)$ and $g(x)$ are identical for all $x \neq 1$. Their limits at 1 are equal.

We see that

$$\lim_{x\to 1}\frac{x^2-1}{x-1}=\lim_{x\to 1}\frac{(x-1)(x+1)}{x-1}=\lim_{x\to 1}(x+1)=2.$$

The limit has the form $\lim_{x\to a} f(x)/g(x)$, where $\lim_{x\to a} f(x)=0$ and $\lim_{x\to a} g(x)=0$. (In this case, we say that $f(x)/g(x)$ has the indeterminate form $0/0$.) The following Problem-Solving Strategy provides a general outline for evaluating limits of this type.

Problem-Solving Strategy: Calculating a Limit When $f(x)/g(x)$ has the Indeterminate Form $0/0$

1. First, we need to make sure that our function has the appropriate form and cannot be evaluated immediately using the limit laws.
2. We then need to find a function that is equal to $h(x)=f(x)/g(x)$ for all $x\neq a$ over some interval containing a. To do this, we may need to try one or more of the following steps:
   a. If $f(x)$ and $g(x)$ are polynomials, we should factor each function and cancel out any common factors.
   b. If the numerator or denominator contains a difference involving a square root, we should try multiplying the numerator and denominator by the conjugate of the expression involving the square root.
   c. If $f(x)/g(x)$ is a complex fraction, we begin by simplifying it.
3. Last, we apply the limit laws.

The next examples demonstrate the use of this Problem-Solving Strategy. Example 2.3.4 illustrates the factor-and-cancel technique; Example 2.3.5 shows multiplying by a conjugate. In Example 2.3.6, we look at simplifying a complex fraction.

Example 2.3.4: Evaluating a Limit by Factoring and Canceling

Evaluate $\lim_{x\to 3}\dfrac{x^2-3x}{2x^2-5x-3}$.

Solution

Step 1. The function $f(x)=\dfrac{x^2-3x}{2x^2-5x-3}$ is undefined for $x=3$. In fact, if we substitute 3 into the function we get $0/0$, which is undefined. Factoring and canceling is a good strategy:

$$\lim_{x\to 3}\frac{x^2-3x}{2x^2-5x-3}=\lim_{x\to 3}\frac{x(x-3)}{(x-3)(2x+1)}$$

Step 2. For all $x\neq 3$, $\dfrac{x^2-3x}{2x^2-5x-3}=\dfrac{x}{2x+1}$. Therefore,

$$\lim_{x\to 3}\frac{x(x-3)}{(x-3)(2x+1)}=\lim_{x\to 3}\frac{x}{2x+1}.$$

Step 3. Evaluate using the limit laws:

$$\lim_{x\to 3}\frac{x}{2x+1}=\frac{3}{7}.$$

Exercise 2.3.4

Evaluate $\lim_{x\to -3}\dfrac{x^2+4x+3}{x^2-9}$.

Hint

Follow the steps in the Problem-Solving Strategy

Answer

$\dfrac{1}{3}$

Example 2.3.5: Evaluating a Limit by Multiplying by a Conjugate

Evaluate $\lim_{x\to -1}\dfrac{\sqrt{x+2}-1}{x+1}$.

Solution

Step 1. $\dfrac{\sqrt{x+2}-1}{x+1}$ has the form $0/0$ at -1. Let's begin by multiplying by $\sqrt{x+2}+1$, the conjugate of $\sqrt{x+2}-1$, on the numerator and denominator:

$$\lim_{x\to -1}\frac{\sqrt{x+2}-1}{x+1}=\lim_{x\to -1}\frac{\sqrt{x+2}-1}{x+1}\cdot\frac{\sqrt{x+2}+1}{\sqrt{x+2}+1}.$$

Step 2. We then multiply out the numerator. We don't multiply out the denominator because we are hoping that the $(x+1)$ in the denominator cancels out in the end:

$$=\lim_{x\to -1}\frac{x+1}{(x+1)(\sqrt{x+2}+1)}.$$

Step 3. Then we cancel:

$$=\lim_{x\to -1}\frac{1}{\sqrt{x+2}+1}.$$

Step 4. Last, we apply the limit laws:

$$\lim_{x\to -1}\frac{1}{\sqrt{x+2}+1}=\frac{1}{2}.$$

Exercise 2.3.5

Evaluate $\lim_{x\to 5}\dfrac{\sqrt{x-1}-2}{x-5}$.

Hint

Follow the steps in the Problem-Solving Strategy

Answer

$\dfrac{1}{4}$

Example 2.3.6: Evaluating a Limit by Simplifying a Complex Fraction

Evaluate $\lim\limits_{x\to 1}\dfrac{\dfrac{1}{x+1}-\dfrac{1}{2}}{x-1}$.

Solution

Step 1. $\dfrac{\dfrac{1}{x+1}-\dfrac{1}{2}}{x-1}$ has the form $0/0$ at 1. We simplify the algebraic fraction by multiplying by $2(x+1)/2(x+1)$:

$$\lim_{x\to 1}\frac{\dfrac{1}{x+1}-\dfrac{1}{2}}{x-1}=\lim_{x\to 1}\frac{\dfrac{1}{x+1}-\dfrac{1}{2}}{x-1}\cdot\frac{2(x+1)}{2(x+1)}.$$

Step 2. Next, we multiply through the numerators. Do not multiply the denominators because we want to be able to cancel the factor $(x-1)$:

$$=\lim_{x\to 1}\frac{2-(x+1)}{2(x-1)(x+1)}.$$

Step 3. Then, we simplify the numerator:

$$=\lim_{x\to 1}\frac{-x+1}{2(x-1)(x+1)}.$$

Step 4. Now we factor out -1 from the numerator:

$$=\lim_{x\to 1}\frac{-(x-1)}{2(x-1)(x+1)}.$$

Step 5. Then, we cancel the common factors of $(x-1)$:

$$=\lim_{x\to 1}\frac{-1}{2(x+1)}.$$

Step 6. Last, we evaluate using the limit laws:

$$\lim_{x\to 1}\frac{-1}{2(x+1)}=-\frac{1}{4}.$$

Exercise 2.3.6

Evaluate $\lim\limits_{x\to -3}\dfrac{\dfrac{1}{x+2}+1}{x+3}$.

Hint

Follow the steps in the Problem-Solving Strategy

Answer

-1

Example 2.3.7 does not fall neatly into any of the patterns established in the previous examples. However, with a little creativity, we can still use these same techniques.

Example 2.3.7: Evaluating a Limit When the Limit Laws Do Not Apply

Evaluate $\lim\limits_{x\to 0}\left(\dfrac{1}{x}+\dfrac{5}{x(x-5)}\right)$.

Solution:

Both $1/x$ and $5/x(x-5)$ fail to have a limit at zero. Since neither of the two functions has a limit at zero, we cannot apply the sum law for limits; we must use a different strategy. In this case, we find the limit by performing addition and then applying one of our previous strategies. Observe that

$$\frac{1}{x}+\frac{5}{x(x-5)}=\frac{x-5+5}{x(x-5)}=\frac{x}{x(x-5)}.$$

Thus,

$$\lim_{x\to 0}\left(\frac{1}{x}+\frac{5}{x(x-5)}\right)=\lim_{x\to 0}\frac{x}{x(x-5)}=\lim_{x\to 0}\frac{1}{x-5}=-\frac{1}{5}.$$

? Exercise 2.3.7

Evaluate $\lim_{x\to 3}\left(\frac{1}{x-3}-\frac{4}{x^2-2x-3}\right)$.

Hint

Use the same technique as Example 2.3.7. Don't forget to factor x^2-2x-3 before getting a common denominator.

Answer

$\frac{1}{4}$

Let's now revisit one-sided limits. Simple modifications in the limit laws allow us to apply them to one-sided limits. For example, to apply the limit laws to a limit of the form $\lim_{x\to a^-}h(x)$, we require the function $h(x)$ to be defined over an open interval of the form (b,a); for a limit of the form $\lim_{x\to a^+}h(x)$, we require the function $h(x)$ to be defined over an open interval of the form (a,c). Example 2.3.8A illustrates this point.

✓ Example 2.3.8A: Evaluating a One-Sided Limit Using the Limit Laws

Evaluate each of the following limits, if possible.

a. $\lim_{x\to 3^-}\sqrt{x-3}$
b. $\lim_{x\to 3^+}\sqrt{x-3}$

Solution

Figure 2.3.2 illustrates the function $f(x)=\sqrt{x-3}$ and aids in our understanding of these limits.

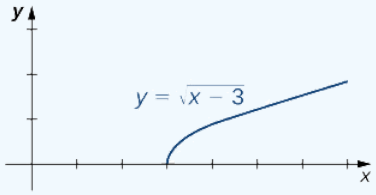


Figure 2.3.2: The graph shows the function $f(x)=\sqrt{x-3}$.

a. The function $f(x)=\sqrt{x-3}$ is defined over the interval $[3,+\infty)$. Since this function is not defined to the left of 3, we cannot apply the limit laws to compute $\lim_{x\to 3^-}\sqrt{x-3}$. In fact, since $f(x)=\sqrt{x-3}$ is undefined to the left of 3, $\lim_{x\to 3^-}\sqrt{x-3}$ does not exist.

b. Since $f(x)=\sqrt{x-3}$ is defined to the right of 3, the limit laws do apply to $\lim_{x\to 3^+}\sqrt{x-3}$. By applying these limit laws we obtain $\lim_{x\to 3^+}\sqrt{x-3}=0$.

In Example 2.3.8B we look at one-sided limits of a piecewise-defined function and use these limits to draw a conclusion about a two-sided limit of the same function.

Example 2.3.8B: Evaluating a Two-Sided Limit Using the Limit Laws

For $f(x) = \begin{cases} 4x-3, & \text{if } x < 2 \\ (x-3)^2, & \text{if } x \geq 2 \end{cases}$, evaluate each of the following limits:

a. $\lim\limits_{x \to 2^-} f(x)$
b. $\lim\limits_{x \to 2^+} f(x)$
c. $\lim\limits_{x \to 2} f(x)$

Solution

Figure 2.3.3 illustrates the function $f(x)$ and aids in our understanding of these limits.

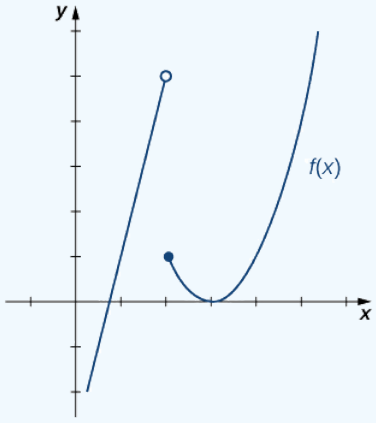


Figure 2.3.3: This graph shows a function $f(x)$.

a. Since $f(x) = 4x - 3$ for all x in $(-\infty, 2)$, replace $f(x)$ in the limit with $4x - 3$ and apply the limit laws:

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} (4x - 3) = 5$$

b. Since $f(x) = (x-3)^2$for all x in $(2, +\infty)$, replace $f(x)$ in the limit with $(x-3)^2$ and apply the limit laws:

$$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^-} (x-3)^2 = 1.$$

c. Since $\lim\limits_{x \to 2^-} f(x) = 5$ and $\lim\limits_{x \to 2^+} f(x) = 1$, we conclude that $\lim\limits_{x \to 2} f(x)$ does not exist.

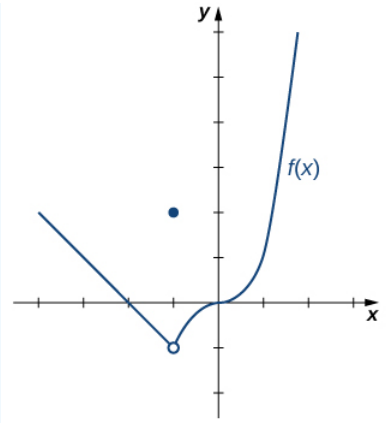
Exercise 2.3.8

Graph $f(x) = \begin{cases} -x-2, & \text{if } x < -1 \\ 2, & \text{if } x = -1 \\ x^3, & \text{if } x > -1 \end{cases}$ and evaluate $\lim\limits_{x \to -1^-} f(x)$.

Hint

Use the method in Example 2.3.8B to evaluate the limit.

Answer



$$\lim_{x \to -1^-} f(x) = -1$$

We now turn our attention to evaluating a limit of the form $\lim_{x \to a} \dfrac{f(x)}{g(x)}$, where $\lim_{x \to a} f(x) = K$, where $K \neq 0$ and $\lim_{x \to a} g(x) = 0$. That is, $f(x)/g(x)$ has the form $K/0, K \neq 0$ at a.

Example 2.3.9: Evaluating a Limit of the Form $K/0,\ K \neq 0$ Using the Limit Laws

Evaluate $\lim_{x \to 2^-} \dfrac{x-3}{x^2-2x}$.

Solution

Step 1. After substituting in $x = 2$, we see that this limit has the form $-1/0$. That is, as x approaches 2 from the left, the numerator approaches -1; and the denominator approaches 0. Consequently, the magnitude of $\dfrac{x-3}{x(x-2)}$ becomes infinite. To get a better idea of what the limit is, we need to factor the denominator:

$$\lim_{x \to 2^-} \frac{x-3}{x^2-2x} = \lim_{x \to 2^-} \frac{x-3}{x(x-2)}$$

Step 2. Since $x-2$ is the only part of the denominator that is zero when 2 is substituted, we then separate $1/(x-2)$ from the rest of the function:

$$= \lim_{x \to 2^-} \frac{x-3}{x} \cdot \frac{1}{x-2}$$

Step 3. Using the Limit Laws, we can write:

$$= \left(\lim_{x \to 2^-} \frac{x-3}{x} \right) \cdot \left(\lim_{x \to 2^-} \frac{1}{x-2} \right).$$

Step 4. $\lim_{x \to 2^-} \dfrac{x-3}{x} = -\dfrac{1}{2}$ and $\lim_{x \to 2^-} \dfrac{1}{x-2} = -\infty$. Therefore, the product of $(x-3)/x$ and $1/(x-2)$ has a limit of $+\infty$:

$$\lim_{x \to 2^-} \frac{x-3}{x^2-2x} = +\infty.$$

Exercise 2.3.9

Evaluate $\lim_{x \to 1} \dfrac{x+2}{(x-1)^2}$.

Solution

Use the methods from Example 2.3.9.

Answer

$+\infty$

The Squeeze Theorem

The techniques we have developed thus far work very well for algebraic functions, but we are still unable to evaluate limits of very basic trigonometric functions. The next theorem, called the **squeeze theorem,** proves very useful for establishing basic trigonometric limits. This theorem allows us to calculate limits by "squeezing" a function, with a limit at a point a that is unknown, between two functions having a common known limit at a. Figure 2.3.4 illustrates this idea.

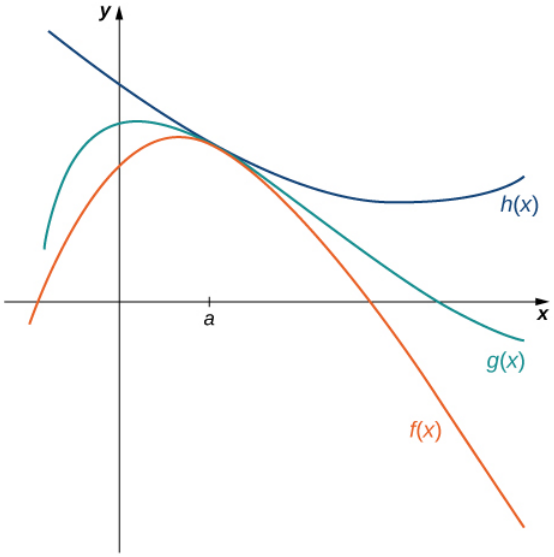


Figure 2.3.4: The Squeeze Theorem applies when $f(x) \le g(x) \le h(x)$ and $\lim_{x \to a} f(x) = \lim_{x \to a} h(x)$.

Theorem 2.3.4: The Squeeze Theorem

Let $f(x), g(x)$, and $h(x)$ be defined for all $x \neq a$ over an open interval containing a. If

$$f(x) \le g(x) \le h(x)$$

for all $x \neq a$ in an open interval containing a and

$$\lim_{x \to a} f(x) = L = \lim_{x \to a} h(x)$$

where L is a real number, then $\lim_{x \to a} g(x) = L$.

Example 2.3.10: Applying the Squeeze Theorem

Apply the squeeze theorem to evaluate $\lim_{x \to 0} x \cos x$.

Solution

Because $-1 \le \cos x \le 1$ for all x, we have $-x \le x \cos x \le x$ for $x \ge 0$ and $-x \ge x \cos x \ge x$ for $x \le 0$ (if x is negative the direction of the inequalities changes when we multiply). Since $\lim_{x \to 0}(-x) = 0 = \lim_{x \to 0} x$, from the squeeze theorem, we obtain $\lim_{x \to 0} x \cos x = 0$. The graphs of $f(x) = -x,\ g(x) = x \cos x$, and $h(x) = x$ are shown in Figure 2.3.5.

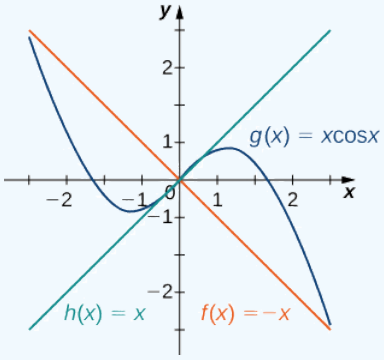


Figure 2.3.5: The graphs of $f(x)$, $g(x)$, and $h(x)$ are shown around the point $x = 0$.

? Exercise 2.3.10

Use the squeeze theorem to evaluate $\lim_{x\to 0} x^2 \sin\frac{1}{x}$.

Hint

Use the fact that $-x^2 \le x^2 \sin(1/x) \le x^2$ to help you find two functions such that $x^2 \sin(1/x)$ is squeezed between them.

Answer

0

We now use the squeeze theorem to tackle several very important limits. Although this discussion is somewhat lengthy, these limits prove invaluable for the development of the material in both the next section and the next chapter. The first of these limits is $\lim_{\theta\to 0} \sin\theta$. Consider the unit circle shown in Figure 2.3.6. In the figure, we see that $\sin\theta$ is the y-coordinate on the unit circle and it corresponds to the line segment shown in blue. The radian measure of angle θ is the length of the arc it subtends on the unit circle. Therefore, we see that for $0 < \theta < \frac{\pi}{2}$, we have $0 < \sin\theta < \theta$.

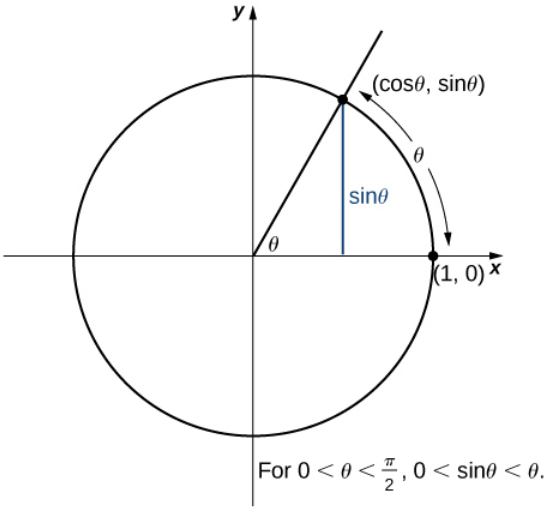


Figure 2.3.6: The sine function is shown as a line on the unit circle.

Because $\lim_{\theta\to 0^+} 0 = 0$ and $\lim_{\theta\to 0^+} \theta = 0$, by using the squeeze theorem we conclude that

$$\lim_{\theta\to 0^+} \sin\theta = 0.$$

To see that $\lim_{\theta\to 0^-} \sin\theta = 0$ as well, observe that for $-\frac{\pi}{2} < \theta < 0, 0 < -\theta < \frac{\pi}{2}$ and hence, $0 < \sin(-\theta) < -\theta$. Consequently, $0 < -\sin\theta < -\theta$. It follows that $0 > \sin\theta > \theta$. An application of the squeeze theorem produces the desired limit. Thus, since $\lim_{\theta\to 0^+} \sin\theta = 0$ and $\lim_{\theta\to 0^-} \sin\theta = 0$,

$$\lim_{\theta \to 0} \sin \theta = 0$$

Next, using the identity $\cos \theta = \sqrt{1 - \sin^2 \theta}$ for $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, we see that

$$\lim_{\theta \to 0} \cos \theta = \lim_{\theta \to 0} \sqrt{1 - \sin^2 \theta} = 1.$$

We now take a look at a limit that plays an important role in later chapters—namely, $\lim_{\theta \to 0} \frac{\sin \theta}{\theta}$. To evaluate this limit, we use the unit circle in Figure 2.3.7. Notice that this figure adds one additional triangle to Figure 2.3.6. We see that the length of the side opposite angle θ in this new triangle is $\tan \theta$. Thus, we see that for $0 < \theta < \frac{\pi}{2}$, we have $\sin \theta < \theta < \tan \theta$.

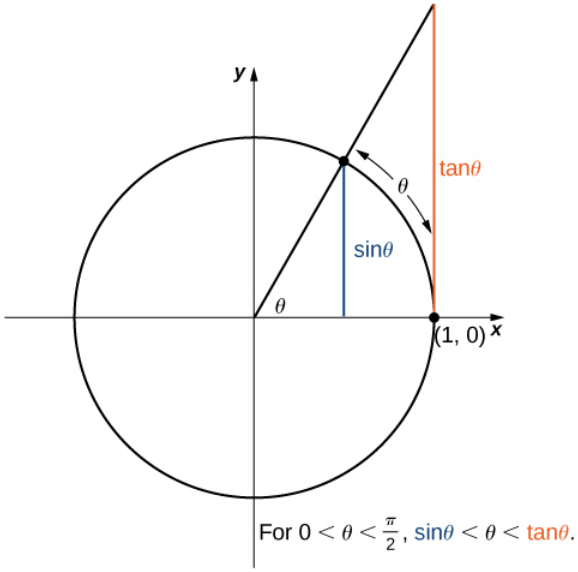


Figure 2.3.7: The sine and tangent functions are shown as lines on the unit circle.

By dividing by $\sin \theta$ in all parts of the inequality, we obtain

$$1 < \frac{\theta}{\sin \theta} < \frac{1}{\cos \theta}.$$

Equivalently, we have

$$1 > \frac{\sin \theta}{\theta} > \cos \theta.$$

Since $\lim_{\theta \to 0^+} 1 = 1 = \lim_{\theta \to 0^+} \cos \theta$, we conclude that $\lim_{\theta \to 0^+} \frac{\sin \theta}{\theta} = 1$, by the squeeze theorem. By applying a manipulation similar to that used in demonstrating that $\lim_{\theta \to 0^-} \sin \theta = 0$, we can show that $\lim_{\theta \to 0^-} \frac{\sin \theta}{\theta} = 1$. Thus,

$$\lim_{\theta \to 0} \frac{\sin \theta}{\theta} = 1.$$

In Example 2.3.11, we use this limit to establish $\lim_{\theta \to 0} \frac{1 - \cos \theta}{\theta} = 0$. This limit also proves useful in later chapters.

Example 2.3.11: Evaluating an Important Trigonometric Limit

Evaluate $\lim_{\theta \to 0} \frac{1 - \cos \theta}{\theta}$.

Solution

In the first step, we multiply by the conjugate so that we can use a trigonometric identity to convert the cosine in the numerator to a sine:

$$\lim_{\theta\to 0}\frac{1-\cos\theta}{\theta}=\lim_{\theta\to 0}\frac{1-\cos\theta}{\theta}\cdot\frac{1+\cos\theta}{1+\cos\theta}$$
$$=\lim_{\theta\to 0}\frac{1-\cos^2\theta}{\theta(1+\cos\theta)}$$
$$=\lim_{\theta\to 0}\frac{\sin^2\theta}{\theta(1+\cos\theta)}$$
$$=\lim_{\theta\to 0}\frac{\sin\theta}{\theta}\cdot\frac{\sin\theta}{1+\cos\theta}$$
$$=\left(\lim_{\theta\to 0}\frac{\sin\theta}{\theta}\right)\cdot\left(\lim_{\theta\to 0}\frac{\sin\theta}{1+\cos\theta}\right)$$
$$=1\cdot\frac{0}{2}=0.$$

Therefore,

$$\lim_{\theta\to 0}\frac{1-\cos\theta}{\theta}=0.$$

Exercise 2.3.11

Evaluate $\lim_{\theta\to 0}\frac{1-\cos\theta}{\sin\theta}$.

Hint

Multiply numerator and denominator by $1+\cos\theta$.

Answer

0

Deriving the Formula for the Area of a Circle

Some of the geometric formulas we take for granted today were first derived by methods that anticipate some of the methods of calculus. The Greek mathematician *Archimedes* (ca. 287−212; BCE) was particularly inventive, using polygons inscribed within circles to approximate the area of the circle as the number of sides of the polygon increased. He never came up with the idea of a limit, but we can use this idea to see what his geometric constructions could have predicted about the limit.

We can estimate the area of a circle by computing the area of an inscribed regular polygon. Think of the regular polygon as being made up of n triangles. By taking the limit as the vertex angle of these triangles goes to zero, you can obtain the area of the circle. To see this, carry out the following steps:

1.Express the height h and the base b of the isosceles triangle in Figure 2.3.8 in terms of θ and r.

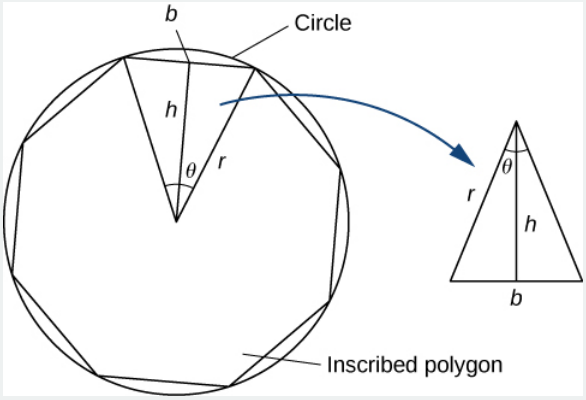


Figure 2.3.8

2. Using the expressions that you obtained in step 1, express the area of the isosceles triangle in terms of θ and r.

(Substitute $\frac{1}{2}\sin\theta$ for $\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$ in your expression.)

3. If an n-sided regular polygon is inscribed in a circle of radius r, find a relationship between θ and n. Solve this for n. Keep in mind there are 2π radians in a circle. (Use radians, not degrees.)

4. Find an expression for the area of the n-sided polygon in terms of r and θ.

5. To find a formula for the area of the circle, find the limit of the expression in step 4 as θ goes to zero. (Hint: $\lim_{\theta\to 0}\frac{\sin\theta}{\theta} = 1$).

The technique of estimating areas of regions by using polygons is revisited in Introduction to Integration.

Key Concepts

- The limit laws allow us to evaluate limits of functions without having to go through step-by-step processes each time.
- For polynomials and rational functions,

$$\lim_{x\to a} f(x) = f(a).$$

- You can evaluate the limit of a function by factoring and canceling, by multiplying by a conjugate, or by simplifying a complex fraction.
- The squeeze theorem allows you to find the limit of a function if the function is always greater than one function and less than another function with limits that are known.

Key Equations

- **Basic Limit Results**

$$\lim_{x\to a} x = a \qquad \lim_{x\to a} c = c$$

- **Important Limits**

$$\lim_{\theta\to 0}\sin\theta = 0$$

$$\lim_{\theta\to 0}\cos\theta = 1$$

$$\lim_{\theta\to 0}\frac{\sin\theta}{\theta} = 1$$

$$\lim_{\theta\to 0}\frac{1-\cos\theta}{\theta} = 0$$

Glossary

constant multiple law for limits

the limit law

$$\lim_{x\to a} cf(x) = c\cdot\lim_{x\to a} f(x) = cL$$

difference law for limits

the limit law

$$\lim_{x\to a}(f(x)-g(x)) = \lim_{x\to a} f(x) - \lim_{x\to a} g(x) = L - M$$

limit laws

the individual properties of limits; for each of the individual laws, let $f(x)$ and $g(x)$ be defined for all $x \neq a$ over some open interval containing a; assume that L and M are real numbers so that $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$; let c be a constant

power law for limits

the limit law

$$\lim_{x \to a} (f(x))^n = (\lim_{x \to a} f(x))^n = L^n$$

for every positive integer n

product law for limits

the limit law

$$\lim_{x \to a}(f(x) \cdot g(x)) = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x) = L \cdot M$$

quotient law for limits

the limit law $\lim_{x \to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)} = \dfrac{L}{M}$ for M≠0

root law for limits

the limit law $\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)} = \sqrt[n]{L}$ for all L if n is odd and for $L \geq 0$ if n is even

squeeze theorem

states that if $f(x) \leq g(x) \leq h(x)$ for all $x \neq a$ over an open interval containing a and $\lim_{x \to a} f(x) = L = \lim_{x \to a} h(x)$ where L is a real number, then $\lim_{x \to a} g(x) = L$

sum law for limits

The limit law $\lim_{x \to a}(f(x) + g(x)) = \lim_{x \to a} f(x) + \lim_{x \to a} g(x) = L + M$

This page titled 2.3: The Limit Laws is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

2.3E: EXERCISES FOR SECTION 2.3

In exercises 1 - 4, use the limit laws to evaluate each limit. Justify each step by indicating the appropriate limit law(s).

1) $\lim\limits_{x\to 0}(4x^2-2x+3)$

Answer

Use constant multiple law and difference law:

$\lim\limits_{x\to 0}(4x^2-2x+3)=4\lim\limits_{x\to 0}x^2-2\lim\limits_{x\to 0}x+\lim\limits_{x\to 0}3=0+0+3=3$

2) $\lim\limits_{x\to 1}\dfrac{x^3+3x^2+5}{4-7x}$

3) $\lim\limits_{x\to -2}\sqrt{x^2-6x+3}$

Answer

Use root law: $\lim\limits_{x\to -2}\sqrt{x^2-6x+3}=\sqrt{\lim\limits_{x\to -2}(x^2-6x+3)}=\sqrt{19}$

4) $\lim\limits_{x\to -1}(9x+1)^2$

In exercises 5 - 10, use direct substitution to evaluate the limit of each continuous function.

5) $\lim\limits_{x\to 7}x^2$

Answer

$\lim\limits_{x\to 7}x^2 \;=\; 49$

6) $\lim\limits_{x\to -2}(4x^2-1)$

7) $\lim\limits_{x\to 0}\dfrac{1}{1+\sin x}$

Answer

$\lim\limits_{x\to 0}\dfrac{1}{1+\sin x} \;=\; 1$

8) $\lim\limits_{x\to 2}e^{2x-x^2}$

9) $\lim\limits_{x\to 1}\dfrac{2-7x}{x+6}$

Answer

$\lim\limits_{x\to 1}\dfrac{2-7x}{x+6} \;=\; -\dfrac{5}{7}$

10) $\lim\limits_{x\to 3}\ln e^{3x}$

In exercises 11 - 20, use direct substitution to show that each limit leads to the indeterminate form $0/0$. Then, evaluate the limit analytically.

11) $\lim\limits_{x\to 4}\dfrac{x^2-16}{x-4}$

Answer

When $x=4$, $\dfrac{x^2-16}{x-4}=\dfrac{16-16}{4-4}=\dfrac{0}{0}$;

then, $\lim\limits_{x\to 4}\dfrac{x^2-16}{x-4}=\lim\limits_{x\to 4}\dfrac{(x+4)(x-4)}{x-4}=\lim\limits_{x\to 4}(x+4)=4+4=8$

12) $\lim\limits_{x\to 2}\dfrac{x-2}{x^2-2x}$

13) $\lim\limits_{x\to 6}\dfrac{3x-18}{2x-12}$

Answer

When $x = 6$, $\dfrac{3x-18}{2x-12} = \dfrac{18-18}{12-12} = \dfrac{0}{0}$;

then, $\lim\limits_{x\to 6} \dfrac{3x-18}{2x-12} = \lim\limits_{x\to 6} \dfrac{3(x-6)}{2(x-6)} = \lim\limits_{x\to 6} \dfrac{3}{2} = \dfrac{3}{2}$

14) $\lim\limits_{h\to 0} \dfrac{(1+h)^2-1}{h}$

15) $\lim\limits_{t\to 9} \dfrac{t-9}{\sqrt{t}-3}$

Answer

When $t = 9$, $\dfrac{t-9}{\sqrt{t}-3} = \dfrac{9-9}{3-3} = \dfrac{0}{0}$;

then, $\lim\limits_{t\to 9} \dfrac{t-9}{\sqrt{t}-3} = \lim\limits_{t\to 9} \dfrac{t-9}{\sqrt{t}-3}\dfrac{\sqrt{t}+3}{\sqrt{t}+3} = \lim\limits_{t\to 9} \dfrac{(t-9)(\sqrt{t}+3)}{t-9} = \lim\limits_{t\to 9}(\sqrt{t}+3) = \sqrt{9}+3 = 6$

16) $\lim\limits_{h\to 0} \dfrac{\dfrac{1}{a+h} - \dfrac{1}{a}}{h}$, where a is a real-valued constant

17) $\lim\limits_{\theta\to\pi} \dfrac{\sin\theta}{\tan\theta}$

Answer

When $\theta = \pi$, $\dfrac{\sin\theta}{\tan\theta} = \dfrac{\sin\pi}{\tan\pi} = \dfrac{0}{0}$;

then, $\lim\limits_{\theta\to\pi} \dfrac{\sin\theta}{\tan\theta} = \lim\limits_{\theta\to\pi} \dfrac{\sin\theta}{\frac{\sin\theta}{\cos\theta}} = \lim\limits_{\theta\to\pi} \cos\theta = \cos\pi = -1$

18) $\lim\limits_{x\to 1} \dfrac{x^3-1}{x^2-1}$

19) $\lim\limits_{x\to 1/2} \dfrac{2x^2+3x-2}{2x-1}$

Answer

When $x = 1/2$, $\dfrac{2x^2+3x-2}{2x-1} = \dfrac{\frac{1}{2}+\frac{3}{2}-2}{1-1} = \dfrac{0}{0}$;

then, $\lim\limits_{x\to 1/2} \dfrac{2x^2+3x-2}{2x-1} = \lim\limits_{x\to 1/2} \dfrac{(2x-1)(x+2)}{2x-1} = \lim\limits_{x\to 1/2}(x+2) = \dfrac{1}{2}+2 = \dfrac{5}{2}$

20) $\lim\limits_{x\to -3} \dfrac{\sqrt{x+4}-1}{x+3}$

In exercises 21 - 24, use direct substitution to obtain an undefined expression. Then, use the method used in Example 9 of this section to simplify the function and determine the limit.

21) $\lim\limits_{x\to -2^-} \dfrac{2x^2+7x-4}{x^2+x-2}$

Answer

$-\infty$

22) $\lim\limits_{x\to -2^+} \dfrac{2x^2+7x-4}{x^2+x-2}$

23) $\lim\limits_{x\to 1^-} \dfrac{2x^2+7x-4}{x^2+x-2}$

Answer

$-\infty$

24) $\lim\limits_{x\to 1^+} \dfrac{2x^2+7x-4}{x^2+x-2}$

In exercises 25 - 32, assume that $\lim_{x\to 6} f(x) = 4$, $\lim_{x\to 6} g(x) = 9$, and $\lim_{x\to 6} h(x) = 6$. Use these three facts and the limit laws to evaluate each limit.

25) $\lim_{x\to 6} 2f(x)g(x)$

Answer

$$\lim_{x\to 6} 2f(x)g(x) = 2\left(\lim_{x\to 6} f(x)\right)\left(\lim_{x\to 6} g(x)\right) = 2(4)(9) = 72$$

26) $\lim_{x\to 6} \dfrac{g(x)-1}{f(x)}$

27) $\lim_{x\to 6} \left(f(x) + \dfrac{1}{3}g(x)\right)$

Answer

$$\lim_{x\to 6} \left(f(x) + \frac{1}{3}g(x)\right) = \lim_{x\to 6} f(x) + \frac{1}{3}\lim_{x\to 6} g(x) = 4 + \frac{1}{3}(9) = 7$$

28) $\lim_{x\to 6} \dfrac{\left(h(x)\right)^3}{2}$

29) $\lim_{x\to 6} \sqrt{g(x) - f(x)}$

Answer

$$\lim_{x\to 6} \sqrt{g(x) - f(x)} = \sqrt{\lim_{x\to 6} g(x) - \lim_{x\to 6} f(x)} = \sqrt{9-4} = \sqrt{5}$$

30) $\lim_{x\to 6} x \cdot h(x)$

31) $\lim_{x\to 6}[(x+1)\cdot f(x)]$

Answer

$$\lim_{x\to 6}[(x+1)f(x)] = \left(\lim_{x\to 6}(x+1)\right)\left(\lim_{x\to 6} f(x)\right) = 7(4) = 28$$

32) $\lim_{x\to 6}(f(x)\cdot g(x) - h(x))$

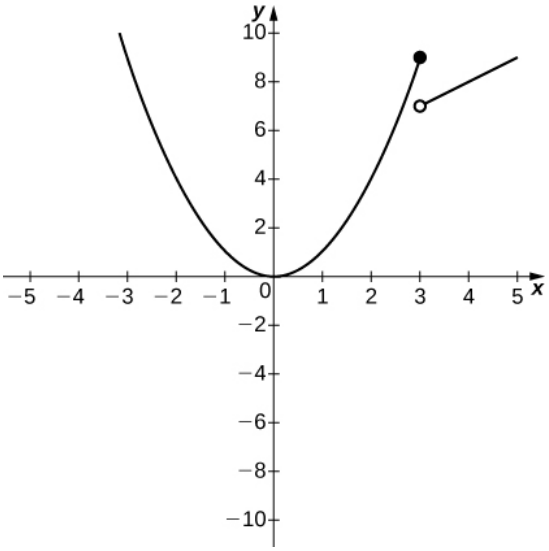
[T] In exercises 33 - 35, use a calculator to draw the graph of each piecewise-defined function and study the graph to evaluate the given limits.

33) $f(x) = \begin{cases} x^2, & x \le 3 \\ x+4, & x > 3 \end{cases}$

a. $\lim_{x\to 3^-} f(x)$

b. $\lim_{x\to 3^+} f(x)$

Answer



a. **9**; b.**7**

34) $g(x)=\begin{cases}x^3-1, & x\le 0\\ 1, & x>0\end{cases}$

a. $\lim\limits_{x\to 0^-} g(x)$

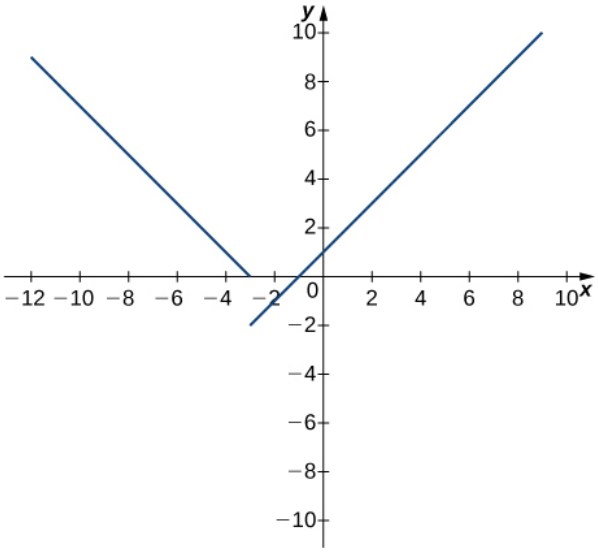
b. $\lim\limits_{x\to 0^+} g(x)$

35) $h(x)=\begin{cases}x^2-2x+1, & x<2\\ 3-x, & x\ge 2\end{cases}$

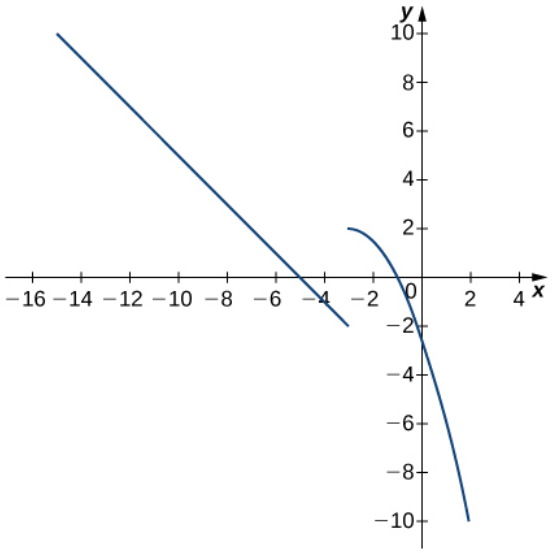
a. $\lim\limits_{x\to 2^-} h(x)$

b. $\lim\limits_{x\to 2^+} h(x)$

In exercises 36 - 43, use the following graphs and the limit laws to evaluate each limit.



$y = f(x)$



$y = g(x)$

36) $\lim_{x \to -3^+} (f(x) + g(x))$

37) $\lim_{x \to -3^-} (f(x) - 3g(x))$

Answer

$$\lim_{x \to -3^-} (f(x) - 3g(x)) = \lim_{x \to -3^-} f(x) - 3 \lim_{x \to -3^-} g(x) = 0 + 6 = 6$$

38) $\lim_{x \to 0} \dfrac{f(x)g(x)}{3}$

39) $\lim_{x \to -5} \dfrac{2 + g(x)}{f(x)}$

Answer

$$\lim_{x \to -5} \frac{2 + g(x)}{f(x)} = \frac{2 + \left(\lim_{x \to -5} g(x)\right)}{\lim_{x \to -5} f(x)} = \frac{2 + 0}{2} = 1$$

40) $\lim_{x \to 1} (f(x))^2$

41) $\lim_{x\to 1}\sqrt[3]{f(x)-g(x)}$

Answer

$$\lim_{x\to 1}\sqrt[3]{f(x)-g(x)}=\sqrt[3]{\lim_{x\to 1}f(x)-\lim_{x\to 1}g(x)}=\sqrt[3]{2+5}=\sqrt[3]{7}$$

42) $\lim_{x\to -7}(x\cdot g(x))$

43) $\lim_{x\to -9}[x\cdot f(x)+2\cdot g(x)]$

Answer

$$\lim_{x\to -9}(xf(x)+2g(x))=\left(\lim_{x\to -9}x\right)\left(\lim_{x\to -9}f(x)\right)+2\lim_{x\to -9}g(x)=(-9)(6)+2(4)=-46$$

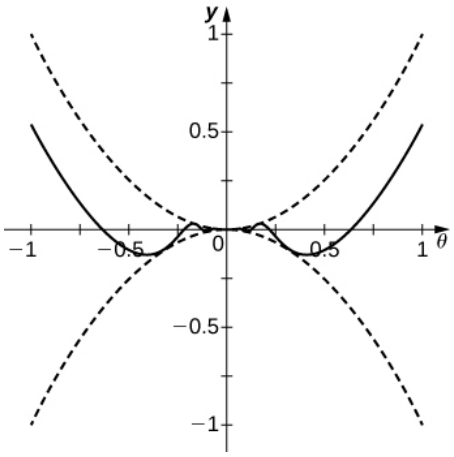
For exercises 44 - 46, evaluate the limit using the squeeze theorem. Use a calculator to graph the functions $f(x)$, $g(x)$, and $h(x)$ when possible.

44) [T] True or False? If $2x-1\le g(x)\le x^2-2x+3$, then $\lim_{x\to 2}g(x)=0$.

45) [T] $\lim_{\theta\to 0}\theta^2\cos\left(\frac{1}{\theta}\right)$

Answer

The limit is zero.



46) $\lim_{x\to 0}f(x)$, where $f(x)=\begin{cases}0, & x \text{ rational}\\ x^2, & x \text{ irrational}\end{cases}$

47) [T] In physics, the magnitude of an electric field generated by a point charge at a distance r in vacuum is governed by Coulomb's law: $E(r)=\frac{q}{4\pi\varepsilon_0 r^2}$, where E represents the magnitude of the electric field, q is the charge of the particle, r is the distance between the particle and where the strength of the field is measured, and $\frac{1}{4\pi\varepsilon_0}$ is Coulomb's constant: $8.988\times 109N\cdot m^2/C^2$.

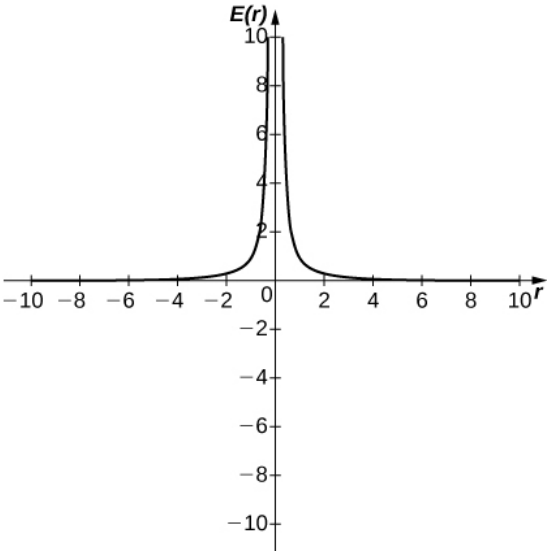
a. Use a graphing calculator to graph $E(r)$ given that the charge of the particle is $q=10^{-10}$.

b. Evaluate $\lim_{r\to 0^+}E(r)$. What is the physical meaning of this quantity? Is it physically relevant? Why are you evaluating from the right?

Answer

a.

https://math.libretexts.org/@go/page/50409



b. ∞. The magnitude of the electric field as you approach the particle q becomes infinite. It does not make physical sense to evaluate negative distance.

48) [T] The density of an object is given by its mass divided by its volume: $\rho = m/V$.

a. Use a calculator to plot the volume as a function of density $(V = m/\rho)$, assuming you are examining something of mass 8 kg $(m = 8)$.

b. Evaluate $\lim_{x \to 0^+} V(\rho)$ and explain the physical meaning.

2.3E: Exercises for Section 2.3 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by LibreTexts.

2.4: Continuity

Learning Objectives

- Explain the three conditions for continuity at a point.
- Describe three kinds of discontinuities.
- Define continuity on an interval.
- State the theorem for limits of composite functions.
- Provide an example of the intermediate value theorem.

Many functions have the property that their graphs can be traced with a pencil without lifting the pencil from the page. Such functions are called *continuous*. Other functions have points at which a break in the graph occurs, but satisfy this property over intervals contained in their domains. They are continuous on these intervals and are said to have a *discontinuity* at a point where a break occurs.

We begin our investigation of continuity by exploring what it means for a function to have *continuity at a point*. Intuitively, a function is continuous at a particular point if there is no break in its graph at that point.

Continuity at a Point

Before we look at a formal definition of what it means for a function to be continuous at a point, let's consider various functions that fail to meet our intuitive notion of what it means to be continuous at a point. We then create a list of conditions that prevent such failures.

Our first function of interest is shown in Figure 2.4.1. We see that the graph of $f(x)$ has a hole at a. In fact, $f(a)$ is undefined. At the very least, for $f(x)$ to be continuous at a, we need the following condition:

i. $f(a)$ is defined

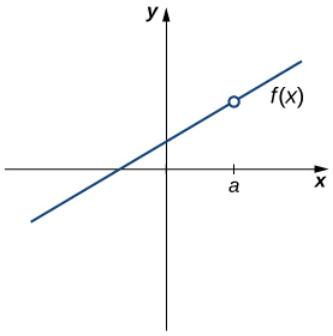


Figure 2.4.1: The function $f(x)$ is not continuous at a because $f(a)$ is undefined.

However, as we see in Figure 2.4.2, this condition alone is insufficient to guarantee continuity at the point a. Although $f(a)$ is defined, the function has a gap at a. In this example, the gap exists because $\lim\limits_{x \to a} f(x)$ does not exist. We must add another condition for continuity at a—namely,

ii. $\lim_{x \to a} f(x)$ exists

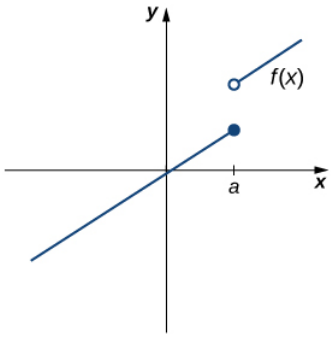


Figure 2.4.2: The function $f(x)$ is not continuous at a because $\lim_{x \to a} f(x)$ does not exist.

However, as we see in Figure 2.4.3, these two conditions by themselves do not guarantee continuity at a point. The function in this figure satisfies both of our first two conditions, but is still not continuous at a. We must add a third condition to our list:

iii. $\lim_{x \to a} f(x) = f(a)$

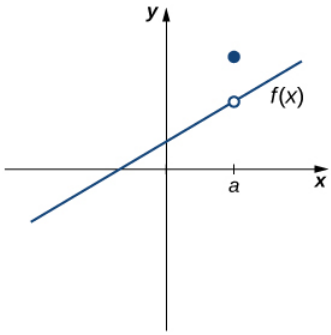


Figure 2.4.3: The function $f(x)$ is not continuous at a because $\lim_{x \to a} f(x) \neq f(a)$.

Now we put our list of conditions together and form a definition of continuity at a point.

Definition: Continuous at a Point

A function $f(x)$ is **continuous at a point** a if and only if the following three conditions are satisfied:

i. $f(a)$ is defined
ii. $\lim_{x \to a} f(x)$ exists
iii. $\lim_{x \to a} f(x) = f(a)$

A function is **discontinuous at a point** a if it fails to be continuous at a.

The following procedure can be used to analyze the continuity of a function at a point using this definition.

Problem-Solving Strategy: Determining Continuity at a Point

1. Check to see if $f(a)$ is defined. If $f(a)$ is undefined, we need go no further. The function is not continuous at a. If $f(a)$ is defined, continue to step 2.
2. Compute $\lim_{x \to a} f(x)$. In some cases, we may need to do this by first computing $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$. If $\lim_{x \to a} f(x)$ does not exist (that is, it is not a real number), then the function is not continuous at a and the problem is solved. If $\lim_{x \to a} f(x)$ exists, then continue to step 3.
3. Compare $f(a)$ and $\lim_{x \to a} f(x)$. If $\lim_{x \to a} f(x) \neq f(a)$, then the function is not continuous at a. If $\lim_{x \to a} f(x) = f(a)$, then the function is continuous at a.

The next three examples demonstrate how to apply this definition to determine whether a function is continuous at a given point. These examples illustrate situations in which each of the conditions for continuity in the definition succeed or fail.

Example 2.4.1A: Determining Continuity at a Point, Condition 1

Using the definition, determine whether the function $f(x) = \dfrac{x^2 - 4}{x - 2}$ is continuous at $x = 2$. Justify the conclusion.

Solution

Let's begin by trying to calculate $f(2)$. We can see that $f(2) = 0/0$, which is undefined. Therefore, $f(x) = \dfrac{x^2 - 4}{x - 2}$ is discontinuous at 2 because $f(2)$ is undefined. The graph of $f(x)$ is shown in Figure 2.4.4.

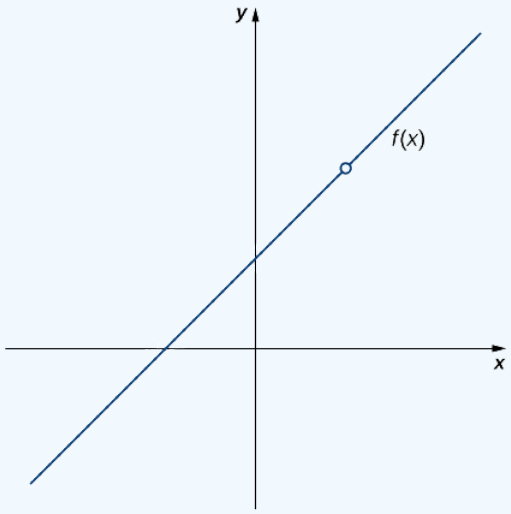


Figure 2.4.4: The function $f(x)$ is discontinuous at 2 because $f(2)$ is undefined.

Example 2.4.1B: Determining Continuity at a Point, Condition 2

Using the definition, determine whether the function $f(x) = \begin{cases} -x^2 + 4, & \text{if } x \leq 3 \\ 4x - 8, & \text{if } x > 3 \end{cases}$ is continuous at $x = 3$. Justify the conclusion.

Solution

Let's begin by trying to calculate $f(3)$.

$$f(3) = -(3^2) + 4 = -5.$$

Thus, $f(3)$ is defined. Next, we calculate $\lim\limits_{x \to 3} f(x)$. To do this, we must compute $\lim\limits_{x \to 3^-} f(x)$ and $\lim\limits_{x \to 3^+} f(x)$:

$$\lim_{x \to 3^-} f(x) = -(3^2) + 4 = -5$$

and

$$\lim_{x \to 3^+} f(x) = 4(3) - 8 = 4.$$

Therefore, $\lim\limits_{x \to 3} f(x)$ does not exist. Thus, $f(x)$ is not continuous at 3. The graph of $f(x)$ is shown in Figure 2.4.5.

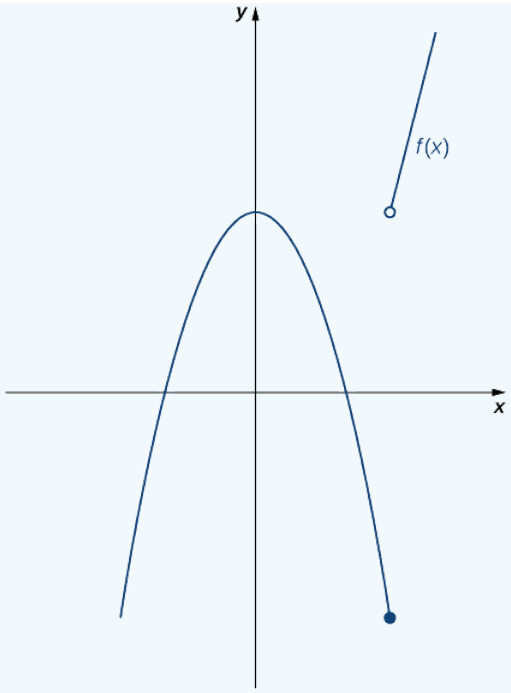


Figure 2.4.5: The function $f(x)$ is not continuous at 3 because $\lim_{x\to 3} f(x)$ does not exist.

Example 2.4.1C: Determining Continuity at a Point, Condition 3

Using the definition, determine whether the function $f(x) = \begin{cases} \frac{\sin x}{x}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$ is continuous at $x = 0$.

Solution

First, observe that

$$f(0) = 1$$

Next,

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{\sin x}{x} = 1.$$

Last, compare $f(0)$ and $\lim_{x\to 0} f(x)$. We see that

$$f(0) = 1 = \lim_{x\to 0} f(x).$$

Since all three of the conditions in the definition of continuity are satisfied, $f(x)$ is continuous at $x = 0$.

Exercise 2.4.1

Using the definition, determine whether the function $f(x) = \begin{cases} 2x+1, & \text{if } x < 1 \\ 2, & \text{if } x = 1 \\ -x+4, & \text{if } x > 1 \end{cases}$ is continuous at $x = 1$. If the function is not continuous at 1, indicate the condition for continuity at a point that fails to hold.

Hint

Check each condition of the definition.

Answer

f is not continuous at 1 because $f(1) = 2 \neq 3 = \lim_{x\to 1} f(x)$.

By applying the definition of continuity and previously established theorems concerning the evaluation of limits, we can state the following theorem.

Theorem 2.4.1: Continuity of Polynomials and Rational Functions

Polynomials and rational functions are continuous at every point in their domains.

Proof

Previously, we showed that if $p(x)$ and $q(x)$ are polynomials, $\lim_{x \to a} p(x) = p(a)$ for every polynomial $p(x)$ and $\lim_{x \to a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}$ as long as $q(a) \neq 0$. Therefore, polynomials and rational functions are continuous on their domains.

□

We now apply Theorem 2.4.1 to determine the points at which a given rational function is continuous.

Example 2.4.2:Continuity of a Rational Function

For what values of x is $f(x) = \frac{x+1}{x-5}$ continuous?

Solution

The rational function $f(x) = \frac{x+1}{x-5}$ is continuous for every value of x except $x = 5$.

Exercise 2.4.2

For what values of x is $f(x) = 3x^4 - 4x^2$ continuous?

Hint

Use the Continuity of Polynomials and Rational Functions stated above.

Answer

$f(x)$ is continuous at every real number.

Types of Discontinuities

As we have seen in Example 2.4.1A and Example 2.4.1B, discontinuities take on several different appearances. We classify the types of discontinuities we have seen thus far as removable discontinuities, infinite discontinuities, or jump discontinuities. Intuitively, a **removable discontinuity** is a discontinuity for which there is a hole in the graph, a **jump discontinuity** is a noninfinite discontinuity for which the sections of the function do not meet up, and an **infinite discontinuity** is a discontinuity located at a vertical asymptote. Figure 2.4.6 illustrates the differences in these types of discontinuities. Although these terms provide a handy way of describing three common types of discontinuities, keep in mind that not all discontinuities fit neatly into these categories.

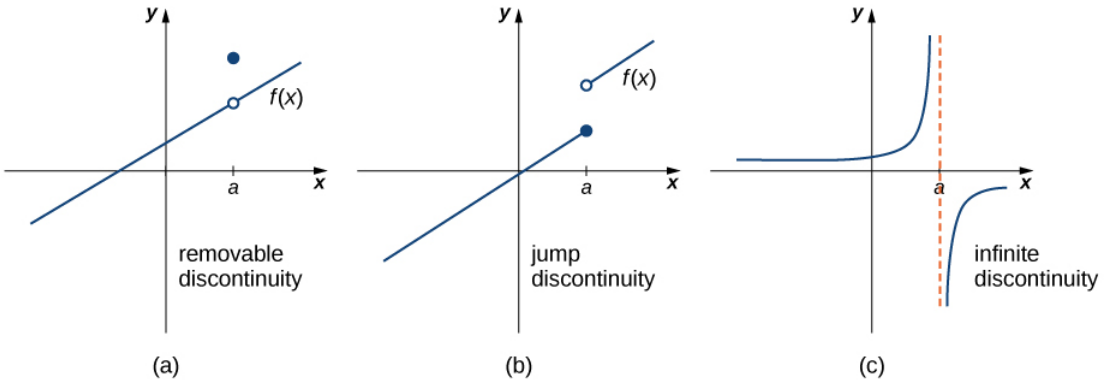


Figure 2.4.6: Discontinuities are classified as (a) removable, (b) jump, or (c) infinite.

These three discontinuities are formally defined as follows:

Definition

If $f(x)$ is discontinuous at a, then

1. f has a **removable discontinuity** at a if $\lim\limits_{x\to a} f(x)$ exists. (Note: When we state that $\lim\limits_{x\to a} f(x)$ exists, we mean that $\lim\limits_{x\to a} f(x) = L$, where L is a real number.)
2. f has a **jump discontinuity** at a if $\lim\limits_{x\to a^-} f(x)$ and $\lim\limits_{x\to a^+} f(x)$ both exist, but $\lim\limits_{x\to a^-} f(x) \neq \lim\limits_{x\to a^+} f(x)$. (Note: When we state that $\lim\limits_{x\to a^-} f(x)$ and $\lim\limits_{x\to a^+} f(x)$ both exist, we mean that both are real-valued and that neither take on the values $\pm\infty$.)
3. f has an **infinite discontinuity** at a if $\lim\limits_{x\to a^-} f(x) = \pm\infty$ or $\lim\limits_{x\to a^+} f(x) = \pm\infty$.

Example 2.4.3: Classifying a Discontinuity

In Example 2.4.1A, we showed that $f(x) = \dfrac{x^2-4}{x-2}$ is discontinuous at $x = 2$. Classify this discontinuity as removable, jump, or infinite.

Solution

To classify the discontinuity at 2 we must evaluate $\lim\limits_{x\to 2} f(x)$:

$$\begin{aligned}\lim_{x\to 2} f(x) &= \lim_{x\to 2} \frac{x^2-4}{x-2} \\ &= \lim_{x\to 2} \frac{(x-2)(x+2)}{x-2} \\ &= \lim_{x\to 2} (x+2) \\ &= 4.\end{aligned}$$

Since f is discontinuous at 2 and $\lim\limits_{x\to 2} f(x)$ exists, f has a removable discontinuity at $x = 2$.

Example 2.4.4: Classifying a Discontinuity

In Example 2.4.1B, we showed that $f(x) = \begin{cases} -x^2+4, & \text{if } x \leq 3 \\ 4x-8, & \text{if } x > 3 \end{cases}$ is discontinuous at $x = 3$. Classify this discontinuity as removable, jump, or infinite.

Solution

Earlier, we showed that f is discontinuous at 3 because $\lim\limits_{x\to 3} f(x)$ does not exist. However, since $\lim\limits_{x\to 3^-} f(x) = -5$ and $\lim\limits_{x\to 3^+} f(x) = 4$ both exist, we conclude that the function has a jump discontinuity at 3.

Example 2.4.5: Classifying a Discontinuity

Determine whether $f(x) = \dfrac{x+2}{x+1}$ is continuous at -1. If the function is discontinuous at -1, classify the discontinuity as removable, jump, or infinite.

Solution

The function value $f(-1)$ is undefined. Therefore, the function is not continuous at -1. To determine the type of discontinuity, we must determine the limit at -1. We see that $\lim\limits_{x\to -1^-} \dfrac{x+2}{x+1} = -\infty$ and $\lim\limits_{x\to -1^+} \dfrac{x+2}{x+1} = +\infty$. Therefore, the function has an infinite discontinuity at -1.

Exercise 2.4.3

For $f(x) = \begin{cases} x^2, & \text{if } x \neq 1 \\ 3, & \text{if } x = 1 \end{cases}$, decide whether f is continuous at 1. If f is not continuous at 1, classify the discontinuity as removable, jump, or infinite.

Hint

Consider the definitions of the various kinds of discontinuity stated above. If the function is discontinuous at 1, look at $\lim\limits_{x\to 1} f(x)$

Answer

Discontinuous at 1; removable

Continuity over an Interval

Now that we have explored the concept of continuity at a point, we extend that idea to **continuity over an interval**. As we develop this idea for different types of intervals, it may be useful to keep in mind the intuitive idea that a function is continuous over an interval if we can use a pencil to trace the function between any two points in the interval without lifting the pencil from the paper. In preparation for defining continuity on an interval, we begin by looking at the definition of what it means for a function to be continuous from the right at a point and continuous from the left at a point.

Definition: Continuity from the Right and from the Left

A function $f(x)$ is said to be **continuous from the right** at a if $\lim\limits_{x\to a^+} f(x) = f(a)$.

A function $f(x)$ is said to be **continuous from the left** at a if $\lim\limits_{x\to a^-} f(x) = f(a)$

A function is continuous over an open interval if it is continuous at every point in the interval. A function $f(x)$ is continuous over a closed interval of the form $[a, b]$ if it is continuous at every point in (a, b) and is continuous from the right at a and is continuous from the left at b. Analogously, a function $f(x)$ is continuous over an interval of the form $(a, b]$ if it is continuous over (a, b) and is continuous from the left at b. Continuity over other types of intervals are defined in a similar fashion.

Requiring that $\lim\limits_{x\to a^+} f(x) = f(a)$ and $\lim\limits_{x\to b^-} f(x) = f(b)$ ensures that we can trace the graph of the function from the point $(a, f(a))$ to the point $(b, f(b))$ without lifting the pencil. If, for example, $\lim\limits_{x\to a^+} f(x) \neq f(a)$, we would need to lift our pencil to jump from $f(a)$ to the graph of the rest of the function over $(a, b]$.

Example 2.4.6: Continuity on an Interval

State the interval(s) over which the function $f(x) = \dfrac{x-1}{x^2+2x}$ is continuous.

Solution

Since $f(x) = \dfrac{x-1}{x^2+2x}$ is a rational function, it is continuous at every point in its domain. The domain of $f(x)$ is the set $(-\infty,-2) \cup (-2,0) \cup (0,+\infty)$. Thus, $f(x)$ is continuous over each of the intervals $(-\infty,-2), (-2,0)$, and $(0,+\infty)$.

Example 2.4.7: Continuity over an Interval

State the interval(s) over which the function $f(x) = \sqrt{4-x^2}$ is continuous.

Solution

From the limit laws, we know that $\lim\limits_{x\to a} \sqrt{4-x^2} = \sqrt{4-a^2}$ for all values of a in $(-2,2)$. We also know that $\lim\limits_{x\to -2^+} \sqrt{4-x^2} = 0$ exists and $\lim\limits_{x\to 2^-} \sqrt{4-x^2} = 0$ exists. Therefore, $f(x)$ is continuous over the interval $[-2,2]$.

Exercise 2.4.4

State the interval(s) over which the function $f(x) = \sqrt{x+3}$ is continuous.

Hint

Use Example 2.4.7 as a guide.

Answer

$[-3,+\infty)$

Theorem 2.4.2 allows us to expand our ability to compute limits. In particular, this theorem ultimately allows us to demonstrate that trigonometric functions are continuous over their domains.

Theorem 2.4.2: Composite Function Theorem

If $f(x)$ is continuous at L and $\lim\limits_{x\to a} g(x) = L$, then

$$\lim_{x\to a} f\big(g(x)\big) = f\Big(\lim_{x\to a} g(x)\Big) = f(L). \tag{2.4.1}$$

Before we move on to Example 2.4.8, recall that earlier, in the section on limit laws, we showed $\lim\limits_{x\to 0} \cos x = 1 = \cos(0)$. Consequently, we know that $f(x) = \cos x$ is continuous at 0. In Example 2.4.8, we see how to combine this result with the composite function theorem.

Example 2.4.8: Limit of a Composite Cosine Function

Evaluate $\lim\limits_{x\to \pi/2} \cos\left(x - \dfrac{\pi}{2}\right)$.

Solution

The given function is a composite of $\cos x$ and $x - \frac{\pi}{2}$. Since $\lim\limits_{x\to \pi/2} \left(x - \dfrac{\pi}{2}\right) = 0$ and $\cos x$ is continuous at 0, we may apply the composite function theorem. Thus,

$$\lim_{x\to \pi/2} \cos\left(x - \frac{\pi}{2}\right) = \cos\left(\lim_{x\to \pi/2} \left(x - \frac{\pi}{2}\right)\right) = \cos(0) = 1.$$

Exercise 2.4.4:

Evaluate $\lim\limits_{x\to \pi} \sin(x-\pi)$.

Hint

$f(x) = \sin x$ is continuous at 0. Use Example 2.4.8 as a guide.

Answer

0

The proof of the next theorem uses the composite function theorem as well as the continuity of $f(x) = \sin x$ and $g(x) = \cos x$ at the point 0 to show that trigonometric functions are continuous over their entire domains.

Theorem 2.4.3: Continuity of Trigonometric Functions

Trigonometric functions are continuous over their entire domains.

Proof

We begin by demonstrating that $\cos x$ is continuous at every real number. To do this, we must show that $\lim_{x \to a} \cos x = \cos a$ for all values of a.

$$\begin{aligned}
\lim_{x \to a} \cos x &= \lim_{x \to a} \cos((x - a) + a) && \text{Rewrite } x = x - a + a. \\
&= \lim_{x \to a} (\cos(x - a) \cos a - \sin(x - a) \sin a) && \text{Apply the identity for the cosine of the sum of two angles.} \\
&= \cos(\lim_{x \to a}(x - a)) \cos a - \sin(\lim_{x \to a}(x - a)) \sin a && \text{Since } \lim_{x \to a}(x - a) = 0, \text{ and } \sin x \text{ and } \cos x \text{ are continuous at } 0. \\
&= \cos(0) \cos a - \sin(0) \sin a && \text{Evaluate } \cos(0) \text{ and } \sin(0) \text{ and simplify.} \\
&= 1 \cdot \cos a - 0 \cdot \sin a = \cos a.
\end{aligned}$$

The proof that $\sin x$ is continuous at every real number is analogous. Because the remaining trigonometric functions may be expressed in terms of $\sin x$ and $\cos x$, their continuity follows from the quotient limit law.

□

As you can see, the composite function theorem is invaluable in demonstrating the continuity of trigonometric functions. As we continue our study of calculus, we revisit this theorem many times.

The Intermediate Value Theorem

Functions that are continuous over intervals of the form $[a, b]$, where a and b are real numbers, exhibit many useful properties. Throughout our study of calculus, we will encounter many powerful theorems concerning such functions. The first of these theorems is the **Intermediate Value Theorem.**

The Intermediate Value Theorem

Let f be continuous over a closed, bounded interval $[a, b]$. If z is any real number between $f(a)$ and $f(b)$, then there is a number c in $[a, b]$ satisfying $f(c) = z$ in Figure 2.4.7.

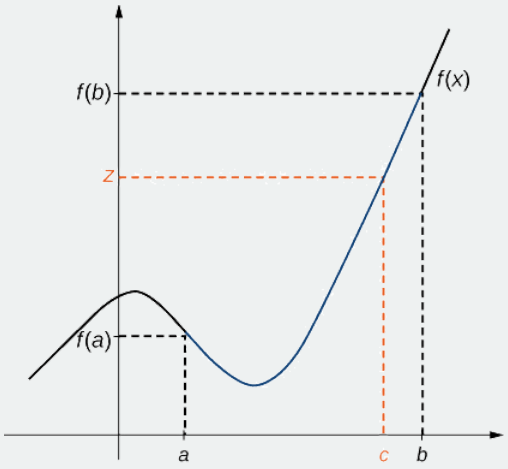


Figure 2.4.7: There is a number $c \in [a, b]$ that satisfies $f(c) = z$.

Example 2.4.9: Application of the Intermediate Value Theorem

Show that $f(x) = x - \cos x$ has at least one zero.

Solution

Since $f(x) = x - \cos x$ is continuous over $(-\infty, +\infty)$, it is continuous over any closed interval of the form $[a, b]$. If you can find an interval $[a, b]$ such that $f(a)$ and $f(b)$ have opposite signs, you can use the Intermediate Value Theorem to conclude there must be a real number c in (a, b) that satisfies $f(c) = 0$. Note that

$$f(0) = 0 - \cos(0) = -1 < 0$$

and

$$f(\tfrac{\pi}{2}) = \tfrac{\pi}{2} - \cos\tfrac{\pi}{2} = \tfrac{\pi}{2} > 0.$$

Using the Intermediate Value Theorem, we can see that there must be a real number c in $[0, \pi/2]$ that satisfies $f(c) = 0$. Therefore, $f(x) = x - \cos x$ has at least one zero.

Example 2.4.10: When Can You Apply the Intermediate Value Theorem?

If $f(x)$ is continuous over $[0, 2]$, $f(0) > 0$ and $f(2) > 0$, can we use the Intermediate Value Theorem to conclude that $f(x)$ has no zeros in the interval $[0, 2]$? Explain.

Solution

No. The Intermediate Value Theorem only allows us to conclude that we can find a value between $f(0)$ and $f(2)$; it doesn't allow us to conclude that we can't find other values. To see this more clearly, consider the function $f(x) = (x - 1)^2$. It satisfies $f(0) = 1 > 0, f(2) = 1 > 0$, and $f(1) = 0$.

Example 2.4.11: When Can You Apply the Intermediate Value Theorem?

For $f(x) = 1/x, f(-1) = -1 < 0$ and $f(1) = 1 > 0$. Can we conclude that $f(x)$ has a zero in the interval $[-1, 1]$?

Solution

No. The function is not continuous over $[-1, 1]$. The Intermediate Value Theorem does not apply here.

Exercise 2.4.5

Show that $f(x) = x^3 - x^2 - 3x + 1$ has a zero over the interval $[0, 1]$.

Hint

Find $f(0)$ and $f(1)$. Apply the Intermediate Value Theorem.

Answer

$f(0) = 1 > 0,\ f(1) = -2 < 0;\ f(x)$ is continuous over $[0, 1]$. It must have a zero on this interval.

Key Concepts

- For a function to be continuous at a point, it must be defined at that point, its limit must exist at the point, and the value of the function at that point must equal the value of the limit at that point.
- Discontinuities may be classified as removable, jump, or infinite.
- A function is continuous over an open interval if it is continuous at every point in the interval. It is continuous over a closed interval if it is continuous at every point in its interior and is continuous at its endpoints.
- The composite function theorem states: If $f(x)$ is continuous at L and $\lim_{x \to a} g(x) = L$, then $\lim_{x \to a} f\big(g(x)\big) = f\big(\lim_{x \to a} g(x)\big) = f(L)$.

- The Intermediate Value Theorem guarantees that if a function is continuous over a closed interval, then the function takes on every value between the values at its endpoints.

Glossary

continuity at a point

A function $f(x)$ is continuous at a point a if and only if the following three conditions are satisfied: (1) $f(a)$ is defined, (2) $\lim\limits_{x \to a} f(x)$ exists, and (3) $\lim x \to a f(x) = f(a)$

continuity from the left

A function is continuous from the left at b if $\lim\limits_{x \to b^-} f(x) = f(b)$

continuity from the right

A function is continuous from the right at a if $\lim\limits_{x \to a^+} f(x) = f(a)$

continuity over an interval

a function that can be traced with a pencil without lifting the pencil; a function is continuous over an open interval if it is continuous at every point in the interval; a function $f(x)$ is continuous over a closed interval of the form $[a, b]$ if it is continuous at every point in (a, b), and it is continuous from the right at a and from the left at b

discontinuity at a point

A function is discontinuous at a point or has a discontinuity at a point if it is not continuous at the point

infinite discontinuity

An infinite discontinuity occurs at a point a if $\lim\limits_{x \to a^-} f(x) = \pm\infty$ or $\lim\limits_{x \to a^+} f(x) = \pm\infty$

Intermediate Value Theorem

Let f be continuous over a closed bounded interval $[a, b]$ if z is any real number between $f(a)$ and $f(b)$, then there is a number c in $[a, b]$ satisfying $f(c) = z$

jump discontinuity

A jump discontinuity occurs at a point a if $\lim\limits_{x \to a^-} f(x)$ and $\lim\limits_{x \to a^+} f(x)$ both exist, but $\lim\limits_{x \to a^-} f(x) \neq \lim\limits_{x \to a^+} f(x)$

removable discontinuity

A removable discontinuity occurs at a point a if $f(x)$ is discontinuous at a, but $\lim\limits_{x \to a} f(x)$ exists

This page titled 2.4: Continuity is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

2.4E: EXERCISES FOR SECTION 2.4

For exercises 1 - 8, determine the point(s), if any, at which each function is discontinuous. Classify any discontinuity as *jump*, *removable*, *infinite*, or *other*.

1) $f(x) = \dfrac{1}{\sqrt{x}}$

Answer

The function is defined for all x in the interval $(0, \infty)$.

2) $f(x) = \dfrac{2}{x^2 + 1}$

3) $f(x) = \dfrac{x}{x^2 - x}$

Answer

Removable discontinuity at $x = 0$; infinite discontinuity at $x = 1$.

4) $g(t) = t^{-1} + 1$

5) $f(x) = \dfrac{5}{e^x - 2}$

Answer

Infinite discontinuity at $x = \ln 2$

6) $f(x) = \dfrac{|x - 2|}{x - 2}$

7) $H(x) = \tan 2x$

Answer

Infinite discontinuities at $x = \dfrac{(2k + 1)\pi}{4}$, for $k = 0, \pm 1, \pm 2, \pm 3, \ldots$

8) $f(t) = \dfrac{t + 3}{t^2 + 5t + 6}$

For exercises 9 - 14, decide if the function continuous at the given point. If it is discontinuous, what type of discontinuity is it?

9) $\dfrac{2x^2 - 5x + 3}{x - 1}$ at $x = 1$

Answer

No. It is a removable discontinuity.

10) $h(\theta) = \dfrac{\sin\theta - \cos\theta}{\tan\theta}$ at $\theta = \pi$

11) $g(u) = \begin{cases} \dfrac{6u^2 + u - 2}{2u - 1}, & \text{if } u \neq \frac{1}{2} \\ \frac{7}{2}, & \text{if } u = \frac{1}{2} \end{cases}$, at $u = \frac{1}{2}$

Answer

Yes. It is continuous.

12) $f(y) = \dfrac{\sin(\pi y)}{\tan(\pi y)}$, at $y = 1$

13) $f(x) = \begin{cases} x^2 - e^x, & \text{if } x < 0 \\ x - 1, & \text{if } x \geq 0 \end{cases}$, at $x = 0$

Answer

Yes. It is continuous.

14) $f(x) = \begin{cases} x\sin(x), & \text{if } x \leq \pi \\ x\tan(x), & \text{if } x > \pi \end{cases}$, at $x = \pi$

In exercises 15 - 19, find the value(s) of k that makes each function continuous over the given interval.

15) $f(x) = \begin{cases} 3x+2, & \text{if } x < k \\ 2x-3, & \text{if } k \le x \le 8 \end{cases}$

Answer

$k = -5$

16) $f(\theta) = \begin{cases} \sin\theta, & \text{if } 0 \le \theta < \frac{\pi}{2} \\ \cos(\theta + k), & \text{if } \frac{\pi}{2} \le \theta \le \pi \end{cases}$

17) $f(x) = \begin{cases} \dfrac{x^2+3x+2}{x+2}, & \text{if } x \neq -2 \\ k, & \text{if } x = -2 \end{cases}$

Answer

$k = -1$

18) $f(x) = \begin{cases} e^{kx}, & \text{if } 0 \le x < 4 \\ x+3, & \text{if } 4 \le x \le 8 \end{cases}$

19) $f(x) = \begin{cases} \sqrt{kx}, & \text{if } 0 \le x \le 3 \\ x+1, & \text{if } 3 < x \le 10 \end{cases}$

Answer

$k = \frac{16}{3}$

In exercises 20 - 21, use the Intermediate Value Theorem (IVT).

20) Let $h(x) = \begin{cases} 3x^2-4, & \text{if } x \le 2 \\ 5+4x, & \text{if } x > 2 \end{cases}$ Over the interval $[0,4]$, there is no value of x such that $h(x) = 10$, although $h(0) < 10$ and $h(4) > 10$. Explain why this does not contradict the IVT.

21) A particle moving along a line for time t has a position function $s(t)$, which is continuous. Assume $s(2) = 5$ and $s(5) = 2$. Another particle moves such that its position is given by $h(t) = s(t) - t$. Explain why there must be a value c for $2 < c < 5$ such that $h(c) = 0$.

Answer

Since both s and $y = t$ are continuous everywhere, then $h(t) = s(t) - t$ is continuous everywhere and, in particular, it is continuous over the closed interval $[2,5]$. Also, $h(2) = 3 > 0$ and $h(5) = -3 < 0$. Therefore, by the IVT, there is a value $x = c$ such that $h(c) = 0$.

22) [T] Use the statement "The cosine of t is equal to t cubed."

a. Write a mathematical equation of the statement.

b. Prove that the equation in part a. has at least one real solution.

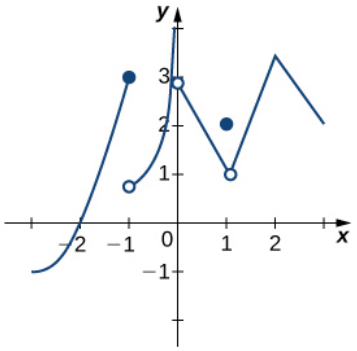
c. Use a calculator to find an interval of length 0.01 that contains a solution.

23) Apply the IVT to determine whether $2^x = x^3$ has a solution in one of the intervals $[1.25, 1.375]$ or $[1.375, 1.5]$. Briefly explain your response for each interval.

Answer

The function $f(x) = 2^x - x^3$ is continuous over the interval $[1.25, 1.375]$ and has opposite signs at the endpoints.

24) Consider the graph of the function $y = f(x)$ shown in the following graph.



a. Find all values for which the function is discontinuous.

https://math.libretexts.org/@go/page/50408

b. For each value in part a., state why the formal definition of continuity does not apply.

c. Classify each discontinuity as either jump, removable, or infinite.

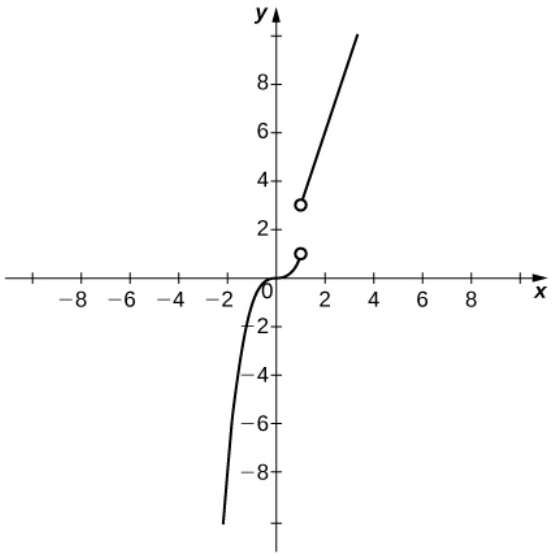
25) Let $f(x) = \begin{cases} 3x, & \text{if } x > 1 \\ x^3, & \text{if } x < 1 \end{cases}$.

a. Sketch the graph of f.

b. Is it possible to find a value k such that $f(1) = k$, which makes $f(x)$ continuous for all real numbers? Briefly explain.

Answer

a.



b. It is not possible to redefine $f(1)$ since the discontinuity is a jump discontinuity.

26) Let $f(x) = \dfrac{x^4 - 1}{x^2 - 1}$ for $x \neq -1, 1$.

a. Sketch the graph of f.

b. Is it possible to find values k_1 and k_2 such that $f(-1) = k$ and $f(1) = k_2$, and that makes $f(x)$ continuous for all real numbers? Briefly explain.

27) Sketch the graph of the function $y = f(x)$ with properties i. through vii.

i. The domain of f is $(-\infty, +\infty)$.

ii. f has an infinite discontinuity at $x = -6$.

iii. $f(-6) = 3$

iv. $\lim\limits_{x \to -3^-} f(x) = \lim\limits_{x \to -3^+} f(x) = 2$

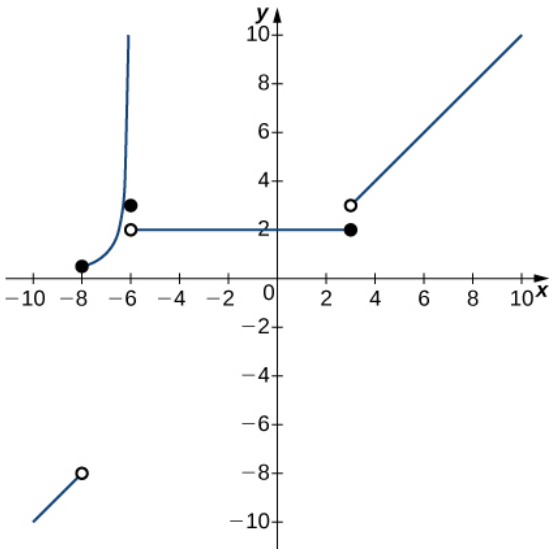
v. $f(-3) = 3$

vi. f is left continuous but not right continuous at $x = 3$.

vii. $\lim\limits_{x \to -\infty} f(x) = -\infty$ and $\lim\limits_{x \to +\infty} f(x) = +\infty$

Answer

Answers may vary; see the following example:



28) Sketch the graph of the function $y = f(x)$ with properties i. through iv.

i. The domain of f is $[0, 5]$.

ii. $\lim_{x \to 1^+} f(x)$ and $\lim_{x \to 1^-} f(x)$ exist and are equal.

iii. $f(x)$ is left continuous but not continuous at $x = 2$, and right continuous but not continuous at $x = 3$.

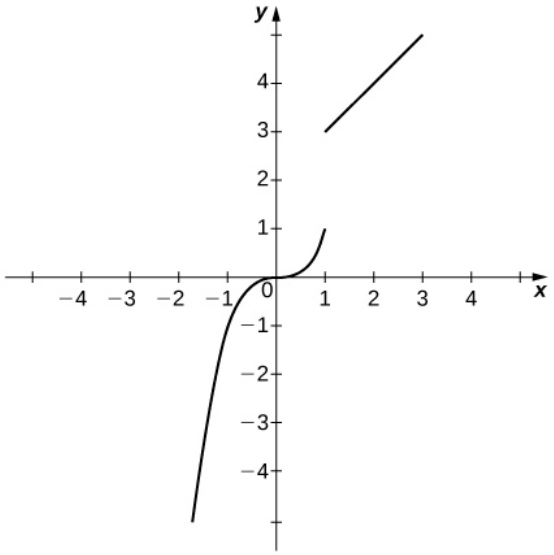
iv. $f(x)$ has a removable discontinuity at $x = 1$, a jump discontinuity at $x = 2$, and the following limits hold: $\lim_{x \to 3^-} f(x) = -\infty$ and $\lim_{x \to 3^+} f(x) = 2$.

In exercises 29 - 30, suppose $y = f(x)$ is defined for all x. For each description, sketch a graph with the indicated property.

29) Discontinuous at $x = 1$ with $\lim_{x \to -1} f(x) = -1$ and $\lim_{x \to 2} f(x) = 4$

Answer

Answers may vary; see the following example:



30) Discontinuous at $x = 2$ but continuous elsewhere with $\lim_{x \to 0} f(x) = \frac{1}{2}$

Determine whether each of the given statements is true. Justify your response with an explanation or counterexample.

31) $f(t) = \frac{2}{e^t - e^{-t}}$ is continuous everywhere.

Answer

False. It is continuous over $(-\infty, 0) \cup (0, \infty)$.

32) If the left- and right-hand limits of $f(x)$ as $x \to a$ exist and are equal, then f cannot be discontinuous at $x = a$.

33) If a function is not continuous at a point, then it is not defined at that point.

Answer

False. Consider $f(x) = \begin{cases} x, & \text{if } x \neq 0 \\ 4, & \text{if } x = 0 \end{cases}$.

34) According to the IVT, $\cos x - \sin x - x = 2$ has a solution over the interval $[-1, 1]$.

35) If $f(x)$ is continuous such that $f(a)$ and $f(b)$ have opposite signs, then $f(x) = 0$ has exactly one solution in $[a, b]$.

Answer

False. Consider $f(x) = \cos(x)$ on $[-\pi, 2\pi]$.

36) The function $f(x) = \dfrac{x^2 - 4x + 3}{x^2 - 1}$ is continuous over the interval $[0, 3]$.

37) If $f(x)$ is continuous everywhere and $f(a), f(b) > 0$, then there is no root of $f(x)$ in the interval $[a, b]$.

Answer

False. The IVT does not work in reverse! Consider $(x - 1)^2$ over the interval $[-2, 2]$.

[T] The following problems consider the scalar form of Coulomb's law, which describes the electrostatic force between two point charges, such as electrons. It is given by the equation $F(r) = k_e \dfrac{|q_1 q_2|}{r^2}$, where k_e is Coulomb's constant, q_i are the magnitudes of the charges of the two particles, and r is the distance between the two particles.

38) To simplify the calculation of a model with many interacting particles, after some threshold value $r = R$, we approximate F as zero.

a. Explain the physical reasoning behind this assumption.

b. What is the force equation?

c. Evaluate the force F using both Coulomb's law and our approximation, assuming two protons with a charge magnitude of 1.6022×10^{-19} **coulombs (C)**, and the Coulomb constant $k_e = 8.988 \times 10^9 Nm^2/C^2$ are 1 m apart. Also, assume $R < 1$ m. How much inaccuracy does our approximation generate? Is our approximation reasonable?

d. Is there any finite value of R for which this system remains continuous at R?

39) Instead of making the force 0 at R, we let the force be $10 - 20$ for $r \geq R$. Assume two protons, which have a magnitude of charge 1.6022×10^{-19} C, and the Coulomb constant $k_e = 8.988 \times 10^9$ Nm^2/C^2. Is there a value R that can make this system continuous? If so, find it.

Answer

$R = 0.0001519$ m

Recall the discussion on spacecraft from the chapter opener. The following problems consider a rocket launch from Earth's surface. The force of gravity on the rocket is given by $F(d) = -mk/d^2$, where m is the mass of the rocket, d is the distance of the rocket from the center of Earth, and k is a constant.

40) [T] Determine the value and units of k given that the mass of the rocket on Earth is 3 million kg. (Hint: The distance from the center of Earth to its surface is 6378 km.)

41) [T] After a certain distance D has passed, the gravitational effect of Earth becomes quite negligible, so we can approximate the force function by $F(d) = \begin{cases} -\dfrac{mk}{d^2}, & \text{if } d < D \\ 10{,}000, & \text{if } d \geq D \end{cases}$. Find the necessary condition D such that the force function remains continuous.

Answer

$D = 63.78$ km

42) As the rocket travels away from Earth's surface, there is a distance D where the rocket sheds some of its mass, since it no longer needs the excess fuel storage. We can write this function as $F(d) = \begin{cases} -\frac{m_1 k}{d^2}, & \text{if } d < D \\ -\frac{m_2 k}{d^2}, & \text{if } d \geq D \end{cases}$. Is there a value of D such that this function is continuous, assuming $m_1 \neq m_2$?

In Exercises 43 - 44, prove each function is continuous everywhere.

43) $f(\theta) = \sin\theta$

Answer

For all values of a, $f(a)$ is defined, $\lim_{\theta \to a} f(\theta)$ exists, and $\lim_{\theta \to a} f(\theta) = f(a)$. Therefore, $f(\theta)$ is continuous everywhere.

44) $g(x) = |x|$

45) Where is $f(x) = \begin{cases} 0, & \text{if } x \text{ is irrational} \\ 1, & \text{if } x \text{ is rational} \end{cases}$ continuous?

Answer

Nowhere

2.4E: Exercises for Section 2.4 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by LibreTexts.

2.5: The Precise Definition of a Limit

Learning Objectives

- Describe the epsilon-delta definition of a limit.
- Apply the epsilon-delta definition to find the limit of a function.
- Describe the epsilon-delta definitions of one-sided limits and infinite limits.
- Use the epsilon-delta definition to prove the limit laws.

By now you have progressed from the very informal definition of a limit in the introduction of this chapter to the intuitive understanding of a limit. At this point, you should have a very strong intuitive sense of what the limit of a function means and how you can find it. In this section, we convert this intuitive idea of a limit into a formal definition using precise mathematical language. The formal definition of a limit is quite possibly one of the most challenging definitions you will encounter early in your study of calculus; however, it is well worth any effort you make to reconcile it with your intuitive notion of a limit. Understanding this definition is the key that opens the door to a better understanding of calculus.

Quantifying Closeness

Before stating the formal definition of a limit, we must introduce a few preliminary ideas. Recall that the distance between two points a and b on a number line is given by $|a-b|$.

- The statement $|f(x)-L|<\varepsilon$ may be interpreted as: The distance between $f(x)$ and L is less than ε.
- The statement $0<|x-a|<\delta$ may be interpreted as: $x\neq a$ and the distance between x and a is less than δ.

It is also important to look at the following equivalences for absolute value:

- The statement $|f(x)-L|<\varepsilon$ is equivalent to the statement $L-\varepsilon<f(x)<L+\varepsilon$.
- The statement $0<|x-a|<\delta$ is equivalent to the statement $a-\delta<x<a+\delta$ and $x\neq a$.

With these clarifications, we can state the formal **epsilon-delta definition of the limit**.

Definition: Finite Limits (Formal)

Let $f(x)$ be defined for all $x\neq a$ over an open interval containing a. Let L be a real number. Then

$$\lim_{x\to a} f(x)=L$$

if, for every $\varepsilon>0$, there exists a $\delta>0$, such that if $0<|x-a|<\delta$, then $|f(x)-L|<\varepsilon$.

This definition may seem rather complex from a mathematical point of view, but it becomes easier to understand if we break it down phrase by phrase. The statement itself involves something called a **universal quantifier** (for every $\varepsilon>0$), an **existential quantifier** (there exists a $\delta>0$), and, last, a **conditional statement** (if $0<|x-a|<\delta$, then $|f(x)-L|<\varepsilon$). Let's take a look at Table 2.5.1, which breaks down the definition and translates each part.

Table 2.5.1

Definition	Translation
1. For every $\varepsilon>0$,	1. For every positive distance ε from L,
2. there exists a $\delta>0$,	2. There is a positive distance δ from a,
3. such that	3. such that
4. if $0<\|x-a\|<\delta$, then $\|f(x)-L\|<\varepsilon$.	4. if x is closer than δ to a and $x\neq a$, then $f(x)$ is closer than ε to L.

We can get a better handle on this definition by looking at the definition geometrically. Figure 2.5.1 shows possible values of δ for various choices of $\varepsilon>0$ for a given function $f(x)$, a number a, and a limit L at a. Notice that as we choose smaller values of ε (the

distance between the function and the limit), we can always find a δ small enough so that if we have chosen an x value within δ of a, then the value of $f(x)$ is within ε of the limit L.

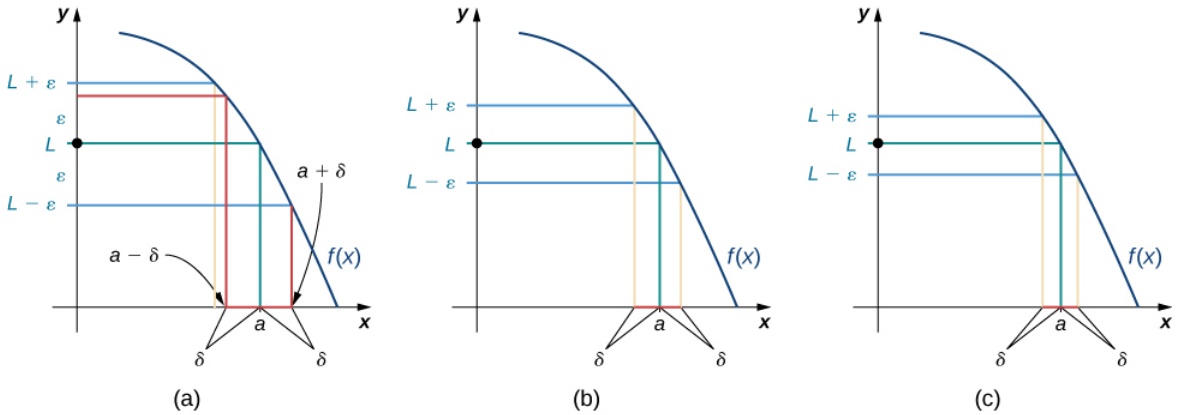


Figure 2.5.1: These graphs show possible values of δ, given successively smaller choices of ε.

Note

Visit the following applet to experiment with finding values of δ for selected values of ε:

- The epsilon-delta definition of limit

Example 2.5.1 shows how you can use this definition to prove a statement about the limit of a specific function at a specified value.

Example 2.5.1: Proving a Statement about the Limit of a Specific Function

Prove that $\lim\limits_{x\to 1}(2x+1)=3$.

Solution

Let $\varepsilon > 0$.

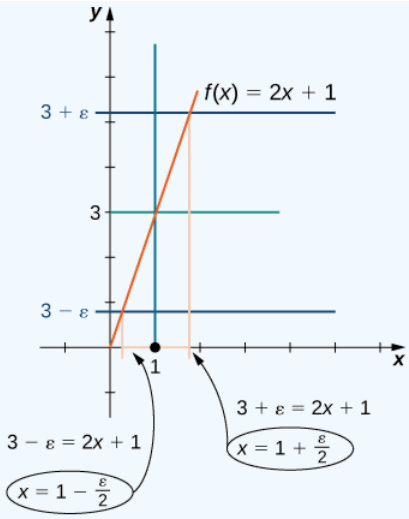
The first part of the definition begins "For every $\varepsilon > 0$."This means we must prove that whatever follows is true no matter what positive value of ε is chosen. By stating "Let $\varepsilon > 0$," we signal our intent to do so.

Choose $\delta = \frac{\varepsilon}{2}$.

The definition continues with "there exists a $\delta > 0$. " The phrase "there exists" in a mathematical statement is always a signal for a scavenger hunt. In other words, we must go and find δ. So, where exactly did $\delta = \varepsilon/2$ come from? There are two basic approaches to tracking down δ. One method is purely algebraic and the other is geometric.

We begin by tackling the problem from an algebraic point of view. Since ultimately we want $|(2x+1)-3| < \varepsilon$, we begin by manipulating this expression: $|(2x+1)-3| < \varepsilon$ is equivalent to $|2x-2| < \varepsilon$, which in turn is equivalent to $|2||x-1| < \varepsilon$. Last, this is equivalent to $|x-1| < \varepsilon/2$. Thus, it would seem that $\delta = \varepsilon/2$ is appropriate.

We may also find δ through geometric methods. Figure 2.5.2 demonstrates how this is done.



δ is the length of the smaller of the two distances marked in brown.

$$\delta = \min\{1 + \tfrac{\varepsilon}{2} - 1,\ 1 - (1 - \tfrac{\varepsilon}{2})\}$$
$$= \min\{\tfrac{\varepsilon}{2}, \tfrac{\varepsilon}{2}\}$$
$$= \tfrac{\varepsilon}{2}$$

Figure 2.5.2: This graph shows how we find δ geometrically.

Assume $0 < |x-1| < \delta$. When δ has been chosen, our goal is to show that if $0 < |x-1| < \delta$, then $|(2x+1)-3| < \varepsilon$. To prove any statement of the form "If this, then that," we begin by assuming "this" and trying to get "that."

Thus,

$|(2x+1)-3| = |2x-2|$ **property of absolute value**

$= |2(x-1)|$

$= |2||x-1|\ |2| = 2$

$= 2|x-1|$

$< 2 \cdot \delta$ **here's where we use the assumption that** $0 < |x-1| < \delta$

$= 2 \cdot \frac{\varepsilon}{2} = \varepsilon$ **here's where we use our choice of** $\delta = \varepsilon/2$

Analysis

In this part of the proof, we started with $|(2x+1)-3|$ and used our assumption $0 < |x-1| < \delta$ in a key part of the chain of inequalities to get $|(2x+1)-3|$ to be less than ε. We could just as easily have manipulated the assumed inequality $0 < |x-1| < \delta$ to arrive at $|(2x+1)-3| < \varepsilon$ as follows:

$0 < |x-1| < \delta \Rightarrow |x-1| < \delta$

$\Rightarrow -\delta < x-1 < \delta$

$\Rightarrow -\frac{\varepsilon}{2} < x-1 < \frac{\varepsilon}{2}$

$\Rightarrow -\varepsilon < 2x-2 < \varepsilon$

$\Rightarrow |2x-2| < \varepsilon$

$\Rightarrow |(2x+1)-3| < \varepsilon.$

Therefore, $\lim\limits_{x \to 1} (2x+1) = 3$. (Having completed the proof, we state what we have accomplished.)

After removing all the remarks, here is a final version of the proof:

Let $\varepsilon > 0$.

Choose $\delta = \varepsilon/2$.

Assume $0 < |x-1| < \delta$.

Thus,

$$
\begin{aligned}
|(2x+1)-3| &= |2x-2| \\
&= |2(x-1)| \\
&= |2||x-1| \\
&= 2|x-1| \\
&< 2 \cdot \delta \\
&= 2 \cdot \frac{\varepsilon}{2} \\
&= \varepsilon.
\end{aligned}
$$

Therefore, $\lim\limits_{x \to 1} (2x+1) = 3$.

The following Problem-Solving Strategy summarizes the type of proof we worked out in Example 2.5.1.

Problem-Solving Strategy: Proving That $\lim\limits_{x \to a} f(x) = L$ for a Specific Function $f(x)$

1. Let's begin the proof with the following statement: Let $\varepsilon > 0$.
2. Next, we need to obtain a value for δ. After we have obtained this value, we make the following statement, filling in the blank with our choice of δ: Choose $\delta =$_____.
3. The next statement in the proof should be (at this point, we fill in our given value for a): Assume $0 < |x-a| < \delta$.
4. Next, based on this assumption, we need to show that $|f(x) - L| < \varepsilon$, where $f(x)$ and L are our function $f(x)$ and our limit L. At some point, we need to use $0 < |x-a| < \delta$.
5. We conclude our proof with the statement: Therefore, $\lim\limits_{x \to a} f(x) = L$.

Example 2.5.2: Proving a Statement about a Limit

Complete the proof that $\lim\limits_{x \to -1} (4x+1) = -3$ by filling in the blanks.

Let _____.

Choose $\delta =$_____.

Assume $0 < |x-$_____$| < \delta$.

Thus, |_____−_____|=_____ε.

Solution

We begin by filling in the blanks where the choices are specified by the definition. Thus, we have

Let $\varepsilon > 0$.

Choose $\delta =$_____.

Assume $0 < |x-(-1)| < \delta$. (or equivalently, $0 < |x+1| < \delta$.)

Thus, $|(4x+1)-(-3)| = |4x+4| = |4||x+1| < 4\delta$_____$\varepsilon$.

Focusing on the final line of the proof, we see that we should choose $\delta = \frac{\varepsilon}{4}$.

We now complete the final write-up of the proof:

Let $\varepsilon > 0$.

Choose $\delta = \frac{\varepsilon}{4}$.

Assume $0 < |x-(-1)| < \delta$ (or equivalently, $0 < |x+1| < \delta$.)

Thus, $|(4x+1)-(-3)| = |4x+4| = |4||x+1| < 4\delta = 4(\varepsilon/4) = \varepsilon$.

Exercise 2.5.1

Complete the proof that $\lim_{x\to 2}(3x-2)=4$ by filling in the blanks.

Let _____.

Choose δ =_____.

Assume $0<|x-$____$|<$____.

Thus,

|_____−____|=_____ε.

Therefore, $\lim_{x\to 2}(3x-2)=4$.

Hint

Follow the outline in the Problem-Solving Strategy that we worked out in full in Example 2.5.2.

Answer

Let $\varepsilon>0$; choose $\delta=\frac{\varepsilon}{3}$; assume $0<|x-2|<\delta$.

Thus, $|(3x-2)-4|=|3x-6|=|3|\cdot|x-2|<3\cdot\delta=3\cdot(\varepsilon/3)=\varepsilon$.

Therefore, $\lim_{x\to 2}(3x-2)=4$.

In Examples 2.5.1 and 2.5.2, the proofs were fairly straightforward, since the functions with which we were working were linear. In Example 2.5.3, we see how to modify the proof to accommodate a nonlinear function.

Example 2.5.3: Proving a Statement about the Limit of a Specific Function (Geometric Approach)

Prove that $\lim_{x\to 2}x^2=4$.

Solution

1. Let $\varepsilon>0$. The first part of the definition begins "For every $\varepsilon>0$," so we must prove that whatever follows is true no matter what positive value of ε is chosen. By stating "Let $\varepsilon>0$," we signal our intent to do so.

2. Without loss of generality, assume $\varepsilon\le 4$. Two questions present themselves: Why do we want $\varepsilon\le 4$ and why is it okay to make this assumption? In answer to the first question: Later on, in the process of solving for δ, we will discover that δ involves the quantity $\sqrt{4-\varepsilon}$. Consequently, we need $\varepsilon\le 4$. In answer to the second question: If we can find $\delta>0$ that "works" for $\varepsilon\le 4$, then it will "work" for any $\varepsilon>4$ as well. Keep in mind that, although it is always okay to put an upper bound on ε, it is never okay to put a lower bound (other than zero) on ε.

3. Choose $\delta=\min\{2-\sqrt{4-\varepsilon},\sqrt{4+\varepsilon}-2\}$. Figure 2.5.3 shows how we made this choice of δ.

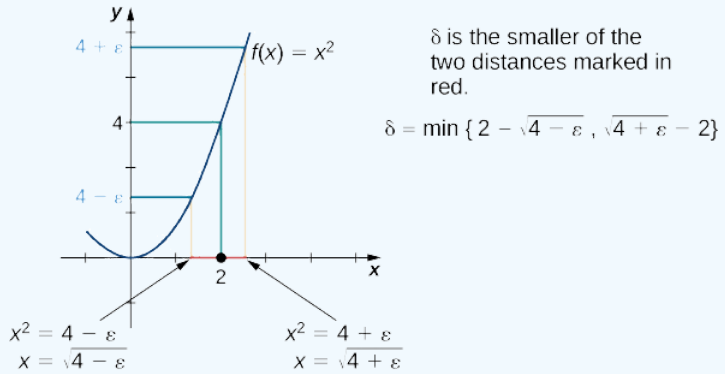


Figure 2.5.3: This graph shows how we find δ geometrically for a given ε for the proof in Example 2.5.3.

4. We must show: If $0<|x-2|<\delta$, then $|x^2-4|<\varepsilon$, so we must begin by assuming

$$0 < |x-2| < \delta.$$

We don't really need $0 < |x-2|$ (in other words, $x \neq 2$) for this proof. Since $0 < |x-2| < \delta \Rightarrow |x-2| < \delta$, it is okay to drop $0 < |x-2|$.

$$|x-2| < \delta.$$

Hence,

$$-\delta < x-2 < \delta.$$

Recall that $\delta = \min\{2-\sqrt{4-\varepsilon}, \sqrt{4+\varepsilon}-2\}$. Thus, $\delta \geq 2-\sqrt{4-\varepsilon}$ and consequently $-(2-\sqrt{4-\varepsilon}) \leq -\delta$. We also use $\delta \leq \sqrt{4+\varepsilon}-2$ here. We might ask at this point: Why did we substitute $2-\sqrt{4-\varepsilon}$ for δ on the left-hand side of the inequality and $\sqrt{4+\varepsilon}-2$ on the right-hand side of the inequality? If we look at Figure 2.5.3, we see that $2-\sqrt{4-\varepsilon}$ corresponds to the distance on the left of 2 on the x-axis and $\sqrt{4+\varepsilon}-2$ corresponds to the distance on the right. Thus,

$$-(2-\sqrt{4-\varepsilon}) \leq -\delta < x-2 < \delta \leq \sqrt{4+\varepsilon}-2.$$

We simplify the expression on the left:

$$-2+\sqrt{4-\varepsilon} < x-2 < \sqrt{4+\varepsilon}-2.$$

Then, we add 2 to all parts of the inequality:

$$\sqrt{4-\varepsilon} < x < \sqrt{4+\varepsilon}.$$

We square all parts of the inequality. It is okay to do so, since all parts of the inequality are positive:

$$4-\varepsilon < x^2 < 4+\varepsilon.$$

We subtract 4 from all parts of the inequality:

$$-\varepsilon < x^2-4 < \varepsilon.$$

Last,

$$|x^2-4| < \varepsilon.$$

5. Therefore,

$$\lim_{x\to 2} x^2 = 4.$$

? Exercise 2.5.2

Find δ corresponding to $\varepsilon > 0$ for a proof that $\lim_{x\to 9}\sqrt{x} = 3$.

Hint

Draw a graph similar to the one in Example 2.5.3.

Answer

Choose $\delta = \min\{9-(3-\varepsilon)^2,\ (3+\varepsilon)^2-9\}$.

The geometric approach to proving that the limit of a function takes on a specific value works quite well for some functions. Also, the insight into the formal definition of the limit that this method provides is invaluable. However, we may also approach limit proofs from a purely algebraic point of view. In many cases, an algebraic approach may not only provide us with additional insight into the definition, it may prove to be simpler as well. Furthermore, an algebraic approach is the primary tool used in proofs of statements about limits. For Example 2.5.4, we take on a purely algebraic approach.

Example 2.5.4: Proving a Statement about the Limit of a Specific Function (Algebraic Approach)

Prove that $\lim_{x\to -1}(x^2-2x+3)=6$.

Solution

Let's use our outline from the Problem-Solving Strategy:

1. Let $\varepsilon > 0$.

2. Choose $\delta = \min\{1, \varepsilon/5\}$. This choice of δ may appear odd at first glance, but it was obtained by taking a look at our ultimate desired inequality: $|\,(x^2-2x+3)-6\,| < \varepsilon$. This inequality is equivalent to $|x+1|\cdot|x-3| < \varepsilon$. At this point, the temptation simply to choose $\delta = \frac{\varepsilon}{x-3}$ is very strong. Unfortunately, our choice of δ must depend on ε only and no other variable. If we can replace $|x-3|$ by a numerical value, our problem can be resolved. This is the place where assuming $\delta \le 1$ comes into play. The choice of $\delta \le 1$ here is arbitrary. We could have just as easily used any other positive number. In some proofs, greater care in this choice may be necessary. Now, since $\delta \le 1$ and $|x+1| < \delta \le 1$, we are able to show that $|x-3| < 5$. Consequently, $|x+1|\cdot|x-3| < |x+1|\cdot 5$. At this point we realize that we also need $\delta \le \varepsilon/5$. Thus, we choose $\delta = \min\{1, \varepsilon/5\}$.

3. Assume $0 < |x+1| < \delta$. Thus,

$$|x+1| < 1 \text{ and } |x+1| < \frac{\varepsilon}{5}.$$

Since $|x+1| < 1$, we may conclude that $-1 < x+1 < 1$. Thus, by subtracting 4 from all parts of the inequality, we obtain $-5 < x-3 < -1$. Consequently, $|x-3| < 5$. This gives us

$$|(x^2-2x+3)-6| = |x+1|\cdot|x-3| < \frac{\varepsilon}{5}\cdot 5 = \varepsilon.$$

Therefore,

$$\lim_{x\to -1}(x^2-2x+3)=6.$$

Exercise 2.5.3

Complete the proof that $\lim_{x\to 1} x^2 = 1$.

Let $\varepsilon > 0$; choose $\delta = \min\{1, \varepsilon/3\}$; assume $0 < |x-1| < \delta$.

Since $|x-1| < 1$, we may conclude that $-1 < x-1 < 1$. Thus, $1 < x+1 < 3$. Hence, $|x+1| < 3$.

Hint

Use Example 2.5.4 as a guide.

Answer

$|\,x^2-1\,| = |x-1|\cdot|x+1| < \varepsilon/3 \cdot 3 = \varepsilon$

You will find that, in general, the more complex a function, the more likely it is that the algebraic approach is the easiest to apply. The algebraic approach is also more useful in proving statements about limits.

Proving Limit Laws

We now demonstrate how to use the epsilon-delta definition of a limit to construct a rigorous proof of one of the limit laws. The **triangle inequality** is used at a key point of the proof, so we first review this key property of absolute value.

Definition: The Triangle Inequality

The **triangle inequality** states that if a and b are any real numbers, then $|a+b| \le |a| + |b|$.

Proof

We prove the following limit law: If $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$, then $\lim_{x\to a} (f(x) + g(x)) = L + M$.

Let $\varepsilon > 0$.

Choose $\delta_1 > 0$ so that if $0 < |x - a| < \delta_1$, then $|f(x) - L| < \varepsilon/2$.

Choose $\delta_2 > 0$ so that if $0 < |x - a| < \delta_2$, then $|g(x) - M| < \varepsilon/2$.

Choose $\delta = \min\{\delta_1, \delta_2\}$.

Assume $0 < |x - a| < \delta$.

Thus,

$$0 < |x - a| < \delta_1 \text{ and } 0 < |x - a| < \delta_2.$$

Hence,

$$\begin{aligned} |(f(x) + g(x)) - (L + M)| &= |(f(x) - L) + (g(x) - M)| \\ &\leq |f(x) - L| + |g(x) - M| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

□

We now explore what it means for a limit not to exist. The limit $\lim_{x\to a} f(x)$ does not exist if there is no real number L for which $\lim_{x\to a} f(x) = L$. Thus, for all real numbers L, $\lim_{x\to a} f(x) \neq L$. To understand what this means, we look at each part of the definition of $\lim_{x\to a} f(x) = L$ together with its opposite. A translation of the definition is given in Table 2.5.2.

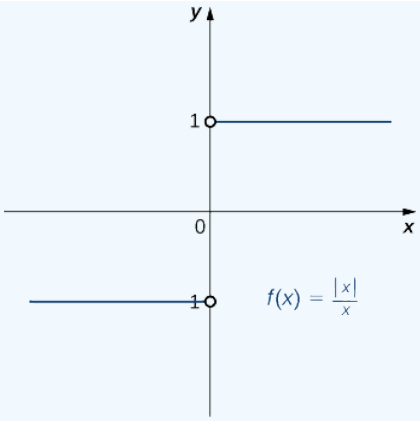
Table 2.5.2. Translation of the Definition of $\lim_{x\to a} f(x) = L$ and its Opposite

Definition	Opposite
1. For every $\varepsilon > 0$,	1. There exists $\varepsilon > 0$ so that
2. there exists a $\delta > 0$, so that	2. for every $\delta > 0$,
3. if $0 < \lvert x - a\rvert < \delta$, then $\lvert f(x) - L\rvert < \varepsilon$.	3. There is an x satisfying $0 < \lvert x - a\rvert < \delta$ so that $\lvert f(x) - L\rvert \geq \varepsilon$.

Finally, we may state what it means for a limit not to exist. The limit $\lim_{x\to a} f(x)$ does not exist if for every real number L, there exists a real number $\varepsilon > 0$ so that for all $\delta > 0$, there is an x satisfying $0 < |x - a| < \delta$, so that $|f(x) - L| \geq \varepsilon$. Let's apply this in Example 2.5.5 to show that a limit does not exist.

Example 2.5.5: Showing That a Limit Does Not Exist

Show that $\lim_{x\to 0} \frac{|x|}{x}$ does not exist. The graph of $f(x) = |x|/x$ is shown here:



Solution

Suppose that L is a candidate for a limit. Choose $\varepsilon = 1/2$.

Let $\delta > 0$. Either $L \geq 0$ or $L < 0$. If $L \geq 0$, then let $x = -\delta/2$.

Thus,

$$|x - 0| = |-\tfrac{\delta}{2} - 0| = \tfrac{\delta}{2} < \delta$$

and

$$\left|\frac{|-\frac{\delta}{2}|}{-\frac{\delta}{2}} - L\right| = |-1 - L| = L + 1 \geq 1 > \tfrac{1}{2} = \varepsilon.$$

On the other hand, if $L < 0$, then let $x = \delta/2$. Thus,

$$|x - 0| = |\tfrac{\delta}{2} - 0| = \tfrac{\delta}{2} < \delta$$

and

$$\left|\frac{|\frac{\delta}{2}|}{\frac{\delta}{2}} - L\right| = |1 - L| = |L| + 1 \geq 1 > \tfrac{1}{2} = \varepsilon.$$

Thus, for any value of L, $\lim\limits_{x \to 0} \dfrac{|x|}{x} \neq L$.

One-Sided Limits

Just as we first gained an intuitive understanding of limits and then moved on to a more rigorous definition of a limit, we now revisit one-sided limits. To do this, we modify the epsilon-delta definition of a limit to give formal epsilon-delta definitions for limits from the right and left at a point. These definitions only require slight modifications from the definition of the limit. In the definition of the limit from the right, the inequality $0 < x - a < \delta$ replaces $0 < |x - a| < \delta$, which ensures that we only consider values of x that are greater than (to the right of) a. Similarly, in the definition of the limit from the left, the inequality $-\delta < x - a < 0$ replaces $0 < |x - a| < \delta$, which ensures that we only consider values of x that are less than (to the left of) a.

Definition: One-Sided Limits (Formal)

Limit from the Right: Let $f(x)$ be defined over an open interval of the form (a, b) where $a < b$. Then

$$\lim_{x \to a^+} f(x) = L$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$, such that if $0 < x - a < \delta$, then $|f(x) - L| < \varepsilon$.

Limit from the Left: Let $f(x)$ be defined over an open interval of the form (b, c) where $b < c$. Then,

$$\lim_{x \to c^-} f(x) = L$$

if for every $\varepsilon > 0$,there exists a $\delta > 0$ such that if $-\delta < x - c < 0$, then $|f(x) - L| < \varepsilon$.

Example 2.5.6: Proving a Statement about a Limit From the Right

Prove that

$$\lim_{x\to 4^+} \sqrt{x-4} = 0.$$

Solution

Let $\varepsilon > 0$.

Choose $\delta = \varepsilon^2$. Since we ultimately want $|\sqrt{x-4} - 0| < \varepsilon$, we manipulate this inequality to get $\sqrt{x-4} < \varepsilon$ or, equivalently, $0 < x - 4 < \varepsilon^2$, making $\delta = \varepsilon^2$ a clear choice. We may also determine δ geometrically, as shown in Figure 2.5.4.

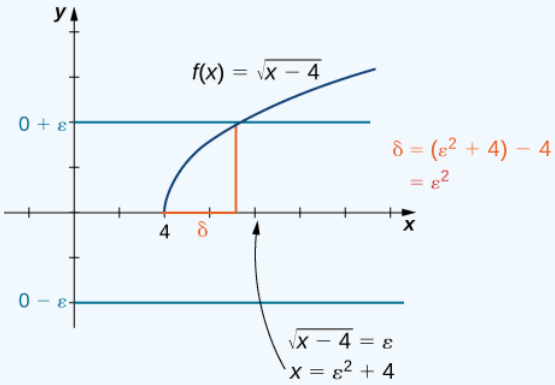


Figure 2.5.4: This graph shows how we find δ for the proof in Example 2.5.6.

Assume $0 < x - 4 < \delta$. Thus, $0 < x - 4 < \varepsilon^2$. Hence, $0 < \sqrt{x-4} < \varepsilon$. Finally, $\left|\sqrt{x-4} - 0\right| < \varepsilon$. Therefore, $\lim_{x\to 4^+} \sqrt{x-4} = 0$.

Exercise 2.5.4

Find δ corresponding to ε for a proof that $\lim_{x\to 1^-} \sqrt{1-x} = 0$.

Hint

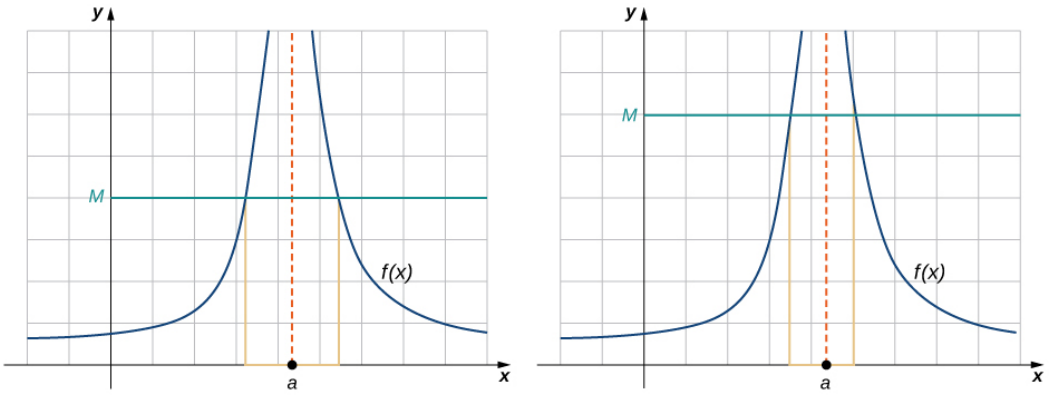
Sketch the graph and use Example 2.5.6 as a solving guide.

Answer

$\delta = \varepsilon^2$

Infinite Limits

We conclude the process of converting our intuitive ideas of various types of limits to rigorous formal definitions by pursuing a formal definition of infinite limits. To have $\lim_{x\to a} f(x) = +\infty$, we want the values of the function $f(x)$ to get larger and larger as x approaches a. Instead of the requirement that $|f(x) - L| < \varepsilon$ for arbitrarily small ε when $0 < |x - a| < \delta$ for small enough δ, we want $f(x) > M$ for arbitrarily large positive M when $0 < |x - a| < \delta$ for small enough δ. Figure 2.5.5 illustrates this idea by showing the value of δ for successively larger values of M.



In each graph, δ is the smaller of the lengths of the two brown intervals.

Figure 2.5.5: These graphs plot values of δ for M to show that $\lim_{x\to a} f(x) = +\infty$.

Definition: Infinite Limits (Formal)

Let $f(x)$ be defined for all $x \neq a$ in an open interval containing a. Then, we have an infinite limit

$$\lim_{x\to a} f(x) = +\infty$$

if for every $M > 0$, there exists $\delta > 0$ such that if $0 < |x - a| < \delta$, then $f(x) > M$.

Let $f(x)$ be defined for all $x \neq a$ in an open interval containing a. Then, we have a negative infinite limit

$$\lim_{x\to a} f(x) = -\infty$$

if for every $M > 0$, there exists $\delta > 0$ such that if $0 < |x - a| < \delta$, then $f(x) < -M$.

Example 2.5.7: Proving a Statement about an Infinite Limit

Prove that $\lim_{x\to 3} \dfrac{1}{(x-3)^2} = \infty$.

Solution

We use a very similar approach to our previous Problem-Solving Strategy. We first find an appropriate $\delta > 0$. Then we write our proof.

Step 1: First we find an appropriate $\delta > 0$.

1. Let M be any real number such that $M > 0$.

2. Let $f(x) = \dfrac{1}{(x-3)^2} > M$. Then we solve for the expression $x - 3$.

Multiplying both sides of the inequality by the positive quantity $(x-3)^2$ and dividing both sides by the positive quantity M gives us:

$$\frac{1}{M} > (x-3)^2$$

Taking the square root of both sides, we have,

$$\sqrt{\frac{1}{M}} > |x-3|. \qquad \left(\text{Remember that } \sqrt{x^2} = |x|.\right)$$

Rewriting this statement gives us, $0 < |x-3| < \sqrt{\dfrac{1}{M}}$. From this we choose $\delta = \sqrt{\dfrac{1}{M}}$.

Step 2: Now we write a proof.

3. Let $\delta = \sqrt{\frac{1}{M}}$ and assume $0 < |x-3| < \delta = \sqrt{\frac{1}{M}}$.

Thus,

$$|x-3| < \sqrt{\frac{1}{M}}.$$

Squaring both sides gives us,

$$(x-3)^2 < \frac{1}{M}.$$

Taking the reciprocal of both sides (and remembering that this will reverse the direction of the inequality),

$$\frac{1}{(x-3)^2} > M.$$

Therefore, we have proven that

$$\lim_{x\to 3} \frac{1}{(x-3)^2} = \infty.$$

A very similar proof will be needed for a limit that is equal to $-\infty$.

Note that a one-sided limit approach will often need to be taken with this type of limit. For example, to prove: $\lim_{x\to 0^+} \frac{1}{x} = \infty$.

Key Concepts

- The intuitive notion of a limit may be converted into a rigorous mathematical definition known as the epsilon-delta definition of the limit.
- The epsilon-delta definition may be used to prove statements about limits.
- The epsilon-delta definition of a limit may be modified to define one-sided limits.
- A similar definition of an infinite limit can be used to prove statements about infinite limits.

Glossary

epsilon-delta definition of the limit

$\lim_{x\to a} f(x) = L$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that if $0 < |x-a| < \delta$, then $|f(x) - L| < \varepsilon$

triangle inequality

If a and b are any real numbers, then $|a+b| \le |a| + |b|$

formal definition of an infinite limit

$\lim_{x\to a} f(x) = \infty$ if for every $M > 0$, there exists a $\delta > 0$ such that if $0 < |x-a| < \delta$, then $f(x) > M$
$\lim_{x\to a} f(x) = -\infty$ if for every $M > 0$, there exists a $\delta > 0$ such that if $0 < |x-a| < \delta$, then $f(x) < -M$

- Gilbert Strang (MIT) and Edwin “Jed” Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.
- Paul Seeburger (Monroe Community College), added Example 2.5.7 and entries for infinite limits under Key Concepts and the Glossary.

This page titled 2.5: The Precise Definition of a Limit is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

2.5E: EXERCISES FOR SECTION 2.5

In exercises 1 - 4, write the appropriate $\varepsilon - \delta$ definition for each of the given statements.

1) $\lim\limits_{x \to a} f(x) = N$

2) $\lim\limits_{t \to b} g(t) = M$

Answer

For every $\varepsilon > 0$, there exists a $\delta > 0$, so that if $0 < |t - b| < \delta$, then $|g(t) - M| < \varepsilon$

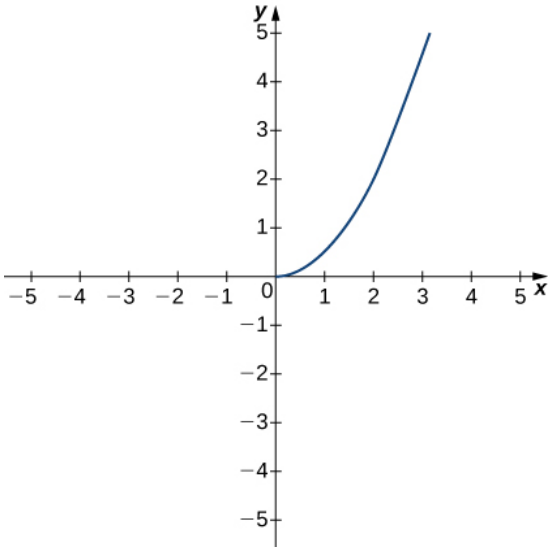
3) $\lim\limits_{x \to c} h(x) = L$

4) $\lim\limits_{x \to a} \varphi(x) = A$

Answer

For every $\varepsilon > 0$, there exists a $\delta > 0$, so that if $0 < |x - a| < \delta$, then $|\varphi(x) - A| < \varepsilon$

The following graph of the function f satisfies $\lim\limits_{x \to 2} f(x) = 2$. In the following exercises, determine a value of $\delta > 0$ that satisfies each statement.



5) If $0 < |x - 2| < \delta$, then $|f(x) - 2| < 1$.

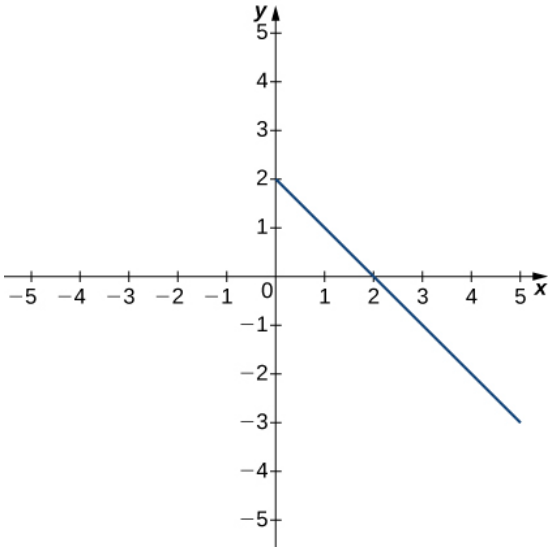
6) If $0 < |x - 2| < \delta$, then $|f(x) - 2| < 0.5$.

Answer

$\delta \leq 0.25$

The following graph of the function f satisfies $\lim\limits_{x \to 3} f(x) = -1$. In the following exercises, determine a value of $\delta > 0$ that satisfies each statement.

https://math.libretexts.org/@go/page/50407



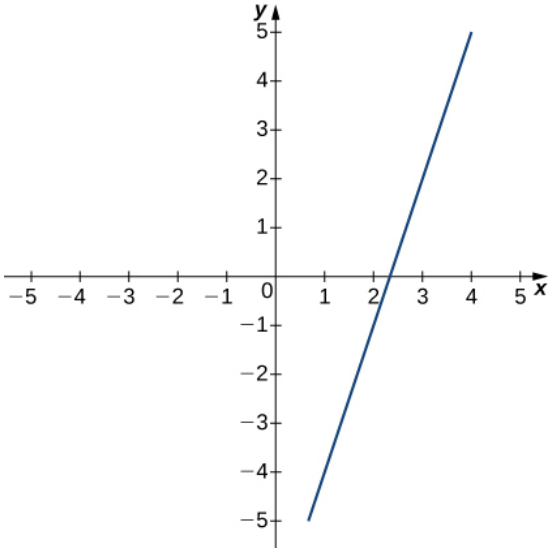
7) If $0 < |x-3| < \delta$, then $|f(x)+1| < 1$.

8) If $0 < |x-3| < \delta$, then $|f(x)+1| < 2$.

Answer

$\delta \leq 2$

The following graph of the function f satisfies $\lim_{x \to 3} f(x) = 2$. In the following exercises, for each value of ε, find a value of $\delta > 0$ such that the precise definition of limit holds true.



9) $\varepsilon = 1.5$

10) $\varepsilon = 3$

Answer

$\delta \leq 1$

[T] In exercises 11 - 12, use a graphing calculator to find a number δ such that the statements hold true.

11) $\left|\sin(2x) - \frac{1}{2}\right| < 0.1$, whenever $\left|x - \frac{\pi}{12}\right| < \delta$

12) $\left|\sqrt{x-4} - 2\right| < 0.1$, whenever $|x-8| < \delta$

Answer

$\delta < 0.3900$

In exercises 13 - 17, use the precise definition of limit to prove the given limits.

13) $\lim\limits_{x\to 2}(5x+8)=18$

14) $\lim\limits_{x\to 3}\dfrac{x^2-9}{x-3}=6$

Answer

Let $\delta=\varepsilon$. If $0<|x-3|<\varepsilon$, then $\left|\dfrac{x^2-9}{x-3}-6\right|=\left|\dfrac{(x+3)(x-3)}{x-3}-6\right|=|x+3-6|=|x-3|<\varepsilon$.

15) $\lim\limits_{x\to 2}\dfrac{2x^2-3x-2}{x-2}=5$

16) $\lim\limits_{x\to 0}x^4=0$

Answer

Let $\delta=\sqrt[4]{\varepsilon}$. If $0<|x|<\sqrt[4]{\varepsilon}$, then $|x^4-0|=x^4<\varepsilon$.

17) $\lim\limits_{x\to 2}(x^2+2x)=8$

In exercises 18 - 20, use the precise definition of limit to prove the given one-sided limits.

18) $\lim\limits_{x\to 5^-}\sqrt{5-x}=0$

Answer

Let $\delta=\varepsilon^2$. If $-\varepsilon^2<x-5<0$, we can multiply through by -1 to get $0<5-x<\varepsilon^2$.
Then $\left|\sqrt{5-x}-0\right|=\sqrt{5-x}<\sqrt{\varepsilon^2}=\varepsilon$.

19) $\lim\limits_{x\to 0^+}f(x)=-2$, where $f(x)=\begin{cases}8x-3, & \text{if } x<0\\ 4x-2, & \text{if } x\geq 0\end{cases}$.

20) $\lim\limits_{x\to 1^-}f(x)=3$, where $f(x)=\begin{cases}5x-2, & \text{if } x<1\\ 7x-1, & \text{if } x\geq 1\end{cases}$.

Answer

Let $\delta=\varepsilon/5$. If $-\varepsilon/5<x-1<0$, we can multiply through by -1 to get $0<1-x<\varepsilon/5$.
Then $|f(x)-3|=|5x-2-3|=|5x-5|=5(1-x)$, since $x<1$ here.
And $5(1-x)<5(\varepsilon/5)=\varepsilon$.

In exercises 21 - 23, use the precise definition of limit to prove the given infinite limits.

21) $\lim\limits_{x\to 0}\dfrac{1}{x^2}=\infty$

22) $\lim\limits_{x\to -1}\dfrac{3}{(x+1)^2}=\infty$

Answer

Let $\delta=\sqrt{\frac{3}{N}}$. If $0<|x+1|<\sqrt{\frac{3}{N}}$, then $f(x)=\frac{3}{(x+1)^2}>N$.

23) $\lim\limits_{x\to 2}-\dfrac{1}{(x-2)^2}=-\infty$

24) An engineer is using a machine to cut a flat square of Aerogel of area $144\,\text{cm}^2$. If there is a maximum error tolerance in the area of $8\,\text{cm}^2$, how accurately must the engineer cut on the side, assuming all sides have the same length? How do these numbers relate to δ, ε, a, and L?

Answer

0.033 cm, $\varepsilon=8$, $\delta=0.33$, $a=12$, $L=144$

25) Use the precise definition of limit to prove that the following limit does not exist: $\lim\limits_{x\to 1}\dfrac{|x-1|}{x-1}$.

26) Using precise definitions of limits, prove that $\lim\limits_{x\to 0}f(x)$ does not exist, given that $f(x)$ is the ceiling function. (Hint: Try any $\delta<1$.)

Answer

Answers may very.

27) Using precise definitions of limits, prove that $\lim_{x \to 0} f(x)$ does not exist: $f(x) = \begin{cases} 1, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$. (Hint: Think about how you can always choose a rational number \(0 <d\), >

28) Using precise definitions of limits, determine $\lim_{x \to 0} f(x)$ for $f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$. (Hint: Break into two cases, x rational and x irrational.)

Answer

0

29) Using the function from the previous exercise, use the precise definition of limits to show that $\lim_{x \to a} f(x)$ does not exist for $a \neq 0$

For exercises 30 - 32, suppose that $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} g(x) = M$ both exist. Use the precise definition of limits to prove the following limit laws:

30) $\lim_{x \to a}(f(x) - g(x)) = L - M$

Answer

$f(x) - g(x) = f(x) + (-1)g(x)$

31) $\lim_{x \to a}[cf(x)] = cL$ for any real constant c (Hint: Consider two cases: $c = 0$ and $c \neq 0$.)

32) $\lim_{x \to a}[f(x)g(x)] = LM$. (Hint: $|f(x)g(x) - LM| = |f(x)g(x) - f(x)M + f(x)M - LM| \leq |f(x)||g(x) - M| + |M||f(x) - L|$.)

Answer

Answers may vary.

2.5E: Exercises for Section 2.5 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by LibreTexts.

2.R: CHAPTER 2 REVIEW EXERCISES

***True or False*. In exercises 1 - 4, justify your answer with a proof or a counterexample.**

1) A function has to be continuous at $x = a$ if the $\lim_{x \to a} f(x)$ exists.

2) You can use the quotient rule to evaluate $\lim_{x \to 0} \frac{\sin x}{x}$.

Answer

False, since we cannot have $\lim_{x \to 0} x = 0$ in the denominator.

3) If there is a vertical asymptote at $x = a$ for the function $f(x)$, then f is undefined at the point $x = a$.

4) If $\lim_{x \to a} f(x)$ does not exist, then f is undefined at the point $x = a$.

Answer

False. A jump discontinuity is possible.

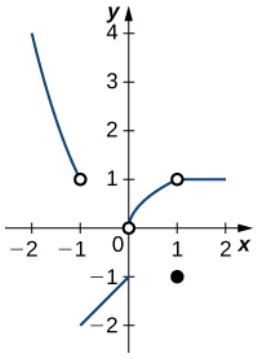
5) Using the graph, find each limit or explain why the limit does not exist.

a. $\lim_{x \to -1} f(x)$

b. $\lim_{x \to 1} f(x)$

c. $\lim_{x \to 0^+} f(x)$

d. $\lim_{x \to 2} f(x)$



In exercises 6 - 15, evaluate the limit algebraically or explain why the limit does not exist.

6) $\lim_{x \to 2} \frac{2x^2 - 3x - 2}{x - 2}$

Answer

5

7) $\lim_{x \to 0} 3x^2 - 2x + 4$

8) $\lim_{x \to 3} \frac{x^3 - 2x^2 - 1}{3x - 2}$

Answer

8/7

9) $\lim_{x \to \pi/2} \frac{\cot x}{\cos x}$

10) $\lim_{x \to -5} \frac{x^2 + 25}{x + 5}$

Answer

DNE

11) $\lim_{x\to 2}\dfrac{3x^2-2x-8}{x^2-4}$

12) $\lim_{x\to 1}\dfrac{x^2-1}{x^3-1}$

Answer

$2/3$

13) $\lim_{x\to 1}\dfrac{x^2-1}{\sqrt{x}-1}$

14) $\lim_{x\to 4}\dfrac{4-x}{\sqrt{x}-2}$

Answer

-4

15) $\lim_{x\to 4}\dfrac{1}{\sqrt{x}-2}$

In exercises 16 - 17, use the squeeze theorem to prove the limit.

16) $\lim_{x\to 0} x^2\cos(2\pi x)=0$

Answer

Since $-1\le\cos(2\pi x)\le 1$, then $-x^2\le x^2\cos(2\pi x)\le x^2$. Since $\lim_{x\to 0}x^2=0=\lim_{x\to 0}-x^2$, it follows that $\lim_{x\to 0}x^2\cos(2\pi x)=0$.

17) $\lim_{x\to 0} x^3\sin\left(\dfrac{\pi}{x}\right)=0$

18) Determine the domain such that the function $f(x)=\sqrt{x-2}+xe^x$ is continuous over its domain.

Answer

$[2,\infty]$

In exercises 19 - 20, determine the value of c such that the function remains continuous. Draw your resulting function to ensure it is continuous.

19) $f(x)=\begin{cases}x^2+1, & \text{if } x>c\\ 2^x, & \text{if } x\le c\end{cases}$

20) $f(x)=\begin{cases}\sqrt{x+1}, & \text{if } x>-1\\ x^2+c, & \text{if } x\le -1\end{cases}$

In exercises 21 - 22, use the precise definition of limit to prove the limit.

21) $\lim_{x\to 1}(8x+16)=24$

22) $\lim_{x\to 0}x^3=0$

Answer

$\delta=\sqrt[3]{\varepsilon}$

23) A ball is thrown into the air and the vertical position is given by $x(t)=-4.9t^2+25t+5$. Use the Intermediate Value Theorem to show that the ball must land on the ground sometime between 5 sec and 6 sec after the throw.

24) A particle moving along a line has a displacement according to the function $x(t)=t^2-2t+4$, where x is measured in meters and t is measured in seconds. Find the average velocity over the time period $t=[0,2]$.

Answer

0 m/sec

25) From the previous exercises, estimate the instantaneous velocity at $t=2$ by checking the average velocity within $t=0.01$ sec.

2.R: Chapter 2 Review Exercises is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by LibreTexts.

CHAPTER OVERVIEW

3: Derivatives

Calculating velocity and changes in velocity are important uses of calculus, but it is far more widespread than that. Calculus is important in all branches of mathematics, science, and engineering, and it is critical to analysis in business and health as well. In this chapter, we explore one of the main tools of calculus, the derivative, and show convenient ways to calculate derivatives. We apply these rules to a variety of functions in this chapter so that we can then explore applications of these techniques.

3.0: Prelude to Derivatives
3.1: Defining the Derivative
3.1E: Exercises for Section 3.1
3.2: The Derivative as a Function
3.2E: Exercises for Section 3.2
3.3: Differentiation Rules
3.3E: Exercises for Section 3.3
3.4: Derivatives as Rates of Change
3.4E: Exercises for Section 3.4
3.5: Derivatives of Trigonometric Functions
3.5E: Exercises for Section 3.5
3.6: The Chain Rule
3.6E: Exercises for Section 3.6
3.7: Derivatives of Inverse Functions
3.7E: Exercises for Section 3.7
3.8: Implicit Differentiation
3.8E: Exercises for Section 3.8
3.9: Derivatives of Exponential and Logarithmic Functions
3.9E: Exercises for Section 3.9
3.R: Chapter 3 Review Exercises

This page titled 3: Derivatives is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

3.0: Prelude to Derivatives

The Hennessey Venom GT is one of the fastest cars in the world. In 2014, it reached a record-setting speed of 270.49 mph. It can go from 0 to 200 mph in 14.51 seconds. The techniques in this chapter can be used to calculate the acceleration the Venom achieves in this feat.)



Figure 3.0.1: The Hennessey Venom GT can go from 0 to 200 mph in 14.51 seconds. (credit: modification of work by Codex41, Flickr)

Calculating velocity and changes in velocity are important uses of calculus, but it is far more widespread than that. Calculus is important in all branches of mathematics, science, and engineering, and it is critical to analysis in business and health as well. In this chapter, we explore one of the main tools of calculus, the derivative, and show convenient ways to calculate derivatives. We apply these rules to a variety of functions in this chapter so that we can then explore applications of these techniques.

This page titled 3.0: Prelude to Derivatives is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

3.1: Defining the Derivative

Learning Objectives

- Recognize the meaning of the tangent to a curve at a point.
- Calculate the slope of a tangent line.
- Identify the derivative as the limit of a difference quotient.
- Calculate the derivative of a given function at a point.
- Describe the velocity as a rate of change.
- Explain the difference between average velocity and instantaneous velocity.
- Estimate the derivative from a table of values.

Now that we have both a conceptual understanding of a limit and the practical ability to compute limits, we have established the foundation for our study of calculus, the branch of mathematics in which we compute derivatives and integrals. Most mathematicians and historians agree that calculus was developed independently by the Englishman Isaac **Newton (1643–1727)** and the German Gottfried **Leibniz (1646–1716)**, whose images appear in Figure 3.1.1. When we credit Newton and Leibniz with developing calculus, we are really referring to the fact that Newton and Leibniz were the first to understand the relationship between the derivative and the integral. Both mathematicians benefited from the work of predecessors, such as Barrow, Fermat, and Cavalieri. The initial relationship between the two mathematicians appears to have been amicable; however, in later years a bitter controversy erupted over whose work took precedence. Although it seems likely that Newton did, indeed, arrive at the ideas behind calculus first, we are indebted to Leibniz for the notation that we commonly use today.

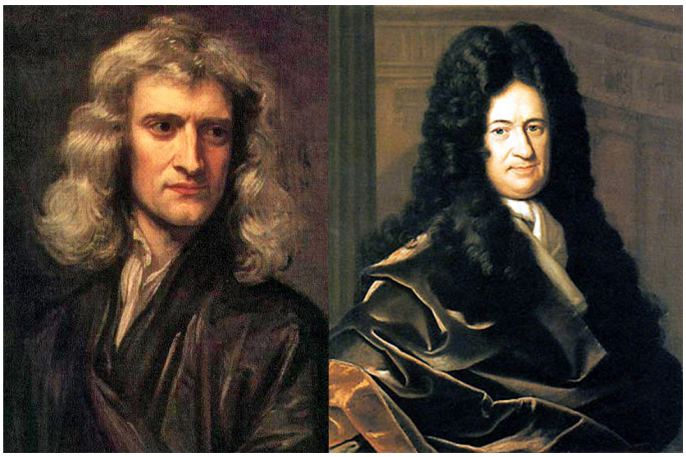


Figure 3.1.1: Newton and Leibniz are credited with developing calculus independently.

Tangent Lines

We begin our study of calculus by revisiting the notion of secant lines and tangent lines. Recall that we used the slope of a secant line to a function at a point $(a, f(a))$ to estimate the rate of change, or the rate at which one variable changes in relation to another variable. We can obtain the slope of the secant by choosing a value of x near a and drawing a line through the points $(a, f(a))$ and $(x, f(x))$, as shown in Figure 3.1.2. The slope of this line is given by an equation in the form of a difference quotient:

$$m_{sec} = \frac{f(x) - f(a)}{x - a}$$

We can also calculate the slope of a secant line to a function at a value a by using this equation and replacing x with $a + h$, where h is a value close to a. We can then calculate the slope of the line through the points $(a, f(a))$ and $(a + h, f(a + h))$. In this case, we find the secant line has a slope given by the following difference quotient with increment h:

$$m_{sec} = \frac{f(a + h) - f(a)}{a + h - a} = \frac{f(a + h) - f(a)}{h}$$

Definition: Difference Quotient

Let f be a function defined on an interval I containing a. If $x \neq a$ is in I, then

$$Q = \frac{f(x) - f(a)}{x - a}$$

is a *difference quotient.*

Also, if $h \neq 0$ is chosen so that $a + h$ is in I, then

$$Q = \frac{f(a + h) - f(a)}{h}$$

is a difference quotient with increment h.

These two expressions for calculating the slope of a secant line are illustrated in Figure 3.1.2. We will see that each of these two methods for finding the slope of a secant line is of value. Depending on the setting, we can choose one or the other. The primary consideration in our choice usually depends on ease of calculation.

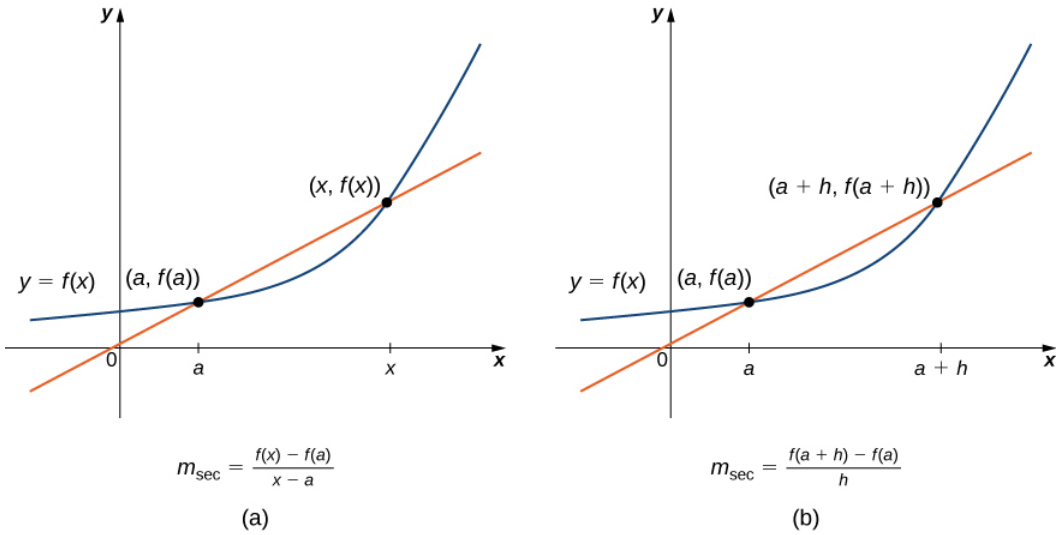


Figure 3.1.2: We can calculate the slope of a secant line in either of two ways.

In Figure 3.1.3a we see that, as the values of x approach a, the slopes of the secant lines provide better estimates of the rate of change of the function at a. Furthermore, the secant lines themselves approach the tangent line to the function at a, which represents the limit of the secant lines. Similarly, Figure 3.1.3b shows that as the values of h get closer to 0, the secant lines also approach the tangent line. The slope of the tangent line at a is the rate of change of the function at a, as shown in Figure 3.1.3c.

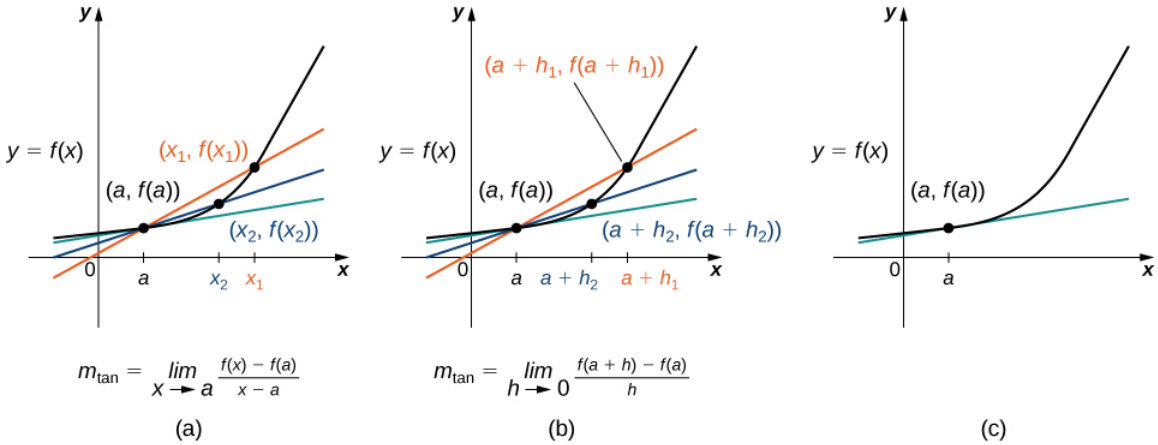


Figure 3.1.3: The secant lines approach the tangent line (shown in green) as the second point approaches the first.

In Figure 3.1.4 we show the graph of $f(x) = \sqrt{x}$ and its tangent line at $(1, 1)$ in a series of tighter intervals about $x = 1$. As the intervals become narrower, the graph of the function and its tangent line appear to coincide, making the values on the tangent line a good approximation to the values of the function for choices of x close to 1. In fact, the graph of $f(x)$ itself appears to be **locally linear** in the immediate vicinity of $x = 1$.

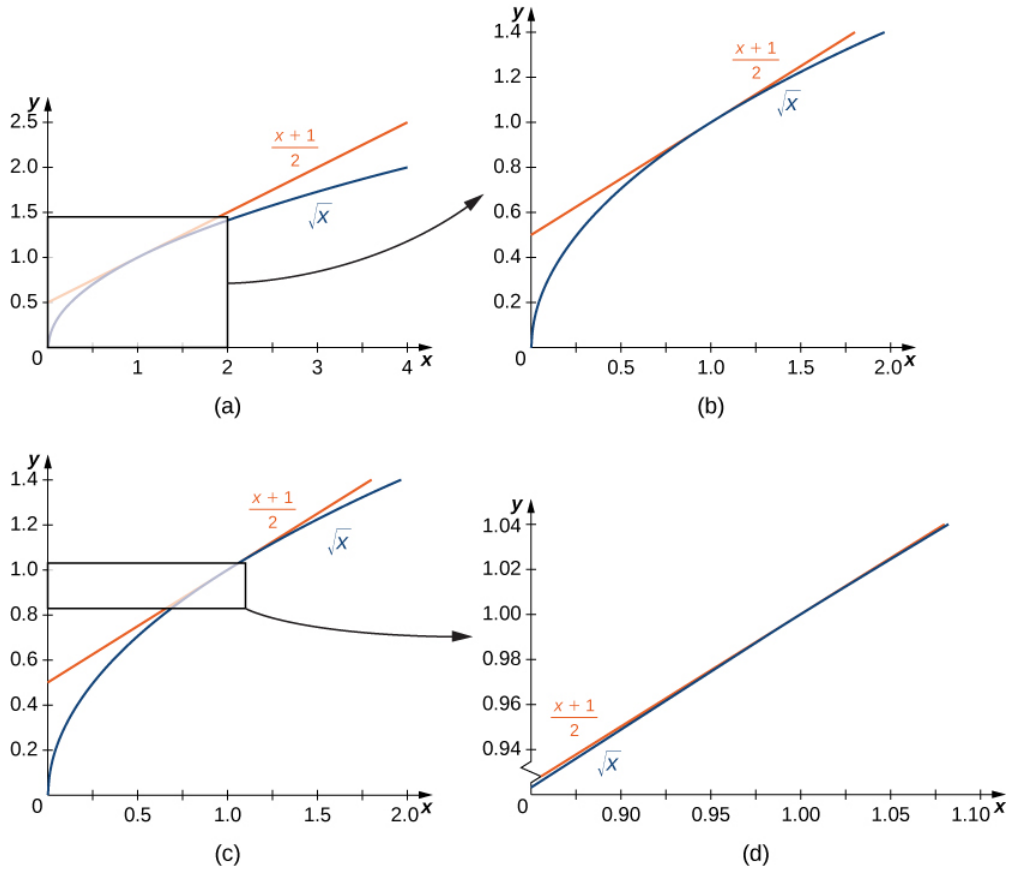


Figure 3.1.4: For values of x close to 1, the graph of $f(x) = \sqrt{x}$ and its tangent line appear to coincide.

Formally we may define the tangent line to the graph of a function as follows.

Definition: Tangent Line

Let $f(x)$ be a function defined in an open interval containing a. The tangent line to $f(x)$ at a is the line passing through the point $(a, f(a))$ having slope

$$m_{tan} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \tag{3.1.1}$$

provided this limit exists.

Equivalently, we may define the *tangent line* to $f(x)$ at a to be the line passing through the point $(a, f(a))$ having slope

$$m_{tan} = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h} \tag{3.1.2}$$

provided this limit exists.

Just as we have used two different expressions to define the slope of a secant line, we use two different forms to define the slope of the tangent line. In this text we use both forms of the definition. As before, the choice of definition will depend on the setting. Now that we have formally defined a tangent line to a function at a point, we can use this definition to find equations of tangent lines.

Example 3.1.1: Finding a Tangent Line

Find the equation of the line tangent to the graph of $f(x) = x^2$ at $x = 3$.

Solution

First find the slope of the tangent line. In this example, use Equation 3.1.1.

$$m_{tan} = \lim_{x \to 3} \frac{f(x) - f(3)}{x - 3} \qquad \text{Apply the definition.}$$

$$= \lim_{x \to 3} \frac{x^2 - 9}{x - 3} \qquad \text{Substitute } f(x) = x^2 \text{ and } f(3) = 9$$

$$= \lim_{x \to 3} \frac{(x - 3)(x + 3)}{x - 3} = \lim_{x \to 3}(x + 3) = 6 \qquad \text{Factor the numerator to evaluate the limit.}$$

Next, find a point on the tangent line. Since the line is tangent to the graph of $f(x)$ at $x = 3$, it passes through the point $(3, f(3))$. We have $f(3) = 9$, so the tangent line passes through the point $(3, 9)$.

Using the point-slope equation of the line with the slope $m = 6$ and the point $(3, 9)$, we obtain the line $y - 9 = 6(x - 3)$. Simplifying, we have $y = 6x - 9$. The graph of $f(x) = x^2$ and its tangent line at 3 are shown in Figure 3.1.5.

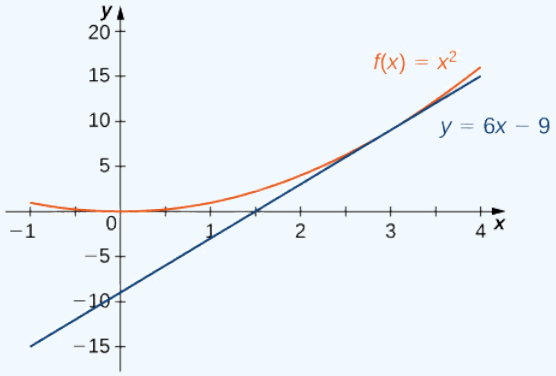


Figure 3.1.5: The tangent line to $f(x)$ at $x = 3$.

Example 3.1.2: The Slope of a Tangent Line Revisited

Use Equation 3.1.2 to find the slope of the line tangent to the graph of $f(x) = x^2$ at $x = 3$.

Solution

The steps are very similar to Example 3.1.1. See Equation 3.1.2 for the definition.

$$m_{tan} = \lim_{h \to 0} \frac{f(3 + h) - f(3)}{h} \qquad \text{Apply the definition.}$$

$$= \lim_{h \to 0} \frac{(3 + h)^2 - 9}{h} \qquad \text{Substitute } f(3 + h) = (3 + h)^2 \text{ and } f(3) = 9$$

$$= \lim_{h \to 0} \frac{9 + 6h + h^2 - 9}{h} \qquad \text{Expand and simplify to evaluate the limit.}$$

$$= \lim_{h \to 0} \frac{h(6 + h)}{h} = \lim_{h \to 0}(6 + h) = 6$$

We obtained the same value for the slope of the tangent line by using the other definition, demonstrating that the formulas can be interchanged.

Example 3.1.3: Finding the Equation of a Tangent Line

Find the equation of the line tangent to the graph of $f(x) = 1/x$ at $x = 2$.

Solution

We can use Equation 3.1.1, but as we have seen, the results are the same if we use Equation 3.1.2.

Misplaced &

We now know that the slope of the tangent line is $-\frac{1}{4}$. To find the equation of the tangent line, we also need a point on the line. We know that $f(2) = \frac{1}{2}$. Since the tangent line passes through the point $(2, \frac{1}{2})$ we can use the point-slope equation of a line to find the equation of the tangent line. Thus the tangent line has the equation $y = -\frac{1}{4}x + 1$. The graphs of $f(x) = \frac{1}{x}$ and $y = -\frac{1}{4}x + 1$ are shown in Figure 3.1.6.

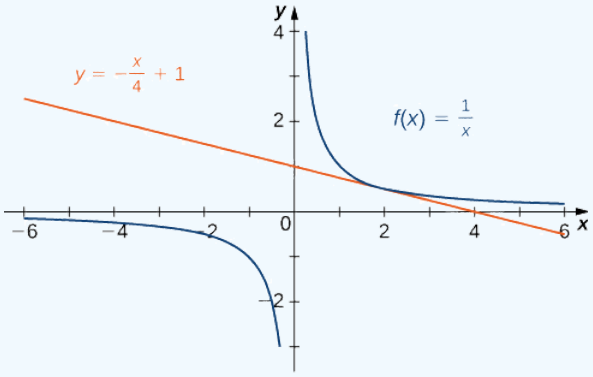


Figure 3.1.6:The line is tangent to $f(x)$ at $x = 2$.

Exercise 3.1.1

Find the slope of the line tangent to the graph of $f(x) = \sqrt{x}$ at $x = 4$.

Hint

Use either Equation 3.1.1 or Equation 3.1.2. Multiply the numerator and the denominator by a conjugate.

Answer

$\frac{1}{4}$

The Derivative of a Function at a Point

The type of limit we compute in order to find the slope of the line tangent to a function at a point occurs in many applications across many disciplines. These applications include velocity and acceleration in physics, marginal profit functions in business, and growth rates in biology. This limit occurs so frequently that we give this value a special name: the **derivative.** The process of finding a derivative is called **differentiation**.

Definition: Derivative

Let $f(x)$ be a function defined in an open interval containing a. The **derivative** of the function $f(x)$ at a, denoted by $f'(a)$, is defined by

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \tag{3.1.3}$$

provided this limit exists.

Alternatively, we may also define the derivative of $f(x)$ at a as

$$f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}. \tag{3.1.4}$$

Example 3.1.4: Estimating a Derivative

For $f(x) = x^2$, use a table to estimate $f'(3)$ using Equation 3.1.3.

Solution

Create a table using values of x just below 3 and just above 3.

x	$\frac{x^2-9}{x-3}$
2.9	5.9
2.99	5.99
2.999	5.999
3.001	6.001
3.01	6.01
3.1	6.1

After examining the table, we see that a good estimate is $f'(3) = 6$.

Exercise 3.1.2

For $f(x) = x^2$, use a table to estimate $f'(3)$ using Equation 3.1.4.

Hint

Evaluate $\frac{(x+h)^2 - x^2}{h}$ at $h = -0.1,\ -0.01,\ -0.001,\ 0.001,\ 0.01,\ 0.1$

Answer

6

Example 3.1.6: Finding a Derivative

For $f(x) = 3x^2 - 4x + 1$, find $f'(2)$ by using Equation 3.1.3.

Solution

Substitute the given function and value directly into the equation.

$$
\begin{aligned}
f'(x) &= \lim_{x\to 2} \frac{f(x) - f(2)}{x-2} && \text{Apply the definition.}\\
&= \lim_{x\to 2} \frac{(3x^2 - 4x + 1) - 5}{x-2} && \text{Substitute } f(x) = 3x^2 - 4x + 1 \text{ and } f(2) = 5.\\
&= \lim_{x\to 2} \frac{(x-2)(3x+2)}{x-2} && \text{Simplify and factor the numerator.}\\
&= \lim_{x\to 2} (3x+2) && \text{Cancel the common factor.}\\
&= 8 && \text{Evaluate the limit.}
\end{aligned}
$$

Example 3.1.7: Revisiting the Derivative

For $f(x) = 3x^2 - 4x + 1$, find $f'(2)$ by using Equation 3.1.4.

Solution

Using this equation, we can substitute two values of the function into the equation, and we should get the same value as in Example 3.1.6.

$$
\begin{aligned}
f'(2) &= \lim_{h\to 0}\frac{f(2+h)-f(2)}{h} && \text{Apply the definition.}\\
&= \lim_{h\to 0}\frac{(3(2+h)^2-4(2+h)+1)-5}{h} && \text{Substitute } f(2)=5 \text{ and } f(2+h)=3(2+h)^2-4(2+h)+1.\\
&= \lim_{h\to 0}\frac{3(4+4h+h^2)-8-4h+1-5}{h} && \text{Expand the numerator.}\\
&= \lim_{h\to 0}\frac{12+12h+3h^2-12-4h}{h} && \text{Distribute and begin simplifying the numerator.}\\
&= \lim_{h\to 0}\frac{3h^2+8h}{h} && \text{Finish simplifying the numerator.}\\
&= \lim_{h\to 0}\frac{h(3h+8)}{h} && \text{Factor the numerator.}\\
&= \lim_{h\to 0}(3h+8) && \text{Cancel the common factor.}\\
&= 8 && \text{Evaluate the limit.}
\end{aligned}
$$

The results are the same whether we use Equation 3.1.3 or Equation 3.1.4.

? Exercise 3.1.4

For $f(x)=x^2+3x+2$, find $f'(1)$.

Hint

Use either Equation 3.1.3, Equation 3.1.4, or try both.

Answer

$f'(1)=5$

Velocities and Rates of Change

Now that we can evaluate a derivative, we can use it in velocity applications. Recall that if $s(t)$ is the position of an object moving along a coordinate axis, the **average velocity** of the object over a time interval $[a,t]$ if $t>a$ or $[t,a]$ if $t<a$ is given by the difference quotient

$$v_{ave}=\frac{s(t)-s(a)}{t-a}. \tag{3.1.5}$$

As the values of t approach a, the values of v_{ave} approach the value we call the **instantaneous velocity** at a. That is, instantaneous velocity at a, denoted $v(a)$, is given by

$$v(a)=s'(a)=\lim_{t\to a}\frac{s(t)-s(a)}{t-a}. \tag{3.1.6}$$

To better understand the relationship between average velocity and instantaneous velocity, see Figure 3.1.7. In this figure, the slope of the tangent line (shown in red) is the instantaneous velocity of the object at time $t=a$ whose position at time t is given by the function $s(t)$. The slope of the secant line (shown in green) is the average velocity of the object over the time interval $[a,t]$.

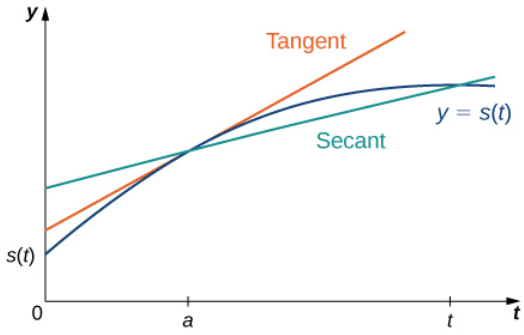


Figure 3.1.7: The slope of the secant line is the average velocity over the interval $[a, t]$. The slope of the tangent line is the instantaneous velocity.

We can use Equation 3.1.6 to calculate the instantaneous velocity, or we can estimate the velocity of a moving object by using a table of values. We can then confirm the estimate by using Equation 3.1.5.

Example 3.1.8: Estimating Velocity

A lead weight on a spring is oscillating up and down. Its position at time t with respect to a fixed horizontal line is given by $s(t) = \sin t$ (Figure 3.1.8). Use a table of values to estimate $v(0)$. Check the estimate by using Equation 3.1.6.

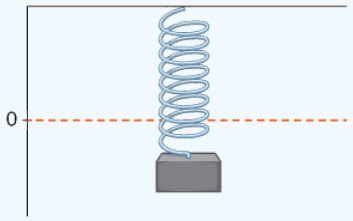


Figure 3.1.8: A lead weight suspended from a spring in vertical oscillatory motion.

Solution

We can estimate the instantaneous velocity at $t = 0$ by computing a table of average velocities using values of t approaching 0, as shown in Table 3.1.2.

Table 3.1.2: Average velocities using values of t approaching 0

t	$\frac{\sin t - \sin 0}{t-0} = \frac{\sin t}{t}$
−0.1	0.998334166
−0.01	0.9999833333
−0.001	0.999999833
0.001	0.999999833
0.01	0.9999833333
0.1	0.998334166

From the table we see that the average velocity over the time interval $[-0.1, 0]$ is 0.998334166, the average velocity over the time interval $[-0.01, 0]$ is 0.9999833333, and so forth. Using this table of values, it appears that a good estimate is $v(0) = 1$.

By using Equation 3.1.6, we can see that

$$v(0) = s'(0) = \lim_{t \to 0} \frac{\sin t - \sin 0}{t - 0} = \lim_{t \to 0} \frac{\sin t}{t} = 1.$$

Thus, in fact, $v(0) = 1$.

Exercise 3.1.5

A rock is dropped from a height of 64 feet. Its height above ground at time t seconds later is given by $s(t) = -16t^2 + 64,\ 0 \leq t \leq 2$. Find its instantaneous velocity 1 second after it is dropped, using Equation 3.1.6.

Hint

$v(t) = s'(t)$. Follow the earlier examples of the derivative using Equation 3.1.6.

Answer

-32 ft/s

As we have seen throughout this section, the slope of a tangent line to a function and instantaneous velocity are related concepts. Each is calculated by computing a derivative and each measures the instantaneous rate of change of a function, or the rate of change of a function at any point along the function.

Definition: Instantaneous Rate of Change

The **instantaneous rate of change** of a function $f(x)$ at a value a is its derivative $f'(a)$.

Example 3.1.9: Chapter Opener: Estimating Rate of Change of Velocity

Reaching a top speed of 270.49 mph, the Hennessey Venom GT is one of the fastest cars in the world. In tests it went from 0 to 60 mph in 3.05 seconds, from 0 to 100 mph in 5.88 seconds, from 0 to 200 mph in 14.51 seconds, and from 0 to 229.9 mph in 19.96 seconds. Use this data to draw a conclusion about the rate of change of velocity (that is, its **acceleration**) as it approaches 229.9 mph. Does the rate at which the car is accelerating appear to be increasing, decreasing, or constant?



Figure 3.1.9: (credit: modification of work by Codex41, Flickr)

Solution: First observe that 60 mph $= 88$ ft/s, 100 mph ≈ 146.67 ft/s, 200 mph ≈ 293.33 ft/s, and 229.9 mph ≈ 337.19 ft/s. We can summarize the information in a table.

Table 3.1.3: $v(t)$ at different values of t

t	$v(t)$
0	0
3.05	88
5.88	147.67
14.51	293.33
19.96	337.19

Now compute the average acceleration of the car in feet per second on intervals of the form $[t, 19.96]$ as t approaches 19.96, as shown in the following table.

Average acceleration

t	$\frac{v(t)-v(19.96)}{t-19.96}=\frac{v(t)-337.19}{t-19.96}$
0.0	16.89
3.05	14.74
5.88	13.46
14.51	8.05

The rate at which the car is accelerating is decreasing as its velocity approaches 229.9 mph (337.19 ft/s).

Example 3.1.10: Rate of Change of Temperature

A homeowner sets the thermostat so that the temperature in the house begins to drop from $70°F$ at 9 p.m., reaches a low of $60°$ during the night, and rises back to $70°$ by 7 a.m. the next morning. Suppose that the temperature in the house is given by $T(t) = 0.4t^2 - 4t + 70$ for $0 \le t \le 10$, where t is the number of hours past 9 p.m. Find the instantaneous rate of change of the temperature at midnight.

Solution

Since midnight is 3 hours past 9 p.m., we want to compute $T'(3)$. Refer to Equation 3.1.3.

$$\begin{aligned} T'(3) &= \lim_{t\to 3}\frac{T(t)-T(3)}{t-3} && \text{Apply the definition.}\\ &= \lim_{t\to 3}\frac{0.4t^2-4t+70-61.6}{t-3} && \text{Substitute } T(t)=0.4t^2-4t+70 \text{ and } T(3)=61.6.\\ &= \lim_{t\to 3}\frac{0.4t^2-4t+8.4}{t-3} && \text{Simplify.}\\ &= \lim_{t\to 3}\frac{0.4(t-3)(t-7)}{t-3} &&\\ &= \lim_{t\to 3}0.4(t-7) && \text{Cancel.}\\ &= -1.6 && \text{Evaluate the limit.} \end{aligned}$$

The instantaneous rate of change of the temperature at midnight is $-1.6°F$ per hour.

Example 3.1.11: Rate of Change of Profit

A toy company can sell x electronic gaming systems at a price of $p = -0.01x + 400$ dollars per gaming system. The cost of manufacturing x systems is given by $C(x) = 100x + 10,000$ dollars. Find the rate of change of profit when $10,000$ games are produced. Should the toy company increase or decrease production?

Solution

The profit $P(x)$ earned by producing x gaming systems is $R(x) - C(x)$, where $R(x)$ is the revenue obtained from the sale of x games. Since the company can sell x games at $p = -0.01x + 400$ per game,

$$R(x) = xp = x(-0.01x + 400) = -0.01x^2 + 400x.$$

Consequently,

$$P(x) = -0.01x^2 + 300x - 10,000.$$

Therefore, evaluating the rate of change of profit gives

$$
\begin{aligned}
P'(10000) &= \lim_{x \to 10000} \frac{P(x) - P(10000)}{x - 10000} \\
&= \lim_{x \to 10000} \frac{-0.01x^2 + 300x - 10000 - 1990000}{x - 10000} \\
&= \lim_{x \to 10000} \frac{-0.01x^2 + 300x - 2000000}{x - 10000} \\
&= 100
\end{aligned}
$$

Since the rate of change of profit $P'(10,000) > 0$ and $P(10,000) > 0$, the company should increase production.

? Exercise 3.1.6

A coffee shop determines that the daily profit on scones obtained by charging s dollars per scone is $P(s) = -20s^2 + 150s - 10$. The coffee shop currently charges $\$3.25$ per scone. Find $P'(3.25)$, the rate of change of profit when the price is $\$3.25$ and decide whether or not the coffee shop should consider raising or lowering its prices on scones.

Hint

Use Example 3.1.11 for a guide.

Answer

$P'(3.25) = 20 > 0$; raise prices

Key Concepts

- The slope of the tangent line to a curve measures the instantaneous rate of change of a curve. We can calculate it by finding the limit of the difference quotient or the difference quotient with increment h.
- The derivative of a function $f(x)$ at a value a is found using either of the definitions for the slope of the tangent line.
- Velocity is the rate of change of position. As such, the velocity $v(t)$ at time t is the derivative of the position $s(t)$ at time t. Average velocity is given by

  $$v_{ave} = \frac{s(t) - s(a)}{t - a}.$$

  Instantaneous velocity is given by

  $$v(a) = s'(a) = \lim_{t \to a} \frac{s(t) - s(a)}{t - a}.$$

- We may estimate a derivative by using a table of values.

Key Equations

- **Difference quotient**

  $$Q = \frac{f(x) - f(a)}{x - a}$$

- **Difference quotient with increment h**

  $$Q = \frac{f(a+h) - f(a)}{a + h - a} = \frac{f(a+h) - f(a)}{h}$$

- **Slope of tangent line**

  $$m_{tan} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

  $$m_{tan} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

- **Derivative of f(x) at a**

  $$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

- **Average velocity**

$$v_{ave} = \frac{s(t) - s(a)}{t - a}$$

- **Instantaneous velocity**

$$v(a) = s'(a) = \lim_{t \to a} \frac{s(t) - s(a)}{t - a}$$

Glossary

derivative

the slope of the tangent line to a function at a point, calculated by taking the limit of the difference quotient, is the derivative

difference quotient

of a function $f(x)$ at a is given by

$$\frac{f(a+h) - f(a)}{h} \text{ or } \frac{f(x) - f(a)}{x - a}$$

differentiation

the process of taking a derivative

instantaneous rate of change

the rate of change of a function at any point along the function a, also called $f'(a)$, or the derivative of the function at a

---

This page titled 3.1: Defining the Derivative is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin “Jed” Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

3.1E: EXERCISES FOR SECTION 3.1

For exercises 1 - 10, use the equation $m_{sec} = \frac{f(x) - f(a)}{x - a}$ to find the slope of the secant line between the values x_1 and x_2 for each function $y = f(x)$.

1) $f(x) = 4x + 7; \quad x_1 = 2, \quad x_2 = 5$

Answer

$m_{sec} = 4$

2) $f(x) = 8x - 3; \quad x_1 = -1, \quad x_2 = 3$

3) $f(x) = x^2 + 2x + 1; \quad x_1 = 3, \quad x_2 = 3.5$

Answer

$m_{sec} = 8.5$

4) $f(x) = -x^2 + x + 2; \quad x_1 = 0.5, \quad x_2 = 1.5$

5) $f(x) = \frac{4}{3x - 1}; \quad x_1 = 1, \quad x_2 = 3$

Answer

$m_{sec} = -\frac{3}{4}$

6) $f(x) = \frac{x - 7}{2x + 1}; \quad x_1 = -2, \quad x_2 = 0$

7) $f(x) = \sqrt{x}; \quad x_1 = 1, \quad x_2 = 16$

Answer

$m_{sec} = 0.2$

8) $f(x) = \sqrt{x - 9}; \quad x_1 = 10, \quad x_2 = 13$

9) $f(x) = x^{1/3} + 1; \quad x_1 = 0, \quad x_2 = 8$

Answer

$m_{sec} = 0.25$

10) $f(x) = 6x^{2/3} + 2x^{1/3}; \quad x_1 = 1, \quad x_2 = 27$

For the functions in exercises 11 - 20,

a. use the equation $m_{tan} = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}$ to find the slope of the tangent line $m_{tan} = f'(a)$, and

b. find the equation of the tangent line to f at $x = a$.

11) $f(x) = 3 - 4x, \quad a = 2$

Answer

a. $m_{tan} = -4$

b. $y = -4x + 3$

12) $f(x) = \frac{x}{5} + 6, \quad a = -1$

13) $f(x) = x^2 + x, \quad a = 1$

Answer

a. $m_{tan} = 3$

b. $y = 3x - 1$

14) $f(x) = 1 - x - x^2, \quad a = 0$

15) $f(x) = \frac{7}{x}, \quad a = 3$

Answer

a. $m_{tan} = \frac{-7}{9}$

b. $y = \frac{-7}{9}x + \frac{14}{3}$

16) $f(x) = \sqrt{x + 8}, \quad a = 1$

17) $f(x) = 2 - 3x^2, \quad a = -2$

Answer

a. $m_{\tan} = 12$
b. $y = 12x + 14$

18) $f(x) = \dfrac{-3}{x-1}, \quad a = 4$

19) $f(x) = \dfrac{2}{x+3}, \quad a = -4$

Answer

a. $m_{\tan} = -2$
b. $y = -2x - 10$

20) $f(x) = \dfrac{3}{x^2}, \quad a = 3$

For the functions $y = f(x)$ in exercises 21 - 30, find $f'(a)$ using the equation $f'(a) = \lim\limits_{x \to a} \dfrac{f(x) - f(a)}{x - a}$.

21) $f(x) = 5x + 4, \quad a = -1$

Answer

$f'(-1) = 5$

22) $f(x) = -7x + 1, \quad a = 3$

23) $f(x) = x^2 + 9x, \quad a = 2$

Answer

$f'(2) = 13$

24) $f(x) = 3x^2 - x + 2, \quad a = 1$

25) $f(x) = \sqrt{x}, \quad a = 4$

Answer

$f'(4) = \frac{1}{4}$

26) $f(x) = \sqrt{x-2}, \quad a = 6$

27) $f(x) = \dfrac{1}{x}, \quad a = 2$

Answer

$f'(2) = -\frac{1}{4}$

28) $f(x) = \dfrac{1}{x-3}, \quad a = -1$

29) $f(x) = \dfrac{1}{x^3}, \quad a = 1$

Answer

$f'(1) = -3$

30) $f(x) = \dfrac{1}{\sqrt{x}}, \quad a = 4$

For the following exercises, given the function $y = f(x)$,

a. find the slope of the secant line PQ for each point $Q(x, f(x))$ with x value given in the table.

b. Use the answers from a. to estimate the value of the slope of the tangent line at P.

c. Use the answer from b. to find the equation of the tangent line to f at point P.

31) [T] $f(x) = x^2 + 3x + 4, \quad P(1, 8)$ (Round to 6 decimal places.)

x	Slope m_{PQ}	x	Slope m_{PQ}
1.1	(i)	0.9	(vii)
1.01	(ii)	0.99	(viii)
1.001	(iii)	0.999	(ix)
1.0001	(iv)	0.9999	(x)
1.00001	(v)	0.99999	(xi)
1.000001	(vi)	0.999999	(xii)

Answer

$a.(i) 5.100000, (ii) 5.010000, (iii) 5.001000, (iv) 5.000100, (v) 5.000010, (vi) 5.000001, (vii) 4.900000, (viii) 4.990000, (ix) 4.999000, (x) 4.999900, (xi) 4.999990, (x) 4.999999$

b. $m_{\tan} = 5$

https://math.libretexts.org/@go/page/50900

c. $y = 5x + 3$

32) [T] $f(x) = \dfrac{x+1}{x^2-1}, \quad P(0,-1)$

x	Slope m_{PQ}	x	Slope m_{PQ}
0.1	(i)	−0.1	(vii)
0.01	(ii)	−0.01	(viii)
0.001	(iii)	−0.001	(ix)
0.0001	(iv)	−0.0001	(x)
0.00001	(v)	−0.00001	(xi)
0.000001	(vi)	−0.000001	(xii)

33) [T] $f(x) = 10e^{0.5x}, \quad P(0,10)$ (Round to 4 decimal places.)

x	Slope m_{PQ}
−0.1	(i)
−0.01	(ii)
−0.001	(iii)
−0.0001	(iv)
−0.00001	(v)
−0.000001	(vi)

Answer

a. $(i) 4.8771,\ (ii) 4.9875,\ (iii) 4.9988,\ (iv) 4.9999,\ (v) 4.9999,\ (vi) 4.9999$

b. $m_{\tan} = 5$

c. $y = 5x + 10$

34) [T] $f(x) = \tan(x), \quad P(\pi, 0)$

x	Slope m_{PQ}
3.1	(i)
3.14	(ii)
3.141	(iii)
3.1415	(iv)
3.14159	(v)
3.141592	(vi)

[T] For the following position functions $y = s(t)$, an object is moving along a straight line, where t is in seconds and s is in meters. Find

a. the simplified expression for the average velocity from $t = 2$ to $t = 2 + h$;

b. the average velocity between $t = 2$ and $t = 2 + h$, where $(i)\ h = 0.1,\ (ii)\ h = 0.01,\ (iii)\ h = 0.001$, and $(iv)\ h = 0.0001$; and

c. use the answer from a. to estimate the instantaneous velocity at $t = 2$ second.

35) $s(t) = \frac{1}{3}t + 5$

Answer

a. $\frac{1}{3}$;

b. $(i)\ \frac{1}{3}$ m/s, $(ii)\ \frac{1}{3}$ m/s, $(iii)\ \frac{1}{3}$ m/s, $(iv)\ \frac{1}{3}$ m/s;

c. $\frac{1}{3}$ m/s

36) $s(t) = t^2 - 2t$

37) $s(t) = 2t^3 + 3$

Answer

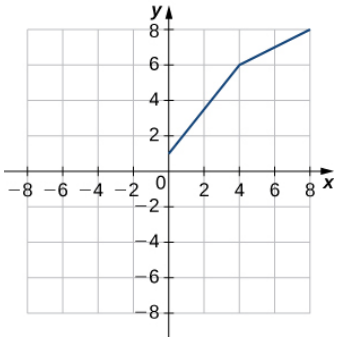
a. $2(h^2 + 6h + 12)$;

b. $(i)\ 25.22$ m/s, $(ii)\ 24.12$ m/s, $(iii)\ 24.01$ m/s, $(iv)\ 24$ m/s;

c. 24 m/s

38) $s(t) = \dfrac{16}{t^2} - \dfrac{4}{t}$

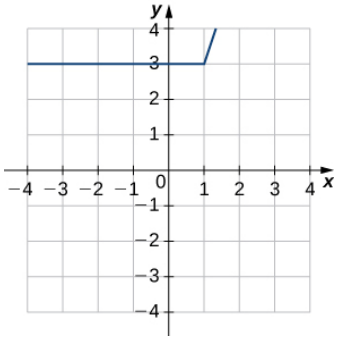
39) Use the following graph to evaluate a. $f'(1)$ and b. $f'(6)$.



Answer

a. 1.25; b. 0.5

40) Use the following graph to evaluate a. $f'(-3)$ and b. $f'(1.5)$.



For the following exercises, use the limit definition of derivative to show that the derivative does not exist at $x = a$ for each of the given functions.

41) $f(x) = x^{1/3}, \quad x = 0$

Answer

$$\lim_{x \to 0^-} \frac{x^{1/3} - 0}{x - 0} = \lim_{x \to 0^-} \frac{1}{x^{2/3}} = \infty$$

42) $f(x) = x^{2/3}, \quad x = 0$

43) $f(x) = \begin{cases} 1, & \text{if } x < 1 \\ x, & \text{if } x \geq 1 \end{cases}, \quad x = 1$

Answer

$$\lim_{x \to 1^-} \frac{1 - 1}{x - 1} = 0 \neq 1 = \lim_{x \to 1^+} \frac{x - 1}{x - 1}$$

44) $f(x) = \dfrac{|x|}{x}, \quad x = 0$

45) [T] The position in feet of a race car along a straight track after t seconds is modeled by the function $s(t) = 8t^2 - \frac{1}{16}t^3$.

a. Find the average velocity of the vehicle over the following time intervals to four decimal places:

i. $[4, 4.1]$

ii. $[4, 4.01]$

iii. $[4, 4.001]$

iv. $[4, 4.0001]$

b. Use a. to draw a conclusion about the instantaneous velocity of the vehicle at $t = 4$ seconds.

Answer

a. $(i) 61.7244 ft/s, \ (ii) 61.0725 ft/s, \ (iii) 61.0072 ft/s, \ (iv) 61.0007 ft/s$

b. At 4 seconds the race car is traveling at a rate/velocity of 61 ft/s.

46) [T] The distance in feet that a ball rolls down an incline is modeled by the function $s(t) = 14t^2$,

where t is seconds after the ball begins rolling.

a. Find the average velocity of the ball over the following time intervals:

i. [5, 5.1]

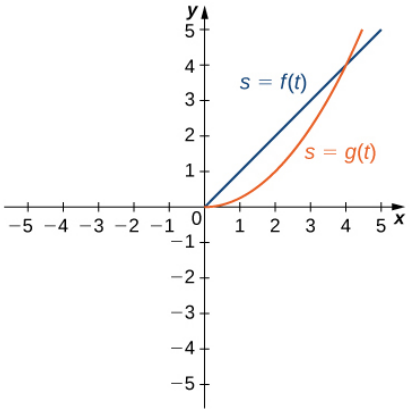
ii. [5, 5.01]

iii. [5, 5.001]

iv. [5, 5.0001]

b. Use the answers from a. to draw a conclusion about the instantaneous velocity of the ball at $t = 5$ seconds.

47) Two vehicles start out traveling side by side along a straight road. Their position functions, shown in the following graph, are given by $s = f(t)$ and $s = g(t)$, where s is measured in feet and t is measured in seconds.



a. Which vehicle has traveled farther at $t = 2$ seconds?

b. What is the approximate velocity of each vehicle at $t = 3$ seconds?

c. Which vehicle is traveling faster at $t = 4$ seconds?

d. What is true about the positions of the vehicles at $t = 4$ seconds?

Answer

a. The vehicle represented by $f(t)$, because it has traveled 2 feet, whereas $g(t)$ has traveled 1 foot.
b. The velocity of $f(t)$ is constant at 1 ft/s, while the velocity of $g(t)$ is approximately 2 ft/s.
c. The vehicle represented by $g(t)$, with a velocity of approximately 4 ft/s.
d. Both have traveled 4 feet in 4 seconds.

48) [T] The total cost $C(x)$, in hundreds of dollars, to produce x jars of mayonnaise is given by $C(x) = 0.000003x^3 + 4x + 300$.

a. Calculate the average cost per jar over the following intervals:

i. [100, 100.1]

ii. [100, 100.01]

iii. [100, 100.001]

iv. [100, 100.0001]

b. Use the answers from a. to estimate the average cost to produce 100 jars of mayonnaise.

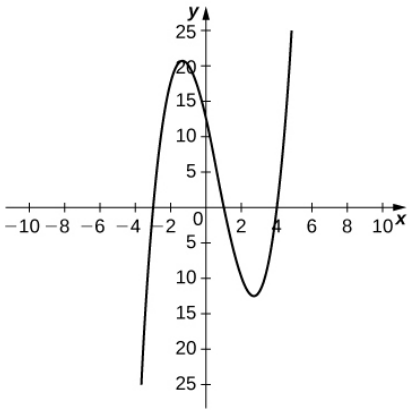
49) [T] For the function $f(x) = x^3 - 2x^2 - 11x + 12$, do the following.

a. Use a graphing calculator to graph f in an appropriate viewing window.

b. Use the ZOOM feature on the calculator to approximate the two values of $x = a$ for which $m_{tan} = f'(a) = 0$.

Answer

a.



b. $a \approx -1.361,\ 2.694$

50) [T] For the function $f(x) = \dfrac{x}{1+x^2}$, do the following.

a. Use a graphing calculator to graph f in an appropriate viewing window.

b. Use the ZOOM feature on the calculator to approximate the values of $x = a$ for which $m_{\tan} = f'(a) = 0$.

51) Suppose that $N(x)$ computes the number of gallons of gas used by a vehicle traveling x miles. Suppose the vehicle gets 30 mpg.

a. Find a mathematical expression for $N(x)$.

b. What is $N(100)$? Explain the physical meaning.

c. What is $N'(100)$? Explain the physical meaning.

Answer

a. $N(x) = \dfrac{x}{30}$

b. ~ 3.3 gallons. When the vehicle travels 100 miles, it has used 3.3 gallons of gas.

c. $\frac{1}{30}$. The rate of gas consumption in gallons per mile that the vehicle is achieving after having traveled 100 miles.

52) [T] For the function $f(x) = x^4 - 5x^2 + 4$, do the following.

a. Use a graphing calculator to graph f in an appropriate viewing window.

b. Use the *nDeriv* function, which numerically finds the derivative, on a graphing calculator to estimate $f'(-2),\ f'(-0.5),\ f'(1.7)$, and $f'(2.718)$.

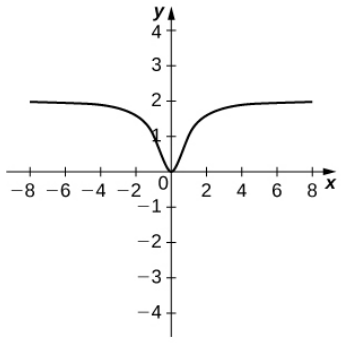
53) [T] For the function $f(x) = \dfrac{x^2}{x^2+1}$, do the following.

a. Use a graphing calculator to graph f in an appropriate viewing window.

b. Use the *nDeriv* function on a graphing calculator to find $f'(-4),\ f'(-2),\ f'(2)$, and $f'(4)$.

Answer

a.



b. $-0.028, -0.16, 0.16, 0.028$

3.1E: Exercises for Section 3.1 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by LibreTexts.

3.2: The Derivative as a Function

Learning Objectives

- Define the derivative function of a given function.
- Graph a derivative function from the graph of a given function.
- State the connection between derivatives and continuity.
- Describe three conditions for when a function does not have a derivative.
- Explain the meaning of a higher-order derivative.

As we have seen, the derivative of a function at a given point gives us the rate of change or slope of the tangent line to the function at that point. If we differentiate a position function at a given time, we obtain the velocity at that time. It seems reasonable to conclude that knowing the derivative of the function at every point would produce valuable information about the behavior of the function. However, the process of finding the derivative at even a handful of values using the techniques of the preceding section would quickly become quite tedious. In this section we define the derivative function and learn a process for finding it.

Derivative Functions

The derivative function gives the derivative of a function at each point in the domain of the original function for which the derivative is defined. We can formally define a derivative function as follows.

Definition: Derivative Function

Let f be a function. The **derivative function**, denoted by f', is the function whose domain consists of those values of x such that the following limit exists:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}. \tag{3.2.1}$$

A function $f(x)$ is said to be **differentiable at** a if $f'(a)$ exists. More generally, a function is said to be **differentiable on** S if it is differentiable at every point in an open set S, and a differentiable function is one in which $f'(x)$ exists on its domain.

In the next few examples we use Equation 3.2.1 to find the derivative of a function.

Example 3.2.1: Finding the Derivative of a Square-Root Function

Find the derivative of $f(x) = \sqrt{x}$.

Solution

Start directly with the definition of the derivative function.

Substitute $f(x+h) = \sqrt{x+h}$ and $f(x) = \sqrt{x}$ into $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$.

$f'(x) = \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$	
$= \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$	Multiply numerator and denominator by $\sqrt{x+h} + \sqrt{x}$ without distributing in the denominator.
$= \lim_{h \to 0} \frac{h}{h\left(\sqrt{x+h} + \sqrt{x}\right)}$	Multiply the numerators and simplify.
$= \lim_{h \to 0} \frac{1}{\left(\sqrt{x+h} + \sqrt{x}\right)}$	Cancel the h.
$= \frac{1}{2\sqrt{x}}$	Evaluate the limit

Example 3.2.2: Finding the Derivative of a Quadratic Function

Find the derivative of the function $f(x) = x^2 - 2x$.

Solution

Follow the same procedure here, but without having to multiply by the conjugate.

Substitute $f(x+h) = (x+h)^2 - 2(x+h)$ and $f(x) = x^2 - 2x$ into $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$.

$f'(x) = \lim_{h \to 0} \dfrac{((x+h)^2 - 2(x+h)) - (x^2 - 2x)}{h}$	
$= \lim_{h \to 0} \dfrac{x^2 + 2xh + h^2 - 2x - 2h - x^2 + 2x}{h}$	Expand $(x+h)^2 - 2(x+h)$.
$= \lim_{h \to 0} \dfrac{2xh - 2h + h^2}{h}$	Simplify
$= \lim_{h \to 0} \dfrac{h(2x - 2 + h)}{h}$	Factor out h from the numerator
$= \lim_{h \to 0} (2x - 2 + h)$	Cancel the common factor of h
$= 2x - 2$	Evaluate the limit

Exercise 3.2.1

Find the derivative of $f(x) = x^2$.

Hint

Use Equation 3.2.1 and follow the example.

Answer

$f'(x) = 2x$

We use a variety of different notations to express the derivative of a function. In Example 3.2.2 we showed that if $f(x) = x^2 - 2x$, then $f'(x) = 2x - 2$. If we had expressed this function in the form $y = x^2 - 2x$, we could have expressed the derivative as $y' = 2x - 2$ or $\dfrac{dy}{dx} = 2x - 2$. We could have conveyed the same information by writing $\dfrac{d}{dx}(x^2 - 2x) = 2x - 2$. Thus, for the function $y = f(x)$, each of the following notations represents the derivative of $f(x)$:

$$f'(x), \quad \frac{dy}{dx}, \quad y', \quad \frac{d}{dx}\big(f(x)\big).$$

In place of $f'(a)$ we may also use $\dfrac{dy}{dx}\Big|_{x=a}$. Use of the $\dfrac{dy}{dx}$ notation (called Leibniz notation) is quite common in engineering and physics. To understand this notation better, recall that the derivative of a function at a point is the limit of the slopes of secant lines as the secant lines approach the tangent line. The slopes of these secant lines are often expressed in the form $\dfrac{\Delta y}{\Delta x}$ where Δy is the difference in the y values corresponding to the difference in the x values, which are expressed as Δx (Figure 3.2.1). Thus the derivative, which can be thought of as the instantaneous rate of change of y with respect to x, is expressed as

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}.$$

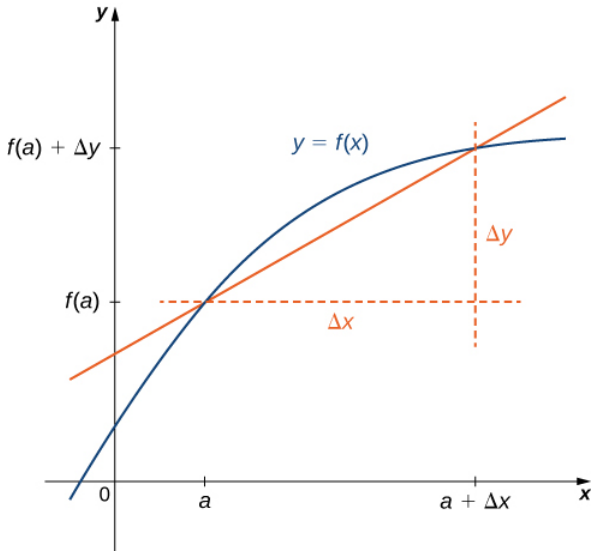


Figure 3.2.1: The derivative is expressed as $\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$.

Graphing a Derivative

We have already discussed how to graph a function, so given the equation of a function or the equation of a derivative function, we could graph it. Given both, we would expect to see a correspondence between the graphs of these two functions, since $f'(x)$ gives the rate of change of a function $f(x)$ (or slope of the tangent line to $f(x)$).

In Example 3.2.1, we found that for $f(x) = \sqrt{x}$, $f'(x) = \frac{1}{2\sqrt{x}}$. If we graph these functions on the same axes, as in Figure 3.2.2, we can use the graphs to understand the relationship between these two functions. First, we notice that $f(x)$ is increasing over its entire domain, which means that the slopes of its tangent lines at all points are positive. Consequently, we expect $f'(x) > 0$ for all values of x in its domain. Furthermore, as x increases, the slopes of the tangent lines to $f(x)$ are decreasing and we expect to see a corresponding decrease in $f'(x)$. We also observe that $f(0)$ is undefined and that $\lim_{x \to 0^+} f'(x) = +\infty$, corresponding to a vertical tangent to $f(x)$ at 0.

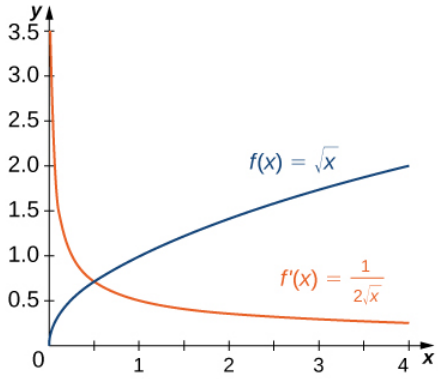


Figure 3.2.2: The derivative $f'(x)$ is positive everywhere because the function $f(x)$ is increasing.

In Example 3.2.2, we found that for $f(x) = x^2 - 2x$, $f'(x) = 2x - 2$. The graphs of these functions are shown in Figure 3.2.3. Observe that $f(x)$ is decreasing for $x < 1$. For these same values of x, $f'(x) < 0$. For values of $x > 1$, $f(x)$ is increasing and $f'(x) > 0$. Also, $f(x)$ has a horizontal tangent at $x = 1$ and $f'(1) = 0$.

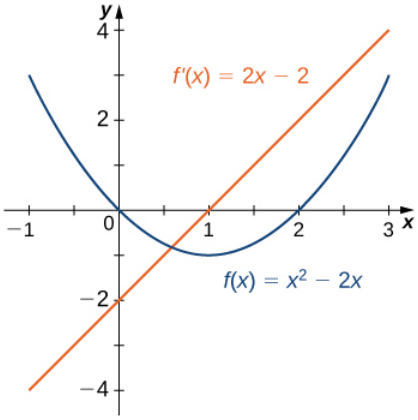
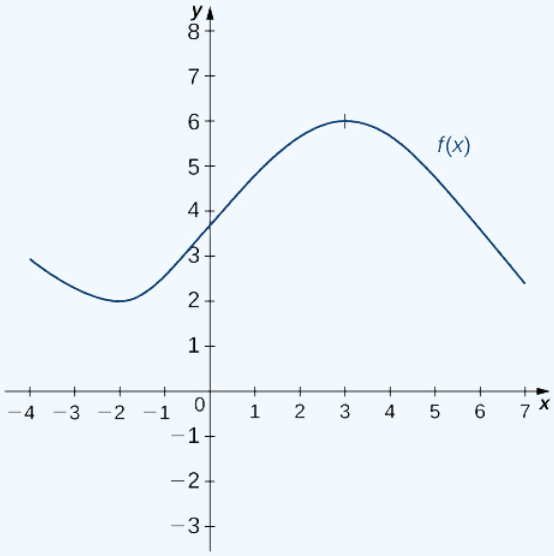


Figure 3.2.3: The derivative $f'(x) < 0$ where the function $f(x)$ is decreasing and $f'(x) > 0$ where $f(x)$ is increasing. The derivative is zero where the function has a horizontal tangent

Example 3.2.3: Sketching a Derivative Using a Function

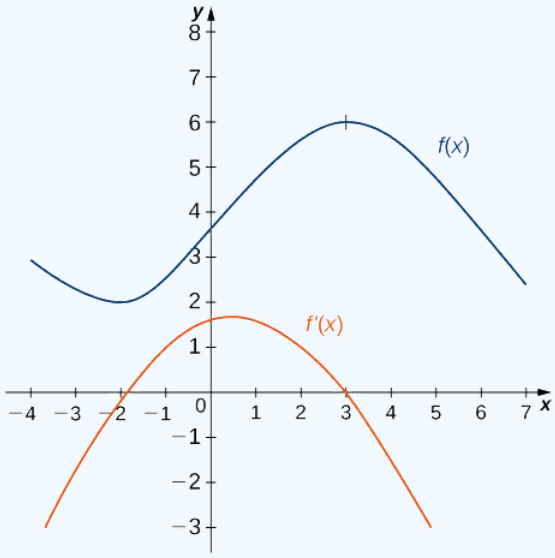
Use the following graph of $f(x)$ to sketch a graph of $f'(x)$.



Solution

The solution is shown in the following graph. Observe that $f(x)$ is increasing and $f'(x) > 0$ on $(-2, 3)$. Also, $f(x)$ is decreasing and $f'(x) < 0$ on $(-\infty, -2)$ and on $(3, +\infty)$. Also note that $f(x)$ has horizontal tangents at -2 and 3, and $f'(-2) = 0$ and $f'(3) = 0$.

https://math.libretexts.org/@go/page/2491



Exercise 3.2.2

Sketch the graph of $f(x) = x^2 - 4$. On what interval is the graph of $f'(x)$ above the x-axis?

Hint

The graph of $f'(x)$ is positive where $f(x)$ is increasing.

Answer

$(0, +\infty)$

Derivatives and Continuity

Now that we can graph a derivative, let's examine the behavior of the graphs. First, we consider the relationship between differentiability and continuity. We will see that if a function is differentiable at a point, it must be continuous there; however, a function that is continuous at a point need not be differentiable at that point. In fact, a function may be continuous at a point and fail to be differentiable at the point for one of several reasons.

Differentiability Implies Continuity

Let $f(x)$ be a function and a be in its domain. If $f(x)$ is differentiable at a, then f is continuous at a.

Proof

If $f(x)$ is differentiable at a, then $f'(a)$ exists and, if we let $h = x - a$, we have $x = a + h$, and as $h = x - a \to 0$, we can see that $x \to a$.

Then

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

can be rewritten as

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

We want to show that $f(x)$ is continuous at a by showing that $\lim_{x \to a} f(x) = f(a)$. Thus,

$$\begin{aligned}\lim_{x\to a} f(x) &= \lim_{x\to a}\ \big(f(x)-f(a)+f(a)\big)\\ &= \lim_{x\to a}\left(\frac{f(x)-f(a)}{x-a}\cdot(x-a)+f(a)\right) \qquad \text{Multiply and divide } (f(x)-f(a)) \text{ by } x-a.\\ &= \left(\lim_{x\to a}\frac{f(x)-f(a)}{x-a}\right)\cdot\left(\lim_{x\to a}(x-a)\right)+\lim_{x\to a} f(a)\\ &= f'(a)\cdot 0+f(a)\\ &= f(a).\end{aligned}$$

Therefore, since $f(a)$ is defined and $\lim_{x\to a} f(x)=f(a)$, we conclude that f is continuous at a.

□

We have just proven that differentiability implies continuity, but now we consider whether continuity implies differentiability. To determine an answer to this question, we examine the function $f(x)=|x|$. This function is continuous everywhere; however, $f'(0)$ is undefined. This observation leads us to believe that continuity does not imply differentiability. Let's explore further. For $f(x)=|x|$,

$$f'(0)=\lim_{x\to 0}\frac{f(x)-f(0)}{x-0}=\lim_{x\to 0}\frac{|x|-|0|}{x-0}=\lim_{x\to 0}\frac{|x|}{x}.$$

This limit does not exist because

$$\lim_{x\to 0^-}\frac{|x|}{x}=-1 \text{ and } \lim_{x\to 0^+}\frac{|x|}{x}=1.$$

See Figure 3.2.4.

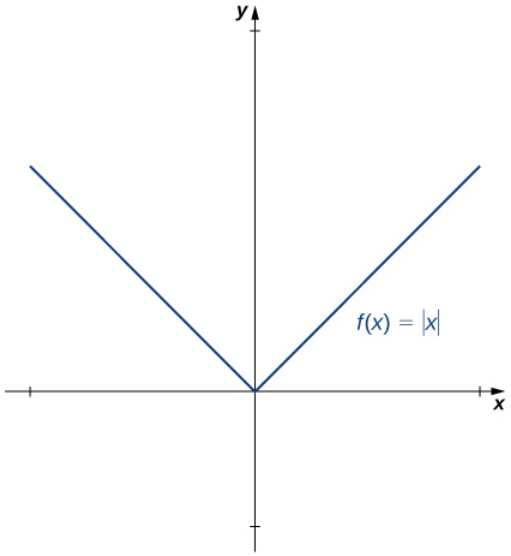


Figure 3.2.4: The function $f(x)=|x|$ is continuous at 0 but is not differentiable at 0.

Let's consider some additional situations in which a continuous function fails to be differentiable. Consider the function $f(x)=\sqrt[3]{x}$:

$$f'(0)=\lim_{x\to 0}\frac{\sqrt[3]{x}-0}{x-0}=\lim_{x\to 0}\frac{1}{\sqrt[3]{x^2}}=+\infty.$$

Thus $f'(0)$ does not exist. A quick look at the graph of $f(x)=\sqrt[3]{x}$ clarifies the situation. The function has a vertical tangent line at 0 (Figure 3.2.5).

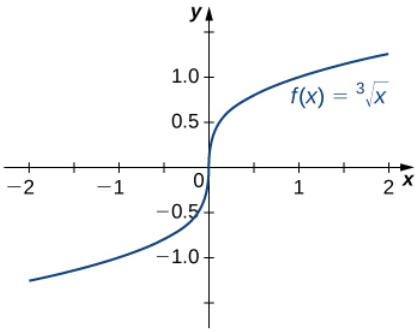


Figure 3.2.5: The function $f(x) = \sqrt[3]{x}$ has a vertical tangent at $x = 0$. It is continuous at 0 but is not differentiable at 0.

The function $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ also has a derivative that exhibits interesting behavior at 0.

We see that

$$f'(0) = \lim_{x \to 0} \frac{x \sin(1/x) - 0}{x - 0} = \lim_{x \to 0} \sin\left(\frac{1}{x}\right).$$

This limit does not exist, essentially because the slopes of the secant lines continuously change direction as they approach zero (Figure 3.2.6).

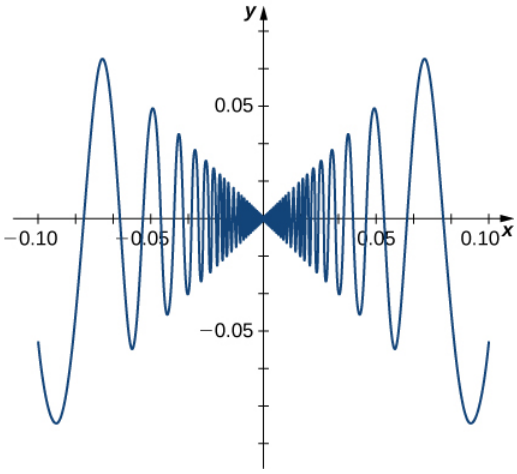


Figure 3.2.6: The function $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ is not differentiable at 0.

In summary:

1. We observe that if a function is not continuous, it cannot be differentiable, since every differentiable function must be continuous. However, if a function is continuous, it may still fail to be differentiable.
2. We saw that $f(x) = |x|$ failed to be differentiable at 0 because the limit of the slopes of the tangent lines on the left and right were not the same. Visually, this resulted in a sharp corner on the graph of the function at 0. From this we conclude that in order to be differentiable at a point, a function must be “smooth” at that point.
3. As we saw in the example of $f(x) = \sqrt[3]{x}$, a function fails to be differentiable at a point where there is a vertical tangent line.
4. As we saw with $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ a function may fail to be differentiable at a point in more complicated ways as well.

Example 3.2.4: A Piecewise Function that is Continuous and Differentiable

A toy company wants to design a track for a toy car that starts out along a parabolic curve and then converts to a straight line (Figure 3.2.7). The function that describes the track is to have the form $f(x) = \begin{cases} \frac{1}{10}x^2 + bx + c, & \text{if } x < -10 \\ -\frac{1}{4}x + \frac{5}{2}, & \text{if } x \geq -10 \end{cases}$ where x and $f(x)$ are in inches. For the car to move smoothly along the track, the function $f(x)$ must be both continuous and differentiable at -10. Find values of b and c that make $f(x)$ both continuous and differentiable.

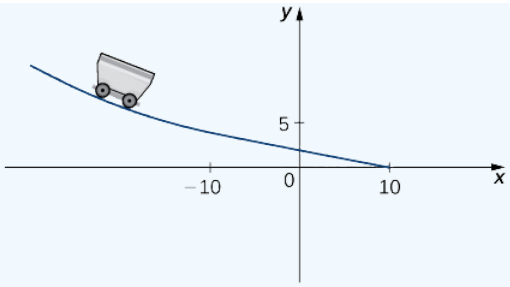


Figure 3.2.7: For the car to move smoothly along the track, the function must be both continuous and differentiable.

Solution

For the function to be continuous at $x = -10$, $\lim\limits_{x \to 10^-} f(x) = f(-10)$. Thus, since

$$\lim_{x \to -10^-} f(x) = \frac{1}{10}(-10)^2 - 10b + c = 10 - 10b + c$$

and $f(-10) = 5$, we must have $10 - 10b + c = 5$. Equivalently, we have $c = 10b - 5$.

For the function to be differentiable at -10,

$$f'(10) = \lim_{x \to -10} \frac{f(x) - f(-10)}{x + 10}$$

must exist. Since $f(x)$ is defined using different rules on the right and the left, we must evaluate this limit from the right and the left and then set them equal to each other:

$$\begin{aligned}
\lim_{x \to -10^-} \frac{f(x) - f(-10)}{x + 10} &= \lim_{x \to -10^-} \frac{\frac{1}{10}x^2 + bx + c - 5}{x + 10} \\
&= \lim_{x \to -10^-} \frac{\frac{1}{10}x^2 + bx + (10b - 5) - 5}{x + 10} && \text{Substitute } c = 10b - 5. \\
&= \lim_{x \to -10^-} \frac{x^2 - 100 + 10bx + 100b}{10(x + 10)} \\
&= \lim_{x \to -10^-} \frac{(x + 10)(x - 10 + 10b)}{10(x + 10)} && \text{Factor by grouping} \\
&= b - 2
\end{aligned}$$

We also have

$$\begin{aligned}
\lim_{x \to -10^+} \frac{f(x) - f(-10)}{x + 10} &= \lim_{x \to -10^+} \frac{-\frac{1}{4}x + \frac{5}{2} - 5}{x + 10} \\
&= \lim_{x \to -10^+} \frac{-(x + 10)}{4(x + 10)} \\
&= -\frac{1}{4}
\end{aligned}$$

This gives us $b - 2 = -\frac{1}{4}$. Thus $b = \frac{7}{4}$ and $c = 10(\frac{7}{4}) - 5 = \frac{25}{2}$.

Exercise 3.2.3

Find values of a and b that make $f(x) = \begin{cases} ax + b, & \text{if } x < 3 \\ x^2, & \text{if } x \geq 3 \end{cases}$ both continuous and differentiable at 3.

Hint

Use Example 3.2.4 as a guide.

Answer

$a = 6$ and $b = -9$

Higher-Order Derivatives

The derivative of a function is itself a function, so we can find the derivative of a derivative. For example, the derivative of a position function is the rate of change of position, or velocity. The derivative of velocity is the rate of change of velocity, which is acceleration. The new function obtained by differentiating the derivative is called the second derivative. Furthermore, we can continue to take derivatives to obtain the third derivative, fourth derivative, and so on. Collectively, these are referred to as **higher-order derivatives**. The notation for the higher-order derivatives of $y = f(x)$ can be expressed in any of the following forms:

$$f''(x),\ f'''(x),\ f^{(4)}(x),\ \ldots\ ,\ f^{(n)}(x)$$

$$y''(x),\ y'''(x),\ y^{(4)}(x),\ \ldots\ ,\ y^{(n)}(x)$$

$$\frac{d^2y}{dx^2},\ \frac{d^3y}{dy^3},\ \frac{d^4y}{dy^4},\ \ldots\ ,\ \frac{d^ny}{dy^n}.$$

It is interesting to note that the notation for $\frac{d^2y}{dx^2}$ may be viewed as an attempt to express $\frac{d}{dx}\left(\frac{dy}{dx}\right)$ more compactly.

Analogously, $\frac{d}{dx}\left(\frac{d}{dx}\left(\frac{dy}{dx}\right)\right) = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = \frac{d^3y}{dx^3}$.

Example 3.2.5: Finding a Second Derivative

For $f(x) = 2x^2 - 3x + 1$, find $f''(x)$.

Solution

First find $f'(x)$.

Substitute $f(x) = 2x^2 - 3x + 1$ and $f(x+h) = 2(x+h)^2 - 3(x+h) + 1$ into $f'(x) = \lim_{h\to 0}\frac{f(x+h) - f(x)}{h}$.

$f'(x) = \lim_{h\to 0}\frac{(2(x+h)^2 - 3(x+h) + 1) - (2x^2 - 3x + 1)}{h}$	
$= \lim_{h\to 0}\frac{4xh + 2h^2 - 3h}{h}$	Simplify the numerator.
$= \lim_{h\to 0}(4x + 2h - 3)$	Factor out the h in the numerator and cancel with the h in the denominator.
$= 4x - 3$	Take the limit.

Next, find $f''(x)$ by taking the derivative of $f'(x) = 4x - 3$.

$f''(x) = \lim_{h\to 0}\frac{f'(x+h) - f'(x)}{h}$	Use $f'(x) = \lim_{h\to 0}\frac{f(x+h) - f(x)}{h}$ with $f'(x)$ in place of $f(x)$.
$= \lim_{h\to 0}\frac{(4(x+h) - 3) - (4x - 3)}{h}$	Substitute $f'(x+h) = 4(x+h) - 3$ and $f'(x) = 4x - 3$.
$= \lim_{h\to 0} 4$	Simplify.
$= 4$	Take the limit.

Exercise 3.2.4

Find $f''(x)$ for $f(x) = x^2$.

Hint

We found $f'(x) = 2x$ in a previous checkpoint. Use Equation 3.2.1 to find the derivative of $f'(x)$

Answer

$f''(x) = 2$

Example 3.2.6: Finding Acceleration

The position of a particle along a coordinate axis at time t (in seconds) is given by $s(t) = 3t^2 - 4t + 1$ (in meters). Find the function that describes its acceleration at time t.

Solution

Since $v(t) = s'(t)$ and $a(t) = v'(t) = s''(t)$, we begin by finding the derivative of $s(t)$:

$$\begin{aligned} s'(t) &= \lim_{h \to 0} \frac{s(t+h) - s(t)}{h} \\ &= \lim_{h \to 0} \frac{3(t+h)^2 - 4(t+h) + 1 - (3t^2 - 4t + 1)}{h} \\ &= 6t - 4. \end{aligned}$$

Next,

$$\begin{aligned} s''(t) &= \lim_{h \to 0} \frac{s'(t+h) - s'(t)}{h} \\ &= \lim_{h \to 0} \frac{6(t+h) - 4 - (6t - 4)}{h} \\ &= 6. \end{aligned}$$

Thus, $a = 6 \text{ m/s}^2$.

Exercise 3.2.5

For $s(t) = t^3$, find $a(t)$.

Hint

Use Example 3.2.6 as a guide.

Answer

$a(t) = 6t$

Key Concepts

- The derivative of a function $f(x)$ is the function whose value at x is $f'(x)$.
- The graph of a derivative of a function $f(x)$ is related to the graph of $f(x)$. Where $f(x)$ has a tangent line with positive slope, $f'(x) > 0$. Where $f(x)$ has a tangent line with negative slope, $f'(x) < 0$. Where $f(x)$ has a horizontal tangent line, $f'(x) = 0$.
- If a function is differentiable at a point, then it is continuous at that point. A function is not differentiable at a point if it is not continuous at the point, if it has a vertical tangent line at the point, or if the graph has a sharp corner or cusp.
- Higher-order derivatives are derivatives of derivatives, from the second derivative to the nth derivative.

Key Equations

- **The derivative function**

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Glossary

derivative function

gives the derivative of a function at each point in the domain of the original function for which the derivative is defined

differentiable at a

a function for which $f'(a)$ exists is differentiable at a

differentiable on S

a function for which $f'(x)$ exists for each x in the open set S is differentiable on S

differentiable function

a function for which $f'(x)$ exists is a differentiable function

higher-order derivative

a derivative of a derivative, from the second derivative to the nth derivative, is called a higher-order derivative

- Gilbert Strang (MIT) and Edwin "Jed" Herman (Harvey Mudd) with many contributing authors. This content by OpenStax is licensed with a CC-BY-SA-NC 4.0 license. Download for free at http://cnx.org.
- Paul Seeburger (Monroe Community College) added explanation of the alternative definition of the derivative used in the proof of that differentiability implies continuity.

This page titled 3.2: The Derivative as a Function is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by Gilbert Strang & Edwin "Jed" Herman (OpenStax) via source content that was edited to the style and standards of the LibreTexts platform.

3.2E: EXERCISES FOR SECTION 3.2

For exercises 1 - 10, use the definition of a derivative to find $f'(x)$.

1) $f(x) = 6$

2) $f(x) = 2 - 3x$

Answer

$f'(x) = -3$

3) $f(x) = \frac{2x}{7} + 1$

4) $f(x) = 4x^2$

Answer

$f'(x) = 8x$

5) $f(x) = 5x - x^2$

6) $f(x) = \sqrt{2x}$

Answer

$f'(x) = \frac{1}{\sqrt{2x}}$

7) $f(x) = \sqrt{x-6}$

8) $f(x) = \frac{9}{x}$

Answer

$f'(x) = \frac{-9}{x^2}$

9) $f(x) = x + \frac{1}{x}$

10) $f(x) = \frac{1}{\sqrt{x}}$

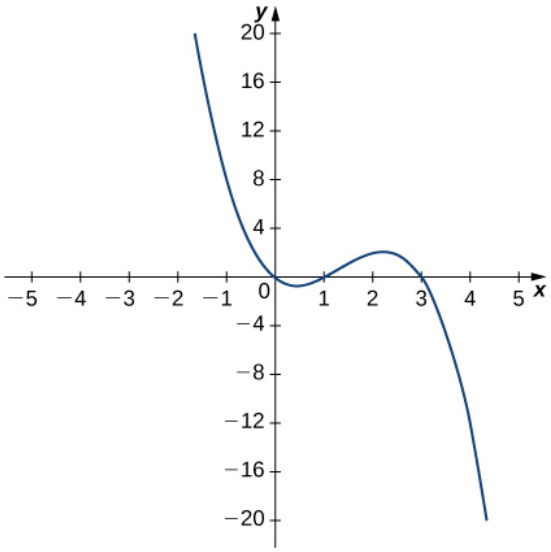
Answer

$f'(x) = \frac{-1}{2x^{3/2}}$

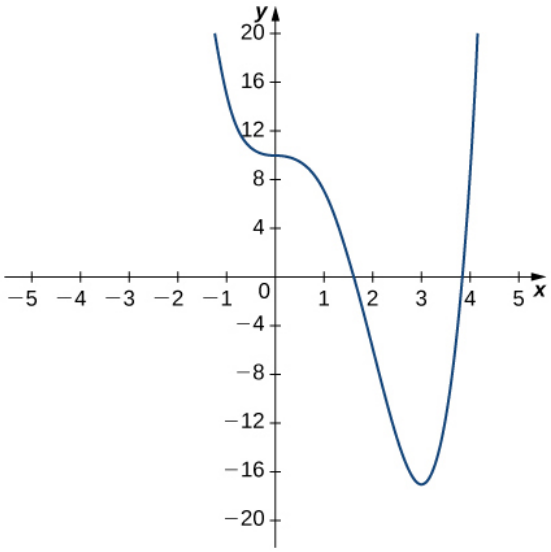
For the exercises 11 - 14, use the graph of $y = f(x)$ to sketch the graph of its derivative $f'(x)$.

11)

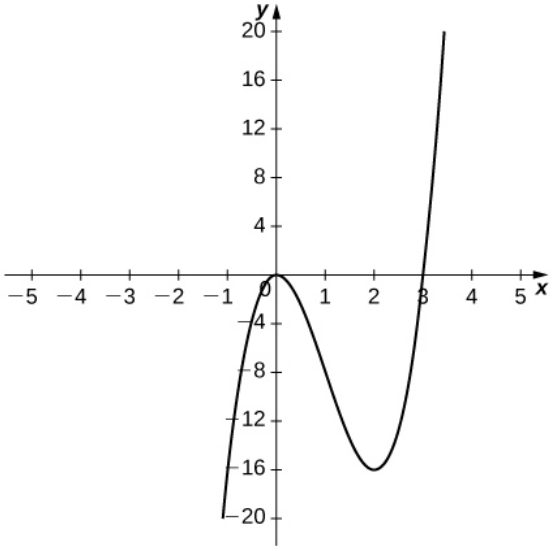
https://math.libretexts.org/@go/page/50903



12)

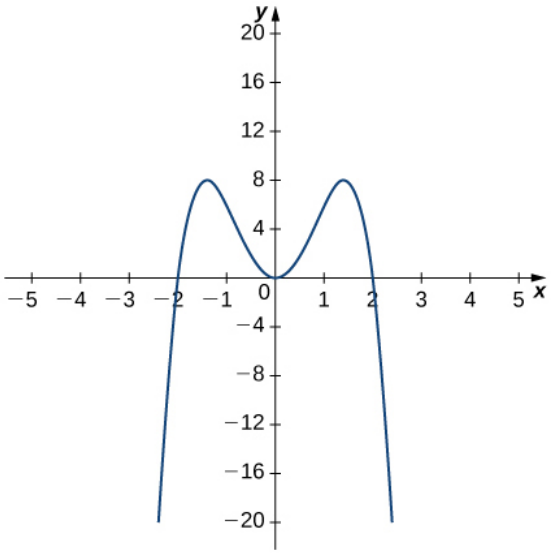


Answer

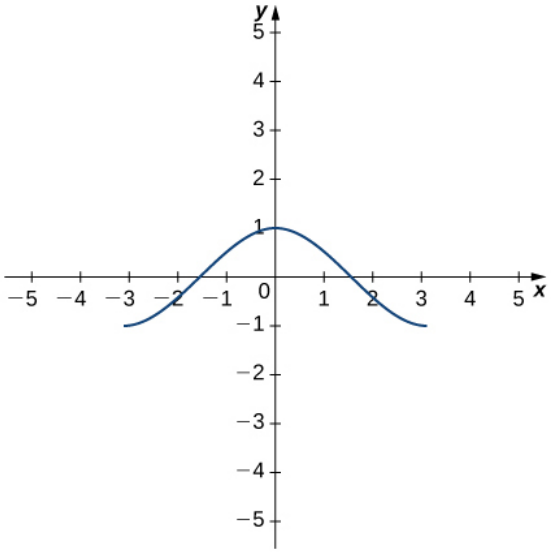


https://math.libretexts.org/@go/page/50903

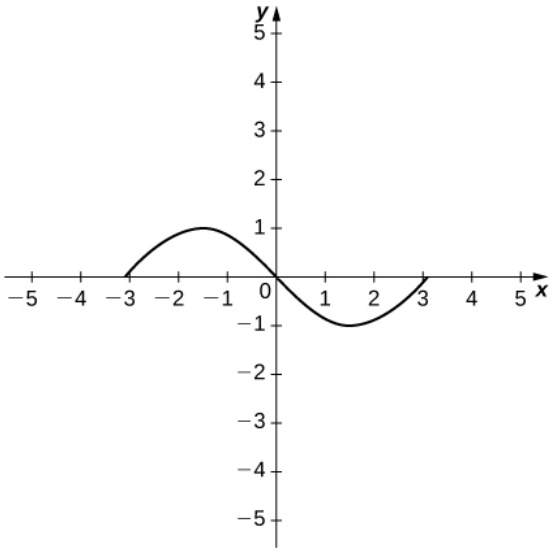
13)


y
x
20
16
12
8
4
−5 −4 −3 −2 −1 0 1 2 3 4 5
−4
−8
−12
−16
−20

14)


y
x
5
4
3
2
1
−5 −4 −3 −2 −1 0 1 2 3 4 5
−1
−2
−3
−4
−5

Answer



For exercises 15 - 20, the given limit represents the derivative of a function $y = f(x)$ at $x = a$. Find $f(x)$ and a.

15) $\lim\limits_{h \to 0} \dfrac{(1+h)^{2/3} - 1}{h}$

16) $\lim\limits_{h \to 0} \dfrac{[3(2+h)^2 + 2] - 14}{h}$

Answer

$f(x) = 3x^2 + 2, \quad a = 2$

17) $\lim\limits_{h \to 0} \dfrac{\cos(\pi + h) + 1}{h}$

18) $\lim\limits_{h \to 0} \dfrac{(2+h)^4 - 16}{h}$

Answer

$f(x) = x^4, \quad a = 2$

19) $\lim\limits_{h \to 0} \dfrac{[2(3+h)^2 - (3+h)] - 15}{h}$

20) $\lim\limits_{h \to 0} \dfrac{e^h - 1}{h}$

Answer

$f(x) = e^x, \quad a = 0$

For the functions in exercises 21 - 24,

a. sketch the graph and

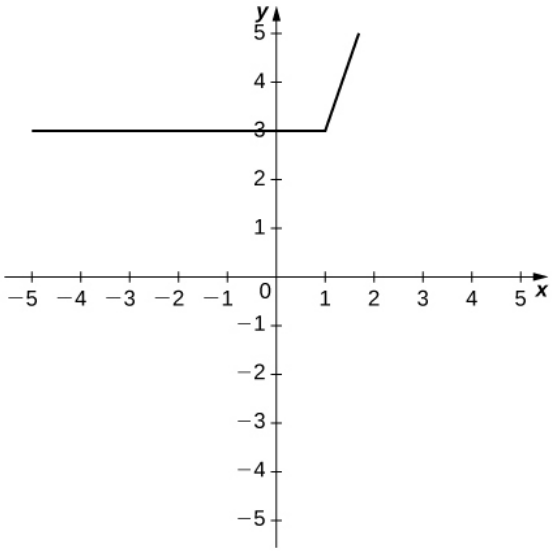
b. use the definition of a derivative to show that the function is not differentiable at $x = 1$.

21) $f(x) = \begin{cases} 2\sqrt{x}, & \text{if } 0 \le x \le 1 \\ 3x - 1, & \text{if } x > 1 \end{cases}$

22) $f(x) = \begin{cases} 3, & \text{if } x < 1 \\ 3x, & \text{if } x \ge 1 \end{cases}$

Answer

a.



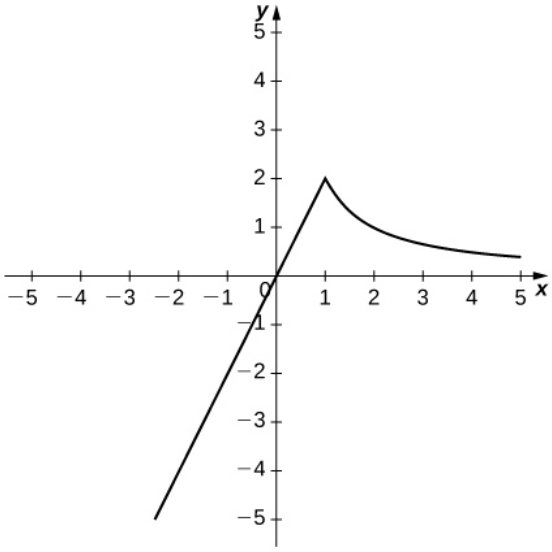
b. $\lim_{h\to 1^-} \frac{3-3}{h} \neq \lim_{h\to 1^+} \frac{3h}{h}$

23) $f(x) = \begin{cases} -x^2+2, & \text{if } x \leq 1 \\ x, & \text{if } x > 1 \end{cases}$

24) $f(x) = \begin{cases} 2x, & \text{if } x \leq 1 \\ \frac{2}{x}, & \text{if } x > 1 \end{cases}$

Answer

a.



b. $\lim_{h\to 1^-} \frac{2h}{h} \neq \lim_{h\to 1^+} \frac{\frac{2}{x+h} - 2x}{h}$.

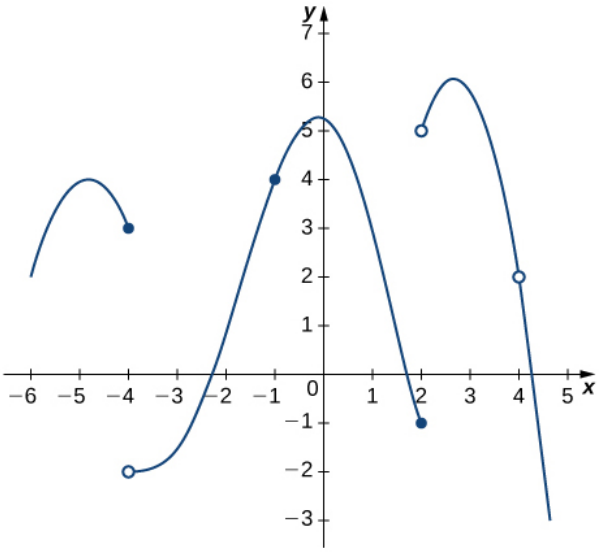
For the graphs in exercises 25 - 26,

a. determine for which values of $x = a$ the $\lim_{x\to a} f(x)$ exists but f is not continuous at $x = a$, and

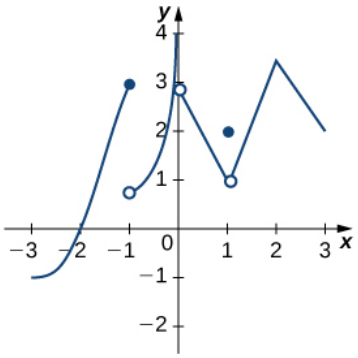
b. determine for which values of $x = a$ the function is continuous but not differentiable at $x = a$.

25)

https://math.libretexts.org/@go/page/50903



26)

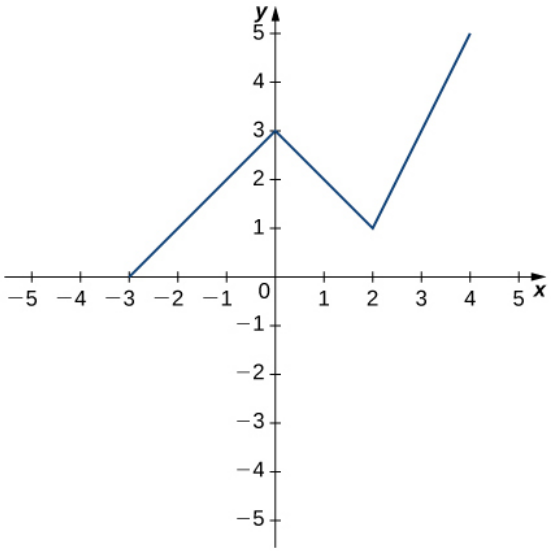


Answer

a. $x = 1$,

b. $x = 2$

27) Use the graph to evaluate a. $f'(-0.5)$, b. $f'(0)$, c. $f'(1)$, d. $f'(2)$, and e. $f'(3)$, if it exists.



For the functions in exercises 28 - 30, use $f''(x) = \lim_{h \to 0} \frac{f'(x+h) - f'(x)}{h}$ to find $f''(x)$.

https://math.libretexts.org/@go/page/50903

28) $f(x) = 2 - 3x$

Answer

$f''(x) = 0$

29) $f(x) = 4x^2$

30) $f(x) = x + \frac{1}{x}$

Answer

$f''(x) = \frac{2}{x^3}$

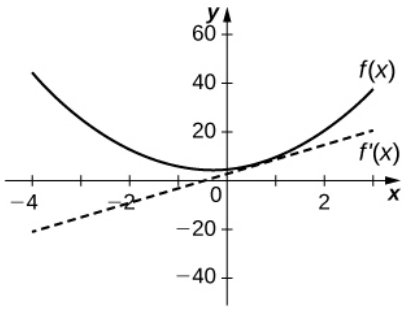
For exercises 31 - 36, use a calculator to graph $f(x)$. Determine the function $f'(x)$, then use a calculator to graph $f'(x)$.

31) [T] $f(x) = -\frac{5}{x}$

32) [T] $f(x) = 3x^2 + 2x + 4$.

Answer

$f'(x) = 6x + 2$

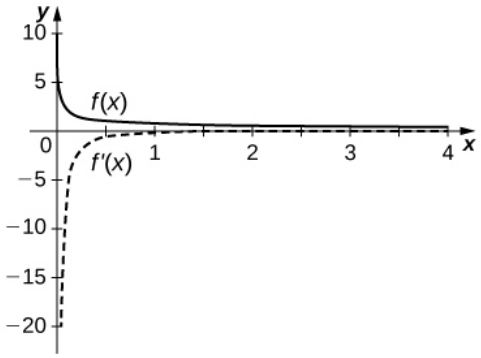


33) [T] $f(x) = \sqrt{x} + 3x$

34) [T] $f(x) = \frac{1}{\sqrt{2x}}$

Answer

$f'(x) = -\frac{1}{(2x)^{3/2}}$



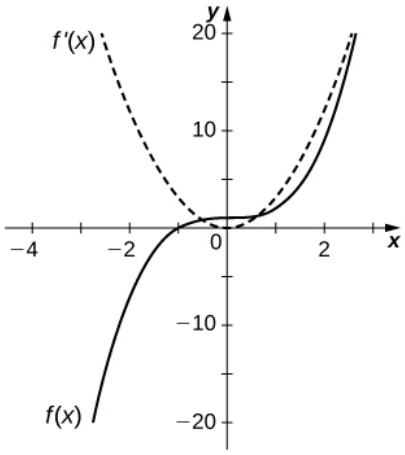
35) [T] $f(x) = 1 + x + \frac{1}{x}$

36) [T] $f(x) = x^3 + 1$

Answer

$f'(x) = 3x^2$

https://math.libretexts.org/@go/page/50903



For exercises 37 - 42, describe what the two expressions represent in terms of each of the given situations. Be sure to include units.

a. $\frac{f(x+h)-f(x)}{h}$

b. $f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

37) $P(x)$ denotes the population of a city at time x in years.

38) $C(x)$ denotes the total amount of money (in thousands of dollars) spent on concessions by x customers at an amusement park.

Answer

a. Average rate at which customers spent on concessions in thousands per customer.

b. Rate (in thousands per customer) at which x customers spent money on concessions in thousands per customer.

39) $R(x)$ denotes the total cost (in thousands of dollars) of manufacturing x clock radios

40) $g(x)$ denotes the grade (in percentage points) received on a test, given x hours of studying.

Answer

a. Average grade received on the test with an average study time between two values.

b. Rate (in percentage points per hour) at which the grade on the test increased or decreased for a given average study time of x hours.

41) $B(x)$denotes the cost (in dollars) of a sociology textbook at university bookstores in the United States in x years since 1990.

42) $p(x)$ denotes atmospheric pressure at an altitude of x feet.

Answer

a. Average change of atmospheric pressure between two different altitudes.

b. Rate (torr per foot) at which atmospheric pressure is increasing or decreasing at x feet.

43) Sketch the graph of a function $y=f(x)$ with all of the following properties:

a. $f'(x)>0$ for $-2\le x<1$

b. $f'(2)=0$

c. $f'(x)>0$ for $x>2$

d. $f(2)=2$ and $f(0)=1$

e. $\lim_{x\to-\infty}f(x)=0$ and $\lim_{x\to\infty}f(x)=\infty$

f. $f'(1)$ does not exist.

44) Suppose temperature T in degrees Fahrenheit at a height x in feet above the ground is given by $y=T(x)$.

a. Give a physical interpretation, with units, of $T'(x)$.

b. If we know that $T'(1000)=-0.1$, explain the physical meaning.

Answer

a. The rate (in degrees per foot) at which temperature is increasing or decreasing for a given height x.

b. The rate of change of temperature as altitude changes at 1000 feet is -0.1 degrees per foot.

45) Suppose the total profit of a company is $y = P(x)$ thousand dollars when x units of an item are sold.

a. What does $\frac{P(b) - P(a)}{b - a}$ for $0 < a < b$ measure, and what are the units?

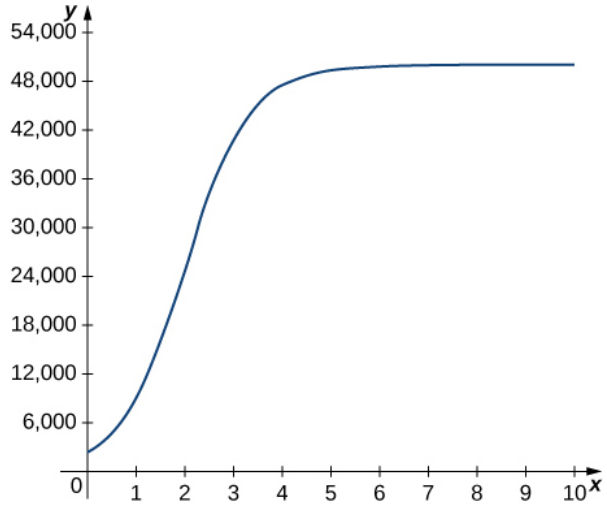
b. What does $P'(x)$ measure, and what are the units?

c. Suppose that $P'(30) = 5$, what is the approximate change in profit if the number of items sold increases from 30 to 31?

46) The graph in the following figure models the number of people $N(t)$ who have come down with the flu t weeks after its initial outbreak in a town with a population of 50,000 citizens.

a. Describe what $N'(t)$ represents and how it behaves as t increases.

b. What does the derivative tell us about how this town is affected by the flu outbreak?



Answer

a. The rate at which the number of people who have come down with the flu is changing t weeks after the initial outbreak.

b. The rate is increasing sharply up to the third week, at which point it slows down and then becomes constant.

For exercises 47 - 52, use the following table, which shows the height h of the Saturn V rocket for the Apollo 11 mission t seconds after launch.

Time (seconds)	Height (meters)
0	0
1	2
2	4
3	13
4	25
5	32

47) What is the physical meaning of $h'(t)$? What are the units?

48) [T] Construct a table of values for $h'(t)$ and graph both $h(t)$ and $h'(t)$ on the same graph. (Hint: for interior points, estimate both the left limit and right limit and average them.)

Answer

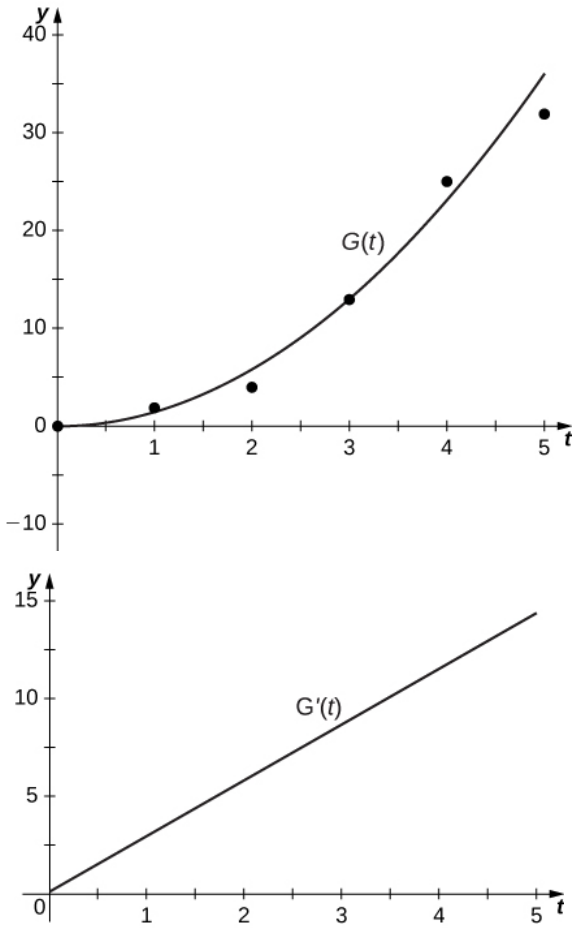
Time (seconds)	$h'(t)$ (m/s)
0	2
1	2
2	5.5
3	10.5
4	9.5
5	7

49) [T] The best linear fit to the data is given by $H(t) = 7.229t - 4.905$, where H is the height of the rocket (in meters) and t is the time elapsed since takeoff. From this equation, determine $H'(t)$. Graph $H(t$ with the given data and, on a separate coordinate plane, graph $H'(t)$.

50) [T] The best quadratic fit to the data is given by $G(t) = 1.429t^2 + 0.0857t - 0.1429$, where G is the height of the rocket (in meters) and t is the time elapsed since takeoff. From this equation, determine $G'(t)$. Graph $G(t)$ with the given data and, on a separate coordinate plane, graph $G'(t)$.

Answer

$G'(t) = 2.858t + 0.0857$



51) [T] The best cubic fit to the data is given by $F(t) = 0.2037t^3 + 2.956t^2 - 2.705t + 0.4683$, where F is the height of the rocket (in m) and t is the time elapsed since take off. From this equation, determine $F'(t)$. Graph $F(t)$ with the given data and, on a separate coordinate plane, graph $F'(t)$. Does the linear, quadratic, or cubic function fit the data best?

52) Using the best linear, quadratic, and cubic fits to the data, determine what $H''(t)$, $G''(t)$ and $F''(t)$ are. What are the physical meanings of $H''(t)$, $G''(t)$ and $F''(t)$, and what are their units?

Answer

$H''(t) = 0$, $G''(t) = 2.858$ and $f''(t) = 1.222t + 5.912$ represent the acceleration of the rocket, with units of meters per second squared $(\mathrm{m/s^2})$.

3.2E: Exercises for Section 3.2 is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by LibreTexts.

3.3: Differentiation Rules

Learning Objectives

- State the constant, constant multiple, and power rules.
- Apply the sum and difference rules to combine derivatives.
- Use the product rule for finding the derivative of a product of functions.
- Use the quotient rule for finding the derivative of a quotient of functions.
- Extend the power rule to functions with negative exponents.
- Combine the differentiation rules to find the derivative of a polynomial or rational function.

Finding derivatives of functions by using the definition of the derivative can be a lengthy and, for certain functions, a rather challenging process. For example, previously we found that

$$\frac{d}{dx}\left(\sqrt{x}\right) = \frac{1}{2\sqrt{x}}$$

by using a process that involved multiplying an expression by a conjugate prior to evaluating a limit.

The process that we could use to evaluate $\frac{d}{dx}\left(\sqrt[3]{x}\right)$ using the definition, while similar, is more complicated.

In this section, we develop rules for finding derivatives that allow us to bypass this process. We begin with the basics.

The Basic Rules

The functions $f(x) = c$ and $g(x) = x^n$ where n is a positive integer are the building blocks from which all polynomials and rational functions are constructed. To find derivatives of polynomials and rational functions efficiently without resorting to the limit definition of the derivative, we must first develop formulas for differentiating these basic functions.

The Constant Rule

We first apply the limit definition of the derivative to find the derivative of the constant function, $f(x) = c$. For this function, both $f(x) = c$ and $f(x+h) = c$, so we obtain the following result:

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h\to 0} \frac{c - c}{h} \\ &= \lim_{h\to 0} \frac{0}{h} \\ &= \lim_{h\to 0} 0 = 0. \end{aligned}$$

The rule for differentiating constant functions is called the **constant rule**. It states that the derivative of a constant function is zero; that is, since a constant function is a horizontal line, the slope, or the rate of change, of a constant function is 0. We restate this rule in the following theorem.

The Constant Rule

Let c be a constant. If $f(x) = c$, then $f'(x) = 0$.

Alternatively, we may express this rule as

$$\frac{d}{dx}(c) = 0.$$

Example 3.3.1: Applying the Constant Rule

Find the derivative of $f(x) = 8$.

Solution

This is just a one-step application of the rule: $f'(8) = 0$.

Exercise 3.3.1

Find the derivative of $g(x) = -3$.

Hint

Use the preceding example as a guide

Answer

0

The Power Rule

We have shown that

$$\frac{d}{dx}\left(x^2\right) = 2x \quad \text{and} \quad \frac{d}{dx}\left(x^{1/2}\right) = \frac{1}{2}x^{-1/2}.$$

At this point, you might see a pattern beginning to develop for derivatives of the form $\frac{d}{dx}\left(x^n\right)$. We continue our examination of derivative formulas by differentiating power functions of the form $f(x) = x^n$ where n is a positive integer. We develop formulas for derivatives of this type of function in stages, beginning with positive integer powers. Before stating and proving the general rule for derivatives of functions of this form, we take a look at a specific case, $\frac{d}{dx}(x^3)$. As we go through this derivation, pay special attention to the portion of the expression in boldface, as the technique used in this case is essentially the same as the technique used to prove the general case.

Example 3.3.2: Differentiating x^3

Find $\frac{d}{dx}\left(x^3\right)$.

Solution:

$\frac{d}{dx}\left(x^3\right) = \lim_{h\to 0}\frac{(x+h)^3 - x^3}{h}$	
$= \lim_{h\to 0}\frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$	Notice that the first term in the expansion of $(x+h)^3$ is x^3 and the second term is $3x^2h$. All other terms contain powers of h that are two or greater
$= \lim_{h\to 0}\frac{3x^2h + 3xh^2 + h^3}{h}$	In this step the x^3 terms have been cancelled, leaving only terms containing h.
$= \lim_{h\to 0}\frac{h(3x^2 + 3xh + h^2)}{h}$	Factor out the common factor of h.
$= \lim_{h\to 0}(3x^2 + 3xh + h^2)$	After cancelling the common factor of h,the only term not containing h is $3x^2$.
$= 3x^2$	Let h go to 0.

Exercise 3.3.2

Find $\frac{d}{dx}(x^4)$.

Hint

Use $(x+h)^4 = x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4$ and follow the procedure outlined in the preceding example.

Answer

$\frac{d}{dx}(x^4) = 4x^3$

As we shall see, the procedure for finding the derivative of the general form $f(x) = x^n$ is very similar. Although it is often unwise to draw general conclusions from specific examples, we note that when we differentiate $f(x) = x^3$, the power on x becomes the coefficient of x^2 in the derivative and the power on x in the derivative decreases by 1. The following theorem states that the **power rule** holds for all positive integer powers of x. We will eventually extend this result to negative integer powers. Later, we will see that this rule may also be extended first to rational powers of x and then to arbitrary powers of x. Be aware, however, that this rule does not apply to functions in which a constant is raised to a variable power, such as $f(x) = 3^x$.

The Power Rule

Let n be a positive integer. If $f(x) = x^n$,then

$$f'(x) = nx^{n-1}.$$

Alternatively, we may express this rule as

$$\frac{d}{dx}(x^n) = nx^{n-1.}$$

Proof

For $f(x) = x^n$ where n is a positive integer, we have

$$f'(x) = \lim_{h\to 0}\frac{(x+h)^n - x^n}{h}.$$

Since

$$(x+h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \binom{n}{3}x^{n-3}h^3 + ... + nxh^{n-1} + h^n,$$

we see that

$$(x+h)^n - x^n = nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \binom{n}{3}x^{n-3}h^3 + ... + nxh^{n-1} + h^n.$$

Next, divide both sides by h:

$$\frac{(x+h)^n - x^n}{h} = \frac{nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \binom{n}{3}x^{n-3}h^3 + ... + nxh^{n-1} + h^n}{h}.$$

Thus,

$$\frac{(x+h)^n - x^n}{h} = nx^{n-1} + \binom{n}{2}x^{n-2}h + \binom{n}{3}x^{n-3}h^2 + ... + nxh^{n-2} + h^{n-1}.$$

Finally,

$$f'(x) = \lim_{h\to 0}(nx^{n-1} + \binom{n}{2}x^{n-2}h + \binom{n}{3}x^{n-3}h^2 + ... + nxh^{n-2} + h^{n-1})$$

$$= nx^{n-1}.$$

□

Example 3.3.3: Applying the Power Rule

Find the derivative of the function $f(x) = x^{10}$ by applying the power rule.

Solution

Using the power rule with $n = 10$, we obtain

$$f'(x) = 10x^{10-1} = 10x^9.$$

Exercise 3.3.3

Find the derivative of $f(x) = x^7$.

Hint

Use the power rule with $n = 7$.

Answer

$f'(x) = 7x^6$

The Sum, Difference, and Constant Multiple Rules

We find our next differentiation rules by looking at derivatives of sums, differences, and constant multiples of functions. Just as when we work with functions, there are rules that make it easier to find derivatives of functions that we add, subtract, or multiply by a constant. These rules are summarized in the following theorem.

Sum, Difference, and Constant Multiple Rules

Let $f(x)$ and $g(x)$ be differentiable functions and k be a constant. Then each of the following equations holds.

Sum Rule. The derivative of the sum of a function f and a function g is the same as the sum of the derivative of f and the derivative of g.

$$\frac{d}{dx}\big(f(x) + g(x)\big) = \frac{d}{dx}\big(f(x)\big) + \frac{d}{dx}\big(g(x)\big);$$

that is,

$$\text{for } s(x) = f(x) + g(x), \quad s'(x) = f'(x) + g'(x).$$

Difference Rule. The derivative of the difference of a function f and a function g is the same as the difference of the derivative of f and the derivative of g :

$$\frac{d}{dx}(f(x) - g(x)) = \frac{d}{dx}(f(x)) - \frac{d}{dx}(g(x));$$

that is,

$$\text{for } d(x) = f(x) - g(x), \quad d'(x) = f'(x) - g'(x).$$

Constant Multiple Rule. The derivative of a constant k multiplied by a function f is the same as the constant multiplied by the derivative:

$$\frac{d}{dx}\big(kf(x)\big) = k\frac{d}{dx}\big(f(x)\big);$$

that is,

$$\text{for } m(x) = kf(x), \quad m'(x) = kf'(x).$$

Proof

We provide only the proof of the sum rule here. The rest follow in a similar manner.

For differentiable functions $f(x)$ and $g(x)$, we set $s(x) = f(x) + g(x)$. Using the limit definition of the derivative we have

$$s'(x) = \lim_{h \to 0} \frac{s(x+h) - s(x)}{h}.$$

By substituting $s(x+h) = f(x+h) + g(x+h)$ and $s(x) = f(x) + g(x)$, we obtain

$$s'(x) = \lim_{h \to 0} \frac{\big(f(x+h) + g(x+h)\big) - \big(f(x) + g(x)\big)}{h}.$$

Rearranging and regrouping the terms, we have

$$s'(x) = \lim_{h \to 0} \left(\frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h} \right).$$

We now apply the sum law for limits and the definition of the derivative to obtain

$$s'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \to 0} \frac{g(x+h) - g(x)}{h} = f'(x) + g'(x).$$

□

Example 3.3.4: Applying the Constant Multiple Rule

Find the derivative of $g(x) = 3x^2$ and compare it to the derivative of $f(x) = x^2$.

Solution

We use the power rule directly:

$$g'(x) = \frac{d}{dx}(3x^2) = 3\frac{d}{dx}(x^2) = 3(2x) = 6x.$$

Since $f(x) = x^2$ has derivative $f'(x) = 2x$, we see that the derivative of $g(x)$ is 3 times the derivative of $f(x)$. This relationship is illustrated in Figure 3.3.1.

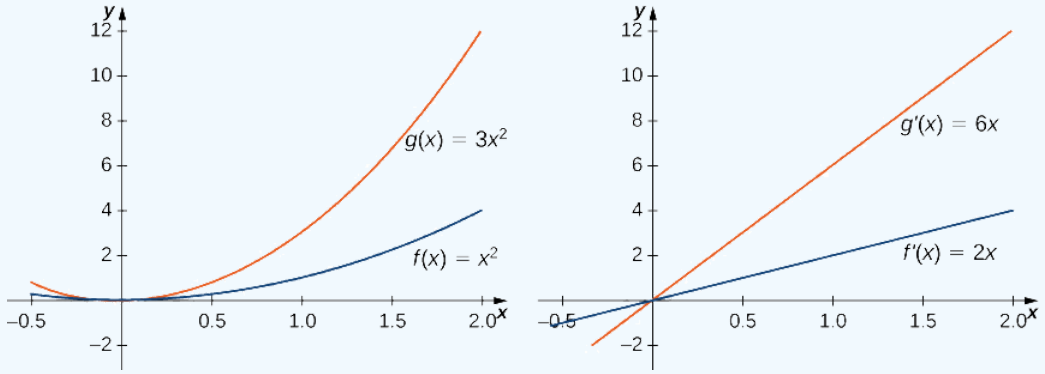


Figure 3.3.1: The derivative of $g(x)$ is 3 times the derivative of $f(x)$.

Example 3.3.5: Applying Basic Derivative Rules

Find the derivative of $f(x) = 2x^5 + 7$.

Solution

We begin by applying the rule for differentiating the sum of two functions, followed by the rules for differentiating constant multiples of functions and the rule for differentiating powers. To better understand the sequence in which the differentiation rules are applied, we use Leibniz notation throughout the solution:

$$
\begin{aligned}
f'(x) &= \frac{d}{dx}\left(2x^5+7\right) \\
&= \frac{d}{dx}(2x^5)+\frac{d}{dx}(7) && \text{Apply the sum rule.} \\
&= 2\frac{d}{dx}(x^5)+\frac{d}{dx}(7) && \text{Apply the constant multiple rule.} \\
&= 2(5x^4)+0 && \text{Apply the power rule and the constant rule.} \\
&= 10x^4 && \text{Simplify.}
\end{aligned}
$$

? Exercise 3.3.4

Find the derivative of $f(x)=2x^3-6x^2+3$.

Hint

Use the preceding example as a guide.

Answer

$f'(x)=6x^2-12x$.

✓ Example 3.3.6: Finding the Equation of a Tangent Line

Find the equation of the line tangent to the graph of $f(x)=x^2-4x+6$ at $x=1$

Solution

To find the equation of the tangent line, we need a point and a slope. To find the point, compute

$$f(1)=1^2-4(1)+6=3.$$

This gives us the point $(1,3)$. Since the slope of the tangent line at 1 is $f'(1)$, we must first find $f'(x)$. Using the definition of a derivative, we have

$$f'(x)=2x-4$$

so the slope of the tangent line is $f'(1)=-2$. Using the point-slope formula, we see that the equation of the tangent line is

$$y-3=-2(x-1).$$

Putting the equation of the line in slope-intercept form, we obtain

$$y=-2x+5.$$

? Exercise 3.3.5

Find the equation of the line tangent to the graph of $f(x)=3x^2-11$ at $x=2$. Use the point-slope form.

Hint

Use the preceding example as a guide.

Answer

$y=12x-23$

The Product Rule

Now that we have examined the basic rules, we can begin looking at some of the more advanced rules. The first one examines the derivative of the product of two functions. Although it might be tempting to assume that the derivative of the product is the product

of the derivatives, similar to the sum and difference rules, the **product rule** does not follow this pattern. To see why we cannot use this pattern, consider the function $f(x) = x^2$, whose derivative is $f'(x) = 2x$ and not $\frac{d}{dx}(x) \cdot \frac{d}{dx}(x) = 1 \cdot 1 = 1$.

Product Rule

Let $f(x)$ and $g(x)$ be differentiable functions. Then

$$\frac{d}{dx}(f(x)g(x)) = \frac{d}{dx}(f(x)) \cdot g(x) + \frac{d}{dx}(g(x)) \cdot f(x).$$

That is,

$$\text{if } p(x) = f(x)g(x), \quad \text{then } p'(x) = f'(x)g(x) + g'(x)f(x).$$

This means that the derivative of a product of two functions is the derivative of the first function times the second function plus the derivative of the second function times the first function.

Proof

We begin by assuming that $f(x)$ and $g(x)$ are differentiable functions. At a key point in this proof we need to use the fact that, since $g(x)$ is differentiable, it is also continuous. In particular, we use the fact that since $g(x)$ is continuous, $\lim_{h \to 0} g(x+h) = g(x)$.

By applying the limit definition of the derivative to $p(x) = f(x)g(x)$, we obtain

$$p'(x) = \lim_{h \to 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}.$$

By adding and subtracting $f(x)g(x+h)$ in the numerator, we have

$$p'(x) = \lim_{h \to 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h}.$$

After breaking apart this quotient and applying the sum law for limits, the derivative becomes

$$p'(x) = \lim_{h \to 0} \frac{f(x+h)g(x+h) - f(x)g(x+h)}{h} + \lim_{h \to 0} \frac{f(x)g(x+h) - f(x)g(x)}{h}.$$

Rearranging, we obtain

$$\begin{aligned} p'(x) &= \lim_{h \to 0} \left(\frac{f(x+h) - f(x)}{h} \cdot g(x+h) \right) + \lim_{h \to 0} \left(\frac{g(x+h) - g(x)}{h} \cdot f(x) \right) \\ &= \left(\lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \right) \cdot \left(\lim_{h \to 0} g(x+h) \right) + \left(\lim_{h \to 0} \frac{g(x+h) - g(x)}{h} \right) \cdot f(x) \end{aligned}$$

By using the continuity of $g(x)$, the definition of the derivatives of $f(x)$ and $g(x)$, and applying the limit laws, we arrive at the product rule,

$$p'(x) = f'(x)g(x) + g'(x)f(x).$$

□

Example 3.3.7: Applying the Product Rule to Constant Functions

For $p(x) = f(x)g(x)$, use the product rule to find $p'(2)$ if $f(2) = 3,\ f'(2) = -4,\ g(2) = 1$, and $g'(2) = 6$.

Solution

Since $p(x) = f(x)g(x)$, $p'(x) = f'(x)g(x) + g'(x)f(x)$, and hence

$p'(2) = f'(2)g(2) + g'(2)f(2) = (-4)(1) + (6)(3) = 14.$

Example 3.3.8: Applying the Product Rule to Binomials

For $p(x) = (x^2 + 2)(3x^3 - 5x)$, find $p'(x)$ by applying the product rule. Check the result by first finding the product and then differentiating.

Solution

If we set $f(x) = x^2 + 2$ and $g(x) = 3x^3 - 5x$, then $f'(x) = 2x$ and $g'(x) = 9x^2 - 5$. Thus,

$$p'(x) = f'(x)g(x) + g'(x)f(x) = (2x)(3x^3 - 5x) + (9x^2 - 5)(x^2 + 2).$$

Simplifying, we have

$$p'(x) = 15x^4 + 3x^2 - 10.$$

To check, we see that $p(x) = 3x^5 + x^3 - 10x$ and, consequently, $p'(x) = 15x^4 + 3x^2 - 10$.

Exercise 3.3.6

Use the product rule to obtain the derivative of $p(x) = 2x^5(4x^2 + x)$.

Hint

Set $f(x) = 2x^5$ and $g(x) = 4x^2 + x$ and use the preceding example as a guide.

Answer

$p'(x) = 10x^4(4x^2 + x) + (8x + 1)(2x^5) = 56x^6 + 12x^5$.

The Quotient Rule

Having developed and practiced the product rule, we now consider differentiating quotients of functions. As we see in the following theorem, the derivative of the quotient is not the quotient of the derivatives; rather, it is the derivative of the function in the numerator times the function in the denominator minus the derivative of the function in the denominator times the function in the numerator, all divided by the square of the function in the denominator. In order to better grasp why we cannot simply take the quotient of the derivatives, keep in mind that

$$\frac{d}{dx}(x^2) = 2x, \text{ not } \frac{\frac{d}{dx}(x^3)}{\frac{d}{dx}(x)} = \frac{3x^2}{1} = 3x^2.$$

The Quotient Rule

Let $f(x)$ and $g(x)$ be differentiable functions. Then

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{\frac{d}{dx}(f(x)) \cdot g(x) - \frac{d}{dx}(g(x)) \cdot f(x)}{\big(g(x)\big)^2}.$$

That is, if

$$q(x) = \frac{f(x)}{g(x)}$$

then

$$q'(x) = \frac{f'(x)g(x) - g'(x)f(x)}{\big(g(x)\big)^2}.$$